AP 去重整合题库

AP Biology Unit 1 ＋ Unit 3 酶相关 ＋ AP Chemistry Unit 2

已剔除全部经二次检查确认的重复题目：
第一部分 AP Biology Unit 1 Chemistry of Life：74 题（本单元无内部重复，沿用原编号 1–74，含答案解析）
第二部分 AP Biology Unit 3 酶相关题目：45 题（原 U3 Q95–126、Q129–141，剔除混入的 2 道呼吸作用题 Q127/Q128，重新编号 1–45，含答案解析）
第三部分 AP Chemistry Unit 2 全部：7 个课时 115 题（答案版，各课时内沿用原编号）＋ Progress Check FRQ 1 大题

说明：Progress Check MCQ 15 题经二次检查确认全部重复自课时题库（Q1–Q15 逐题对应见查重报告），本册不再收录；化学各课时另有一份同内容的纯题目版（未收录）。

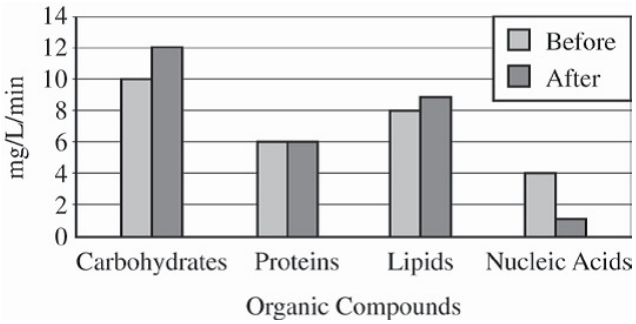
第一部分 AP Biology Unit 1: Chemistry of Life

74 题 ｜ 含答案解析 ｜ 沿用题库原编号 1–74（本单元无内部重复）

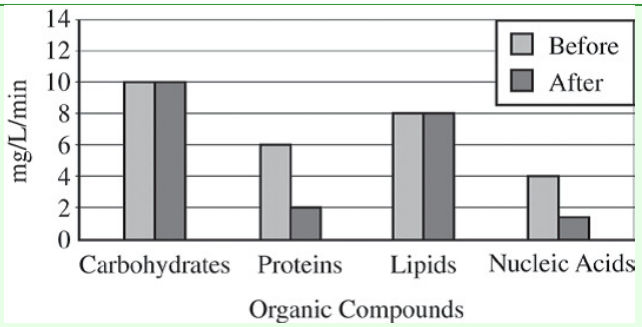
1 Chemistry of Life MCQ

1. A culture of *Spirogyra* (an autotrophic alga) is maintained in a water solution containing dissolved carbon dioxide and a source of phosphates but lacking nitrogen compounds. A researcher determines the rates of synthesis of several organic compounds found in the *Spirogyra* before and after several weeks in the water solution. Which of the following graphs best illustrates a likely result of the experiment?

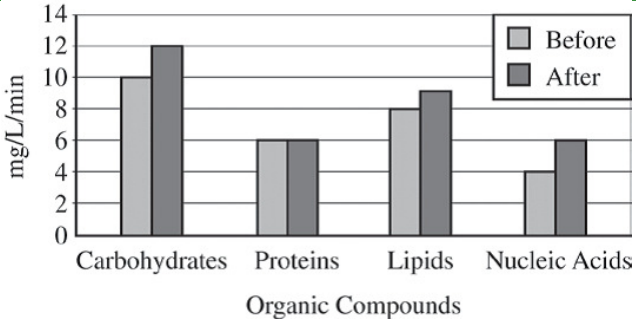
(A)



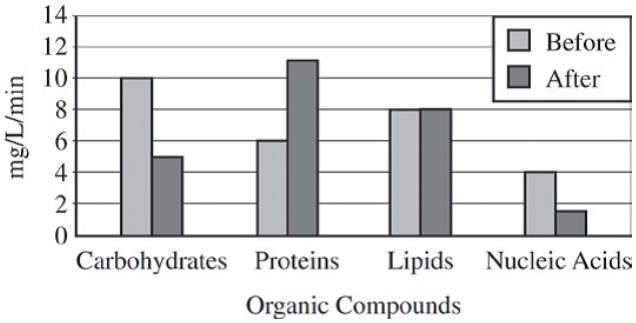
(B) ✓



(C)



(D)



1 Chemistry of Life MCQ

Answer B

This option is correct. Without a source of nitrogen, the *Spirogyra* cannot synthesize proteins or nucleic acids, both of which contain nitrogen. This answer reflects that the student is able to represent graphically the exchange of molecules between an organism and its environment and the use of these molecules in synthesis.

2. A feature of organic compounds NOT found in inorganic compounds is the presence of

(A) ionizing chemical groups

(B) electrons

(C) carbon atoms covalently bonded to each other ✓

(D) oxygen

(E) hydrogen bonds

3. tRNA molecules deliver amino acids to ribosomes where the amino acid next

(A) forms an ionic bond with another amino acid

(B) is deaminated and catabolized

(C) becomes an anticodon

(D) is covalently bonded to a growing polypeptide ✓

(E) forms hydrogen bonds with esters

4. The CFTR protein is made up of 1,480 amino acids linked together in a chain. Some humans produce a version of the CFTR protein in which phenylalanine (an amino acid) has been deleted from position 508 of the amino acid chain.

Which of the following best predicts how the amino acid deletion will affect the structure of the CFTR protein?

(A) It will have no observable effect on the structure of the CFTR protein.

(B) It will affect the primary structure of the CFTR protein, but the other levels of protein structure will not be affected.

(C) It will affect the secondary and tertiary structures of the CFTR protein, but the primary structure will not be affected.

(D) It will affect the primary, secondary, and tertiary structures of the CFTR protein. ✓

Answer D

Correct. A change in the primary structure of the CFTR protein will lead to changes in the secondary

and tertiary structures. As a result, the protein will not fold properly and will not function normally.

5. Which of the following describes a key difference among the 20 amino acids that are used to make proteins?

(A) Only some amino acids have an R-group.

(B) Only some amino acids have a carboxyl group (COOH).

(C) Some amino acids are hydrophobic. ✓

(D) Some amino acids contain the element phosphorus.

Answer C

Correct. Due to having nonpolar R-groups, 10 of the 20 amino acids are hydrophobic. Interactions between hydrophobic amino acids play an important role in determining protein structure and function.

6. Scientists examined the folded structure of a purified protein resuspended in water and found that amino acids with nonpolar R groups were primarily buried in the middle of the protein, whereas amino acids with polar R groups were primarily on the surface of the protein. Which of the following best explains the location of the amino acids in the folded protein?

(A) Polar R groups on the surface of the protein can form ionic bonds with the charged ends of the water molecules.

(B) Polar R groups are too bulky to fit in the middle of the protein and are pushed toward the protein's surface.

(C) Nonpolar R groups that cannot form hydrogen bonds with water are pushed into the middle of the protein. ✓

(D) Nonpolar R groups from different parts of the protein form covalent bonds with each other to maintain the protein's structure.

1 Chemistry of Life MCQ

7. Figure 1 shows three amino acids that are part of a polypeptide chain. Figure 2 shows the same section of the chain after a mutation has occurred.

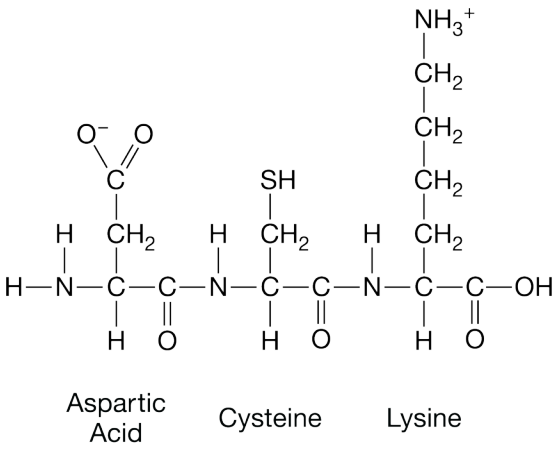


Figure 1: Original Amino Acid Chain

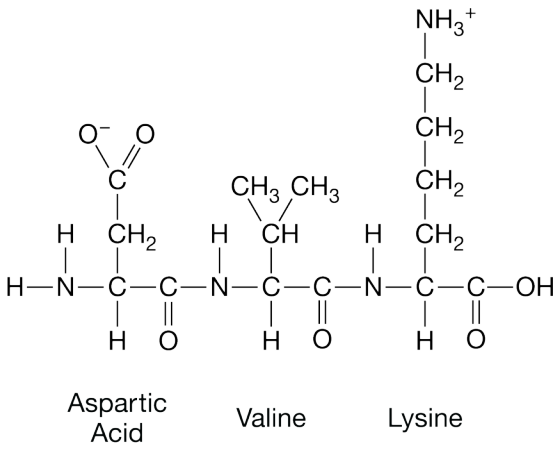


Figure 2: Mutated Amino Acid Chain

How might this change affect the structure and function of the protein?

(A) The R-group of the new amino acid, valine, has different chemical properties than the R-group of cysteine. This will cause the protein to misfold and not function properly in the cell. ✓

(B) The new amino acid, valine, has replaced cysteine in the new protein. Since the number of amino acids has remained the same, there will be no change in the three-dimensional folding, or function, of the protein.

(C) Since this is a linear section, it does not influence protein folding. Thus, there will be no change in protein structure or function.

(D) Since the new amino acid is bounded on one side by an amino acid with a negatively charged R-group and by an amino acid on the other side with a positively charged R-group, the charges will balance and the protein will fold as usual.

Answer A

Correct. Three-dimensional folding of a protein is due to interactions among the R groups of the amino acids. Cysteine has a sulfhydryl group, which may form a disulfide bridge with another part of the polypeptide chain. Valine has no sulfhydryl group and is nonpolar, which will affect how the polypeptide will fold.

1 Chemistry of Life MCQ

Students investigated the effect of light on the carbon cycle in aquatic ecosystems by performing the controlled experiment summarized below. The students placed equal amounts of water (pH 7.0) from a large aquarium in glass beakers. The students transferred aquatic plants from the aquarium to several of the beakers, and then they placed equal numbers of the beakers in the light or the dark (Figure 1: groups I and II). Similarly, the students transferred goldfish from the same aquarium to other beakers, and then they placed equal numbers of those beakers in the light or dark (Figure 1: groups III and IV). Finally, the students placed an equal number of beakers containing water only in the light or dark (Figure 1: groups V and VI).

After exposing the samples to light or dark for one hour, the students recorded the pH of the water in each beaker. Carbon dioxide dissolved in water will lower the pH of an aqueous solution. In the experiment, the students used changes in pH to monitor changes in the amount of carbon dioxide in the water. For each treatment group, the students calculated the mean pH and standard error, as documented in the table below.

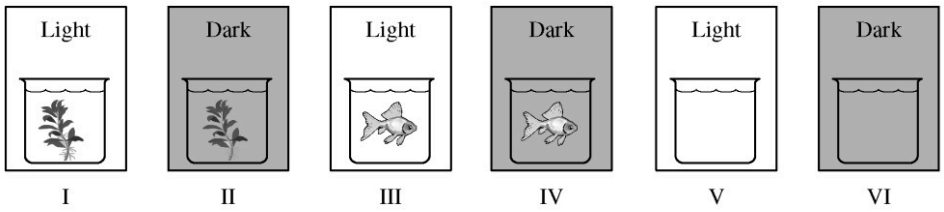


Figure 1. Treatment groups

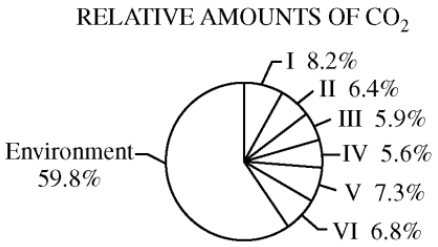
MEAN pH OF TREATMENT GROUPS AFTER 1 HOUR

Treatment group (n = 10)	I	II	III	IV	V	VI
Mean pH	8.2	6.4	5.9	5.6	7.3	6.8
Standard error of the mean	0.3	0.3	0.2	0.3	0.5	0.4

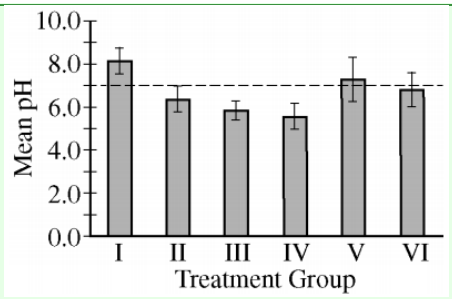
8. Which of the following graphs is the most appropriate representation of the experimental results documented in the table?

1 Chemistry of Life MCQ

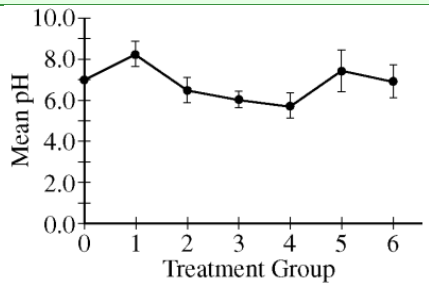
(A)



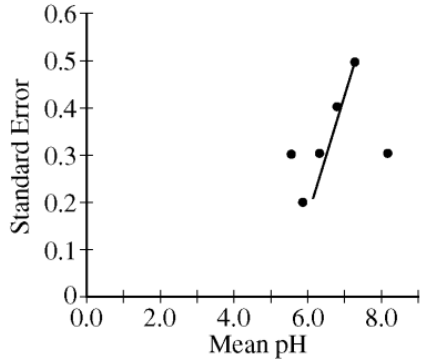
(B)



(C)



(D)



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9. Bacteriophages are viruses that infect bacteria. In an experiment, bacteriophages were labeled with either radioactive phosphorus or radioactive sulfur. The labeled bacteriophages were incubated with bacteria for a brief amount of time and then removed. The infected bacteria cells were found to contain significant amounts of radioactive phosphorus but not radioactive sulfur.

Based on the results of the experiment, which of the following types of molecules did the bacteriophages most likely inject into the bacteria cells?

1 Chemistry of Life MCQ

(A) Simple carbohydrate

(B) Amino acid

(C) DNA ✓

(D) Polypeptide

10.

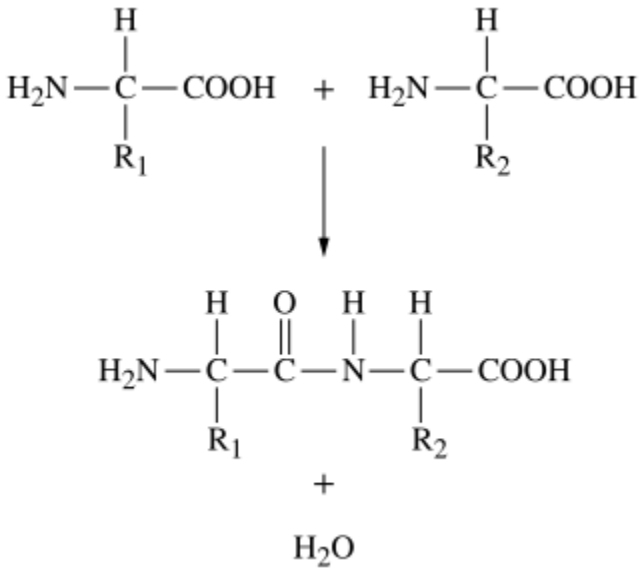


Figure 1. A representation of a reaction involved in the formation of a biological molecule

Which of the following best explains the reaction represented in Figure 1 ?

(A) It is a dehydration synthesis reaction involved in the formation of a lipid.

(B) The reaction is the hydrolysis of a nucleic acid and results in the formation of a nucleotide.

(C) The reaction is the hydrolysis of a nucleic acid and results in the formation of a phospholipid.

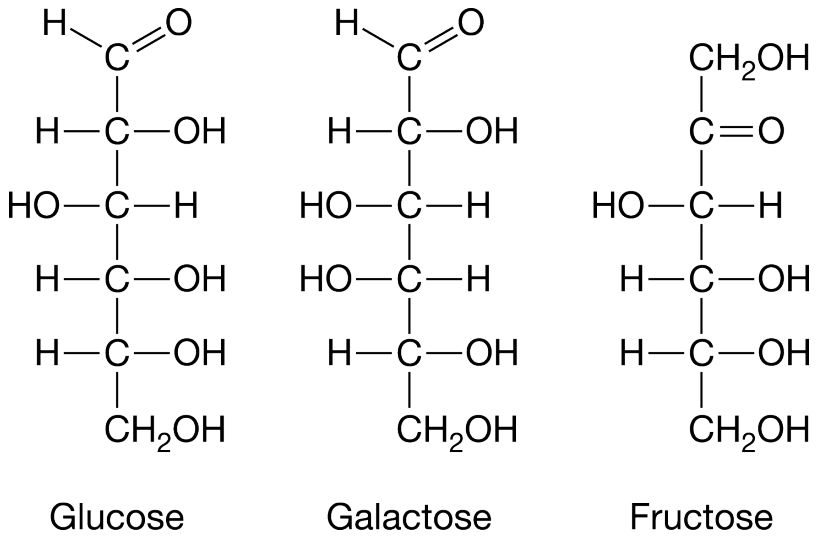
(D) It is a dehydration synthesis reaction involved in the formation of a polypeptide. ✓

Answer D

Correct. Two amino acids undergo a dehydration synthesis reaction, resulting in the formation of a peptide bond between them, the process involved in the growth of a polypeptide.

1 Chemistry of Life MCQ

11. The carbohydrates glucose, galactose, and fructose have the same chemical formula $(C_6H_{12}O_6)$ but different structural formulas, as represented in the figure.



Which of the following statements about glucose, galactose, and fructose is most likely true?

(A) The carbohydrates have the same properties because they have the same number of carbon, hydrogen, and oxygen atoms.

(B) The carbohydrates have the same properties because they each have a single carbon-oxygen double bond.

(C) The carbohydrates have different properties because they have different arrangements of carbon, hydrogen, and oxygen atoms. ✓

(D) The carbohydrates have different properties because they have different numbers of carbon-carbon bonds.

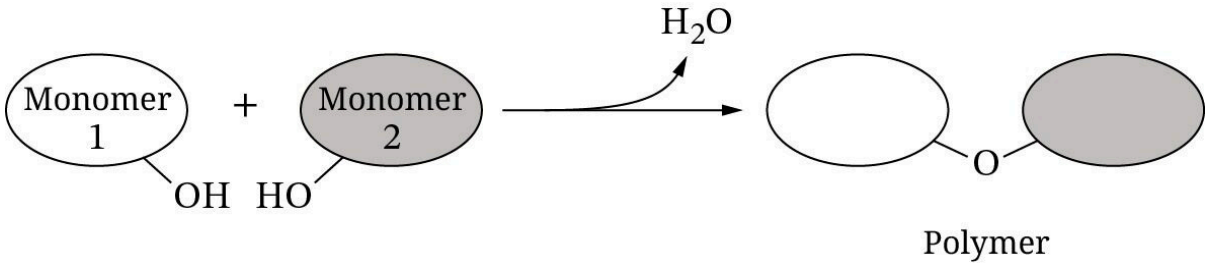
Answer C

Correct. Compounds that have the same chemical formula but different structural formulas usually have different properties.

1 Chemistry of Life MCQ

12. Figure 1 shows the formation of a carbohydrate polymer.

Figure 1. Formation of a carbohydrate polymer



Which of the following best describes how the carbohydrate polymer is being built in Figure 1?

(A) A water molecule is added to form a hydrogen bond between two monomers.

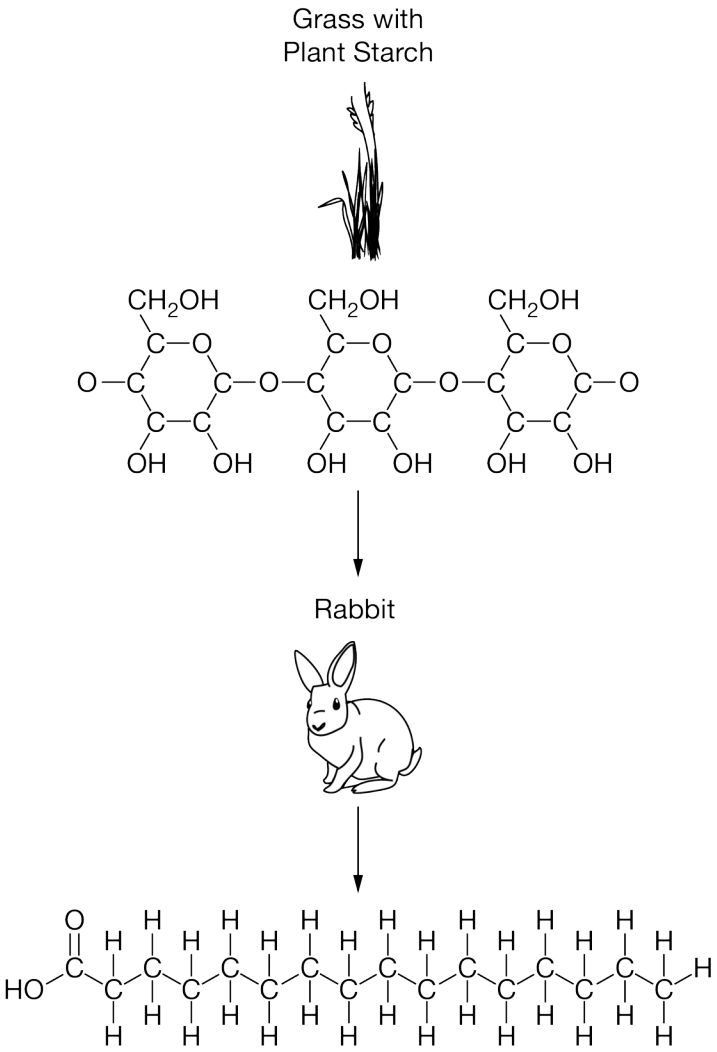
(B) A water molecule is added to form a covalent bond between two monomers

(C) A water molecule is removed to form a hydrogen bond between two monomers.

(D) A water molecule is removed to form a covalent bond between two monomers. ✓

1 Chemistry of Life MCQ

13.



Which of the following statements best describes how organisms such as rabbits obtain the carbon necessary for building biological molecules?

(A) Rabbits eat plants and use energy absorbed from the plants to make carbon atoms from electrons, protons, and neutrons in the air.

(B) Rabbits eat plants and break down plant molecules to obtain carbon and other atoms that they rearrange into new carbon-containing molecules. ✓

(C) Rabbits eat plants and use water absorbed from the plants to hydrolyze CO_2, which the rabbits breathe in from the air and use as a carbon source.

(D) Rabbits eat plants and make carbon-containing molecules by using carbon atoms that the plants absorbed from the soil and stored in the cells of their leaves.

1 Chemistry of Life MCQ

Answer B

Correct. Rabbits break down the plant molecules to obtain the individual atoms or small molecules that can be built into larger, complex biological molecules.

Different polysaccharides are used by plants for energy storage and structural support. The molecular structures for two common polysaccharides are shown in Figure 1. Starch is used by plants for energy storage, and cellulose provides structural support for cell walls. The monomer used to construct both molecules is glucose.

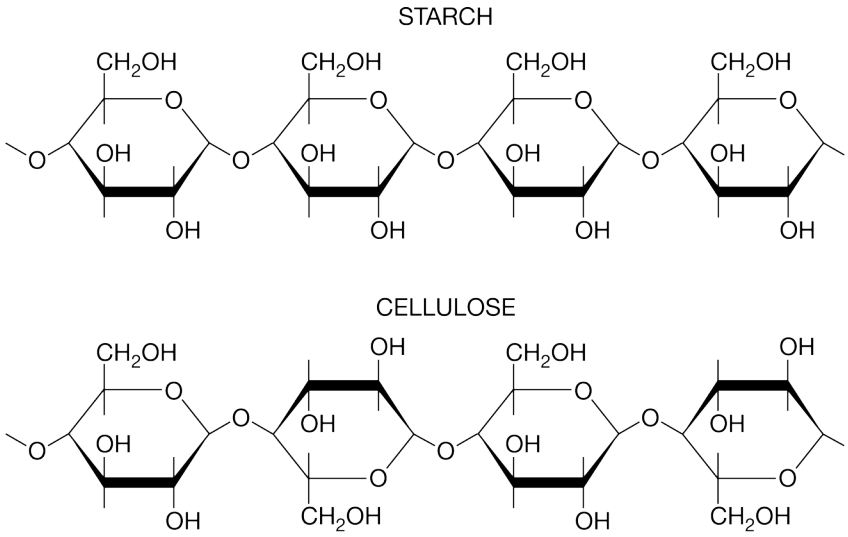


Figure 1. Comparison of segments of starch and cellulose

A study determined the effect of two different digestive enzymes, A and B, on these two polysaccharides. Table 1 presents the data from the study.

Table 1. Effect of Enzymes A and B on Cellulose and Starch

Test Tube	Polysaccharide Added	Enzyme Added	Glucose Detected after 5 Minutes at 37°C
1	Cellulose	A	No
2	Cellulose	B	Yes
3	Starch	A	Yes
4	Starch	B	No

Mammals do not produce digestive enzyme B. However, sheep and cattle are two types of mammals that contain microorganisms in their digestive tract that produce enzyme B.

1 Chemistry of Life MCQ

14. Which of the following would most likely occur if cattle lost the ability to maintain a colony of microorganisms in their digestive tract?

(A) Cattle would no longer be able to synthesize cellulose.

(B) Cattle would have to convert cellulose to starch before digesting it.

(C) Cattle would have to start producing enzyme B without the help of the bacteria.

(D) Cattle would no longer be able to use cellulose as a primary source of glucose. ✓

Answer D

Correct. Without the enzyme B produced by microorganisms in their digestive tract, cellulose would pass through the digestive tract without being digested.

15. Which of the following best describes the process that adds a monosaccharide to an existing polysaccharide?

(A) The monosaccharide is completely broken down by a specific enzyme and then the atoms are reorganized and made into a polysaccharide.

(B) Ionic bonds are formed between adjacent carbon atoms of the monosaccharide and the polysaccharide by adding water $(\mathrm{H_2O})$ and a specific enzyme.

(C) A specific enzyme removes the hydrogen (H) from the monosaccharide and the hydroxide (OH) from the polysaccharide, creating a bond between the two and creating a water $(\mathrm{H_2O})$ molecule. ✓

(D) A specific enzyme removes two hydroxides (OH), one from the monosaccharide, and one from the polysaccharide, creating a bond between the two monosaccharides and creating a hydrogen peroxide ($\mathrm{H_2O_2}$) molecule.

Answer C

Correct. This is a description of dehydration synthesis, which joins multiple monosaccharides to create a polysaccharide and produces water $(\mathrm{H_2O})$ molecules.

16. Based on Figure 1, which of the following best compares the atomic structures of starch and cellulose?

(A) Starch is composed of carbon, hydrogen, and oxygen, while cellulose also contains nitrogen.

(B) Starch and cellulose are composed of repeating glucose monomers; however, in cellulose every other glucose monomer is rotated 180 degrees. ✓

(C) Starch is composed of monomers that each have a $\mathrm{CH_2OH}$ group, while cellulose only has a $\mathrm{CH_2OH}$ group on every other monomer.

(D) Starch and cellulose are composed of identical monomers and therefore have identical structures.

1 Chemistry of Life MCQ

Answer B

Correct. Both starch and cellulose are both composed of repeating glucose molecules; however, the orientation of every other glucose in cellulose is upside down compared with the ones on either side.

17. Which of the following statements best describes the different functions of starch and cellulose in plants?

(A) The differences in the assembly and organization of the monomers of these two polymers result in different chemical properties. ✓

(B) Since starch and cellulose are composed of identical monomers, the cellular environment where they are located controls their function.

(C) The monomers of cellulose are connected by covalent bonds, making it idea for structural support.

(D) The monomers of starch are connected by ionic bonds, making it ideal for energy storage for plants.

Answer A

Correct. The identical orientations of the glucose monomers in starch create a polysaccharide with alpha bonds that is easy to break down into glucose for energy use. The alternating orientations of the glucose monomers in cellulose create beta bonds that produce a rigid polymer that is difficult to digest for energy use.

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18. Cholinesterase is an enzyme that catalyzes the hydrolysis of the signaling molecule acetylcholine. Which of the following best explains how cholinesterase carries out its function?

(A) Cholinesterase breaks covalent bonds within acetylcholine by the addition of a water molecule. ✓

(B) Cholinesterase breaks covalent bonds within acetylcholine by removing a water molecule.

(C) Cholinesterase forms covalent bonds between acetylcholine and another molecule by the addition of a water molecule.

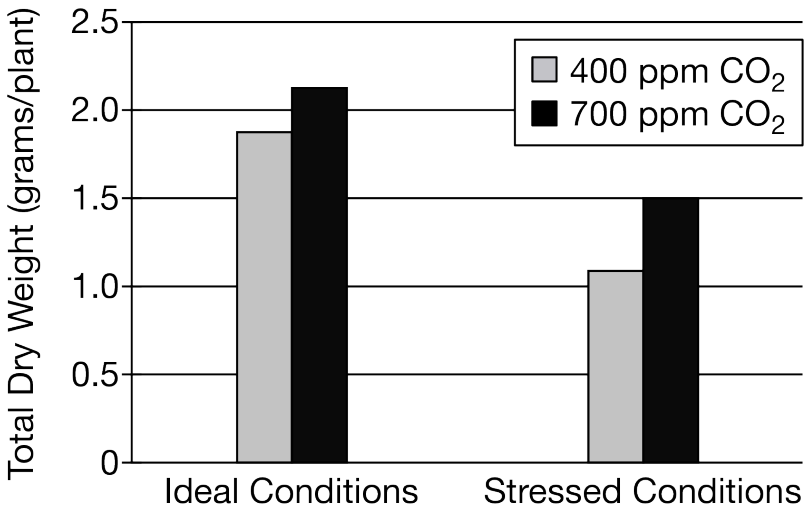
(D) Cholinesterase forms covalent bonds between acetylcholine and another molecule by removing a water molecule.

Answer A

Correct. Hydrolase enzymes catalyze the cleavage of covalent bonds by adding a water molecule to the bond.

1 Chemistry of Life MCQ

19. The figure shows the results of an experiment to investigate the effects of an enriched CO_2 environment on plant growth. Identical plants were separated into different groups and grown either in a standard CO_2 environment (400 ppm CO_2) or in an enriched CO_2 environment (700 ppm CO_2). Of the plants in each environment, half were grown under ideal conditions and half were grown under stressed conditions.



Based on the figure, which statement best describes the observed relationship between atmospheric CO_2 enrichment and plant growth under ideal and stressed conditions?

(A) The increase in atmospheric CO_2 had no observable effect on plant growth under either ideal or stressed conditions.

(B) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under ideal conditions than under stressed conditions.

(C) The increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions. ✓

(D) The increase in atmospheric CO_2 resulted in an inhibition of plant growth under both ideal and stressed conditions.

Answer C

Correct. Based on the figure, the increase in atmospheric CO_2 resulted in a greater increase in plant growth under stressed conditions than under ideal conditions.

20. Which of the following is responsible for the cohesive property of water?

1 Chemistry of Life MCQ

(A) Hydrogen bonds between the oxygen atoms of two adjacent water molecules

(B) Covalent bonds between the hydrogen atoms of two adjacent water molecules

(C) Hydrogen bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule ✓

(D) Covalent bonds between the oxygen atom of one water molecule and a hydrogen atom of another water molecule

(E) Hydrogen bonds between water molecules and other types of molecules

21. Which of the following communities is likely to be most stable?

(A) One where a keystone predator has been removed

(B) One with uniformly spaced vegetation

(C) One with high species diversity ✓

(D) One that lacks decomposers

22. Researchers compared similar proteins from related organisms in different habitats. They found that the proteins from organisms living in harsh environments had a greater number of cysteine amino acids than did proteins from organisms not living in harsh environments. The structure of cysteine is shown. Bonds can form between the sulfur atom of different cysteine amino acids (S-S bonds).

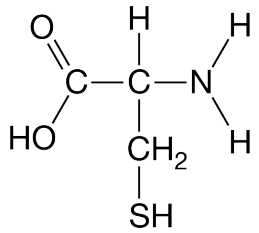


Figure 1. Chemical structure of cysteine

Which of the following best describes the effect of a greater number of cysteine amino acids on the stability of the proteins?

(A) The change has no effect on the stability of the protein because only one type of amino acid is involved.

(B) The change leads to increased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins. ✓

(C) The change leads to decreased protein stability because of an increased number of S-S bonds in the tertiary structure of the proteins.

(D) The change leads to increased protein stability only when the added cysteine amino acids are next to other cysteine amino acids in the primary structure.

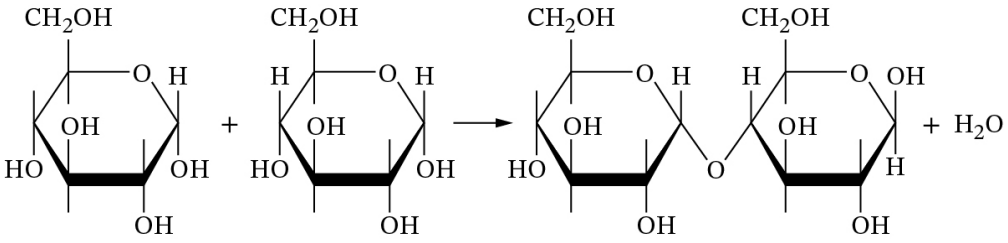
1 Chemistry of Life MCQ

Answer B

Correct. An increased number of S-S bonds are possible with the addition of more cysteine in the proteins. The S-S covalent bonds should add more structural stability to the proteins.

23. Figure 1 shows the formation of maltose (a disaccharide) from two glucose molecules.

Figure 1. Diagram of the formation of maltose



Which of the following statements best describes the molecular changes that occur during the reaction shown in Figure 1?

(A) Two hydroxyl groups are removed from each glucose molecule, and one water molecule is formed.

(B) A water molecule is added to the hydroxyl groups to form the covalent bond between the two glucose molecules.

(C) A hydroxyl group from one glucose molecule and a hydrogen atom from the other glucose molecule are removed, and one water molecule is formed. ✓

(D) A hydrogen atom in one glucose molecule and a hydroxyl group in the other glucose molecule form a hydrogen bond.

1 Chemistry of Life MCQ

24. A polypeptide is polymer of amino acids held together by peptide bonds. The process of dehydration synthesis creates these peptide bonds, as shown in Figure 1.

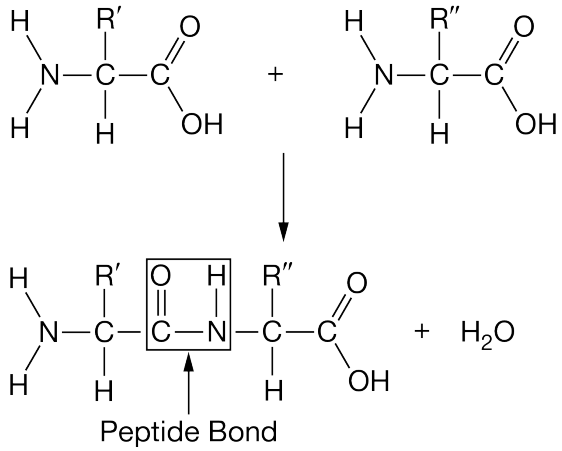


Figure 1. Amino acids are linked through the formation of peptide bonds.

As shown in Figure 1, an amino acid must have which of the following properties in order to be incorporated into a polypeptide?

(A) The ability to remain stable in the presence of water molecules

(B) An R-group that is compatible with the R-group of the last amino acid incorporated

(C) A central carbon atom that reacts with a nitrogen atom to form the peptide bond

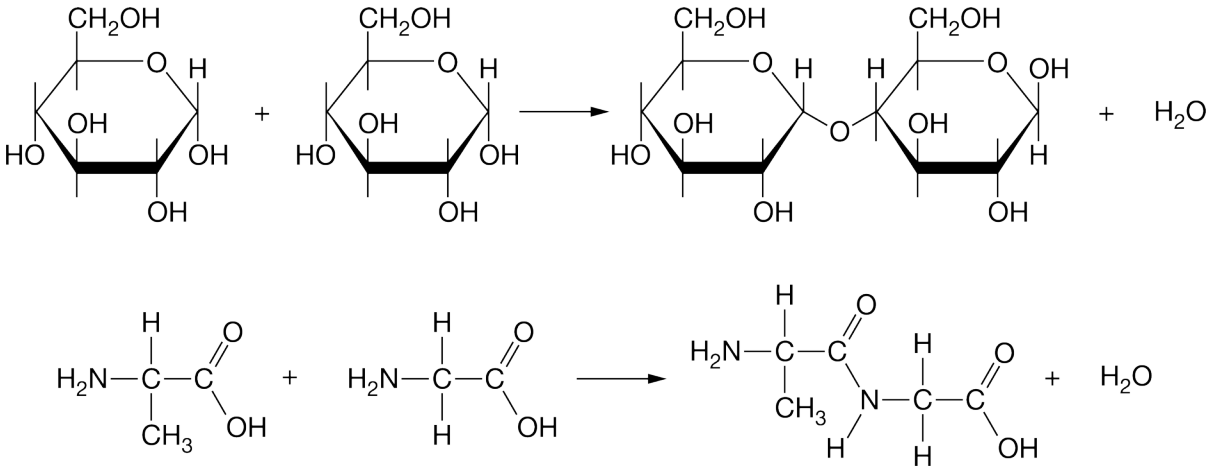
(D) The ability to form a covalent bond with both its NH_2 group and its COOH group ✓

Answer D

Correct. An amino acid needs to participate in the formation of two peptide bonds in order to be part of a polypeptide. The first bond may occur when the NH_2 group interacts with the COOH group of another amino acid. The second bond occurs when the COOH group interacts with the NH_2 group of a third amino acid.

1 Chemistry of Life MCQ

25.



Which of the following is common feature of the illustrated reactions showing the linking of monomers to form macromolecules?

(A) Two identical monomers are joined by a covalent bond.

(B) Two different monomers are joined by a covalent bond.

(C) Monomers are joined by a covalent bond, and a water molecule is produced. ✓

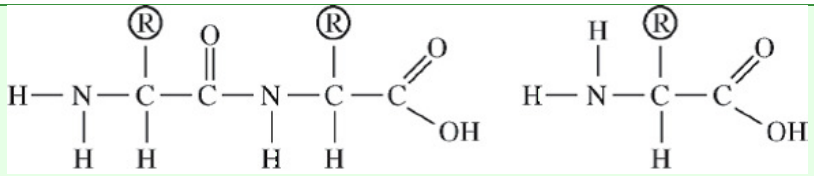
(D) Monomers are joined by ionic bonds, and a water molecule is produced.

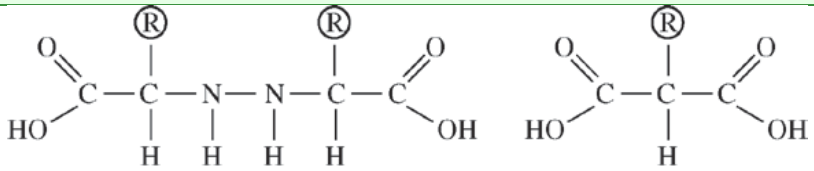
Answer C

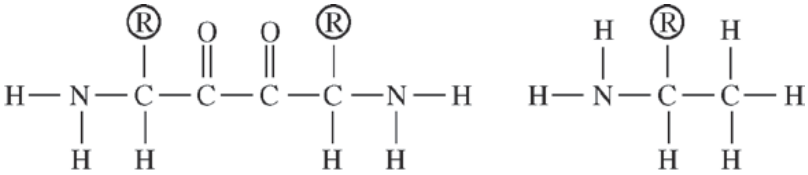
Correct. The monomers of the two reactions illustrated are joined by covalent bonds with the production of a water molecule.

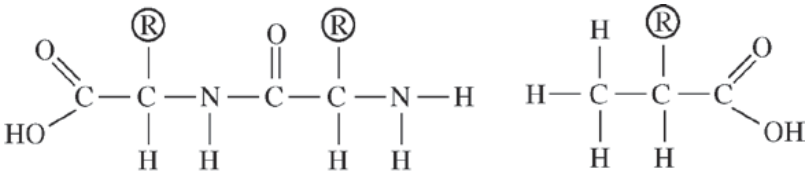
26. Which of the following correctly illustrates a dipeptide and an amino acid in the optimal position to form a tripeptide?

1 Chemistry of Life MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

This option is correct. It demonstrates an ability to refine and/or interpret a representation to explain the synthesis of a biological polymer. Peptides or proteins are polymers of amino acid monomers arranged in a unique linear sequence. Each of the 20 amino acids consists of a carbon atom surrounded by an amine group (NH_2), a carboxyl group (COOH), a hydrogen, and an R (variable) group. When two amino acids are positioned so that the carboxyl group of one amino acid backbone is adjacent to the amine group of another amino acid backbone, they can join by a dehydration reaction.

1 Chemistry of Life MCQ

27.

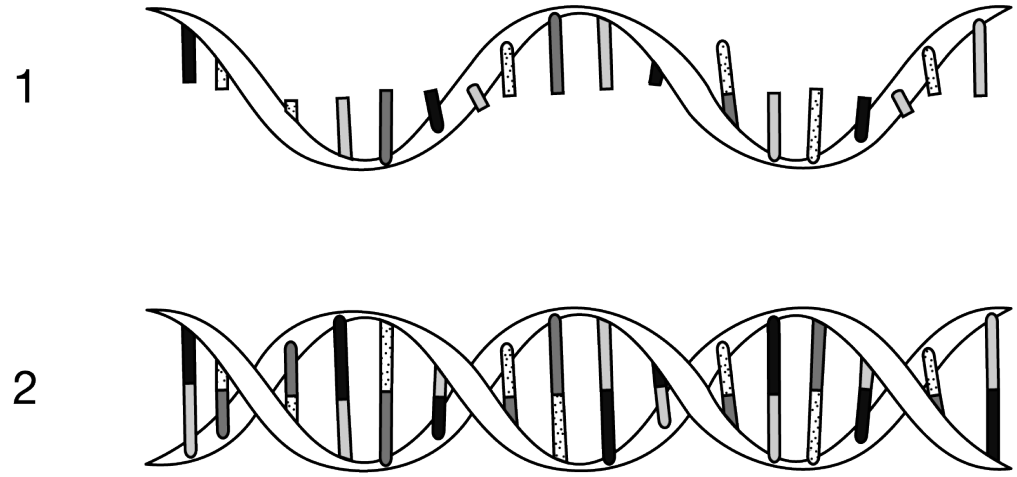


Figure 1. Molecule 1 represents RNA, and molecule 2 represents DNA.

Which of the following best describes a structural similarity between the two molecules shown in Figure 1 that is relevant to their function?

(A) Both molecules are composed of the same four nucleotides, which allows each molecule to be produced from the same pool of available nucleotides.

(B) Both molecules are composed of the same type of five-carbon sugar, which allows each molecule to act as a building block for the production of polysaccharides.

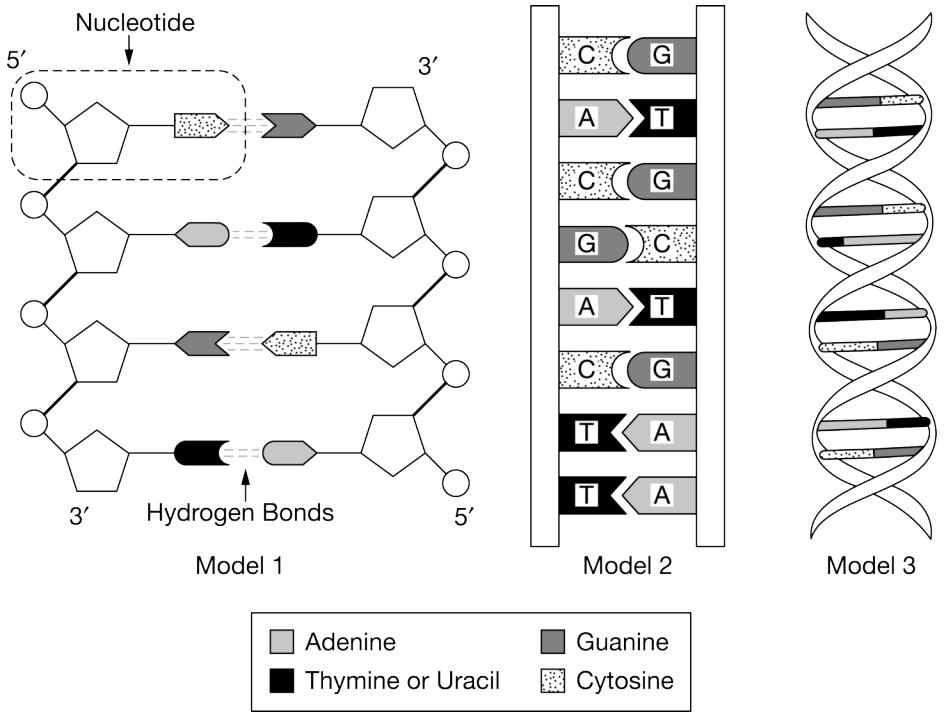
(C) Both molecules contain nucleotides that form base pairs with other nucleotides, which allows each molecule to act as a template in the synthesis of other nucleic acid molecules. ✓

(D) Both molecules contain nitrogenous bases and phosphate groups, which allows each molecule to be used as a monomer in the synthesis of proteins and lipids.

Answer C

Correct. Nucleotides form base pairs with other nucleotides. The base pairing allows a strand of RNA or DNA to act as a template in the synthesis of other nucleic acid molecules. Examples include the cellular processes of DNA replication (DNA is used as a template to make DNA), transcription (DNA is used as a template to make RNA), and reverse transcription (RNA is used as a template to make DNA).

1 Chemistry of Life MCQ



28.

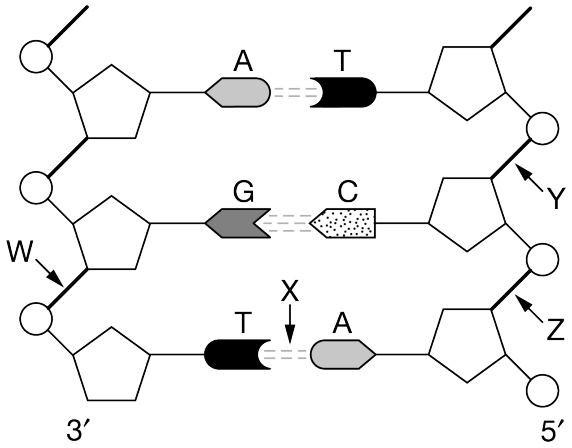


Figure 1. Four different bonds (W, X, Y, and Z) in a DNA molecule

Figure 1 represents a segment of DNA. Radiation can damage the nucleotides in a DNA molecule. To repair some types of damage, a single nucleotide can be removed from a DNA molecule and replaced with an undamaged nucleotide. Which of the four labeled bonds in Figure 1 could be broken to remove and replace the cytosine nucleotide without affecting the biological information coded in the DNA molecule?

(A) Bond X only

(B) Bond W only

(C) Bonds Y and Z at the same time ✓

(D) Bonds W and Z at the same time

1 Chemistry of Life MCQ

Answer C

Correct. By breaking bonds Y and Z at the same time, the cytosine nucleotide could be removed from the DNA molecule and replaced with an undamaged cytosine without changing the biological information stored in the DNA.

29. A student wants to modify model 1 so that it represents an RNA double helix instead of a DNA double helix. Of the following possible changes, which would be most effective in making model 1 look more like RNA than DNA ?

(A) Changing the sequence of the base pairs

(B) Changing the deoxyriboses to riboses by adding $-OH$ groups ✓

(C) Changing the shapes of the nitrogenous bases to match those shown in model 2

(D) Changing the sugar-phosphate backbone to a ribbon, as shown in model 3

Answer B

Correct. RNA contains ribose, whereas DNA contains deoxyribose. A ribose sugar has an $-OH$ group linked to the $2'$ carbon that a deoxyribose sugar does not have.

30. Which feature of model 1 best illustrates how biological information is coded in a DNA molecule?

(A) The $5'$ and $3'$ labels at the ends of each strand

(B) The labeling of the hydrogen bonds between base pairs

(C) The lines connecting sugars and phosphate groups that represent covalent bonds

(D) The linear sequence of the base pairs ✓

Answer D

Correct. The sequence of base pairs in a DNA molecule plays a central role in the coding of biological information.

1 Chemistry of Life MCQ

31. Figure 1 shows a short segment of a double-stranded nucleic acid molecule.

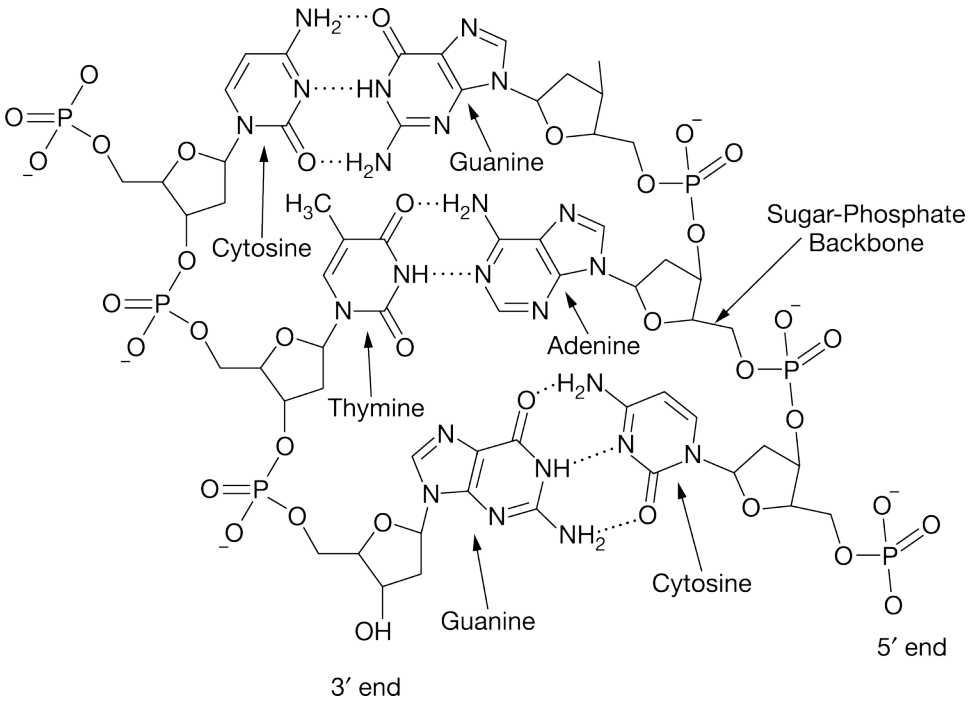


Figure 1. A short segment of a double-stranded nucleic acid molecule

Which of the following statements is correct about the molecule shown in Figure 1 ?

(A) It is RNA because of the relative direction of the two strands.

(B) It is RNA because of the number of different nucleotides found in the molecule.

(C) It is DNA because of the nature of the hydrogen bonds between guanine and cytosine.

(D) It is DNA because of the nucleotides present. ✓

Answer D

Correct. The figure indicates that one of the nucleotides is thymine. Thymine is found in DNA and not in RNA.

1 Chemistry of Life MCQ

32. Figure 1 represents a nucleic acid fragment that is made up of four nucleotides linked together in a chain.

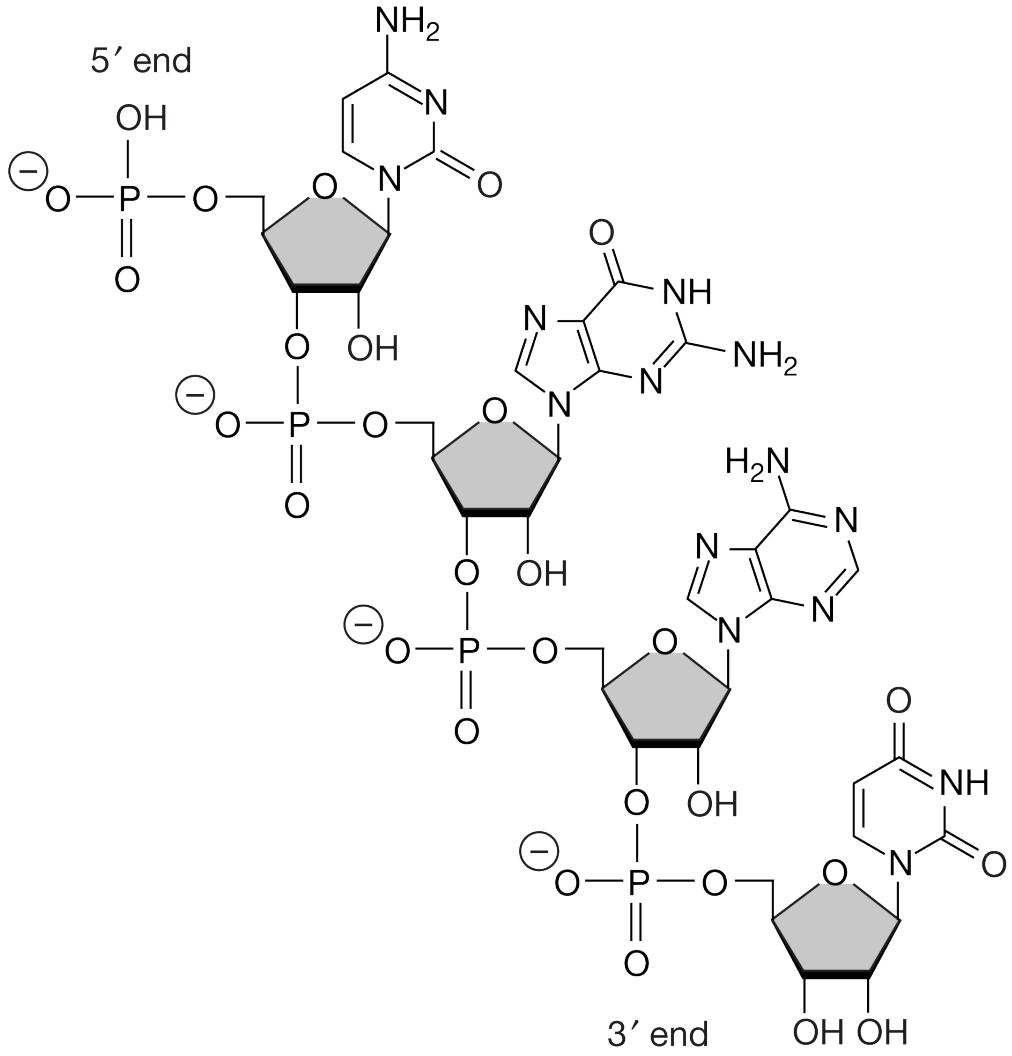


Figure 1. Nucleic acid fragment

Which of the following characteristics of Figure 1 best shows that the fragment is RNA and not DNA?

(A) The 5′ to 3′ orientation of the nucleotide chain

(B) The identity of each nitrogenous base ✓

(C) The charges on the phosphate groups

(D) The type of bond linking the nucleotides together

1 Chemistry of Life MCQ

Answer B

Correct. RNA contains uracil, whereas DNA contains thymine.

33.

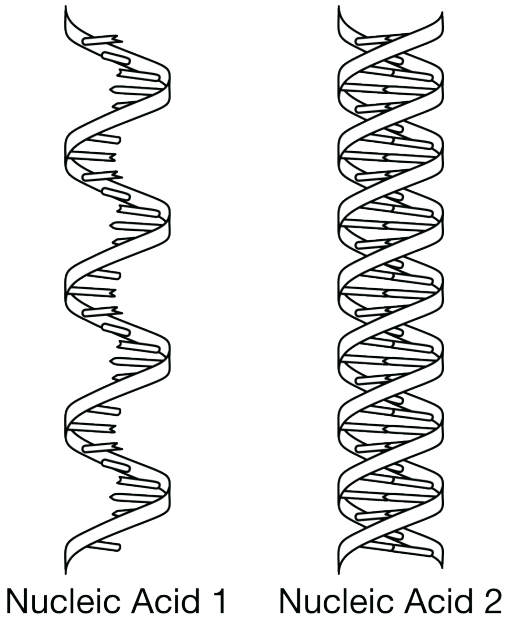


Figure 1. Nucleic acid segments

DNA and RNA are nucleic acids that can store biological information based on the sequence of their nucleotide monomers. Figure 1 shows a short segment of each of the two types of nucleic acids.

Which of the following best describes a structural difference between DNA and RNA?

(A) DNA contains four types of nitrogenous bases, whereas RNA contains only two types of nitrogenous bases.

(B) The backbone of DNA contains deoxyribose, whereas the backbone of RNA contains ribose. ✓

(C) A DNA molecule is composed of two parallel strands with the same $5'$ to $3'$ directionality, whereas an RNA molecule is composed of only one $5'$ to $3'$ strand.

(D) Phosphate groups provide rigidity to DNA, but RNA is flexible and contains no phosphate groups.

Answer B

Correct. The backbones of DNA and RNA are composed of an alternating sugar and base. In DNA, the sugar is deoxyribose, whereas the sugar in RNA is ribose.

1 Chemistry of Life MCQ



Figure 1. Domain structure of two proteins

Investigators studied the interactions between two different proteins, protein I and protein II. Each protein has two structural domains (Figure 1) that represent different functional parts of the protein. Different combinations of the protein domains were tested for their ability to bind to a known DNA sequence, interact with each other, and activate transcription. The results are shown in Table 1.

TABLE 1. INTERACTION AND FUNCTION OF COMBINATIONS OF STRUCTURAL VARIANTS OF PROTEINS I AND II

Test	Domain(s) of Protein I Used	Domain(s) of Protein II Used	DNA Binding	Protein-Protein Interaction	Transcription Activation
1	FG	LM	+	+	+
2	FG	None	+	−	−
3	None	LM	−	−	−
4	F	LM	+	−	−
5	G	LM	−	+	−
6	FG	L	+	−	−
7	FG	M	+	+	+

34. Which of the following claims is best supported by the data from test 5 in Table 1 ?

(A) Domain L is required for transcription activation.

(B) Domain F is not required for protein-protein interaction. ✓

(C) Protein-protein interaction is necessary for DNA binding.

(D) Protein-protein interaction is sufficient for transcription activation.

1 Chemistry of Life MCQ

Answer B

Correct. Based on the data for test 5 in which domain F is omitted but protein-protein interaction occurs, the G domain but not the F domain of protein I interacts with protein II. This is also confirmed in test 4 in which the F domain but not the G domain of protein I is present and in which protein-protein interaction does not occur.

35. Based on the data, which structural domain is most likely required for DNA binding?

(A) Domain F ✓
(B) Domain G
(C) Domain L
(D) Domain M

Answer A

Correct. Based on the data in Table 1, DNA binding occurs anytime both domains of protein I are present. The data for test 4 indicate that DNA binding also occurs when only domain F of protein I is present. The data for test 5 confirm that the presence of domain G is insufficient to allow DNA binding. Therefore, domain F is responsible for DNA binding by protein I.

36. Which of the following claims is best supported by the data in Table 1 ?

(A) Protein-protein interaction always results in transcription activation.
(B) DNA binding is required for protein-protein interaction.
(C) Protein-protein interaction is necessary for, but not adequate for, transcriptional activation. ✓
(D) DNA binding always results in transcription activation.

Answer C

Correct. Based on the data for tests 1 and 7, transcription activation occurs when both DNA binding and protein-protein interaction occur. The data for test 5 demonstrate that protein-protein interaction alone is insufficient for transcription activation to occur.

1 Chemistry of Life MCQ

37. Which of the following is the most likely result of testing a combination of domains G and M?

(A) No functionality

(B) DNA binding only

(C) Transcriptional activation only

(D) Protein-protein interaction only ✓

Answer D

Correct. Based on the data from tests 4 and 5, domain G (and not domain F) enables protein-protein interaction. Based on the data from tests 7 and 8, domain M (and not domain L) enables protein-protein interaction. Therefore, protein-protein interaction is expected to occur when domains G and M are combined.

---

38. A chemical binds to a protein composed of a single polypeptide chain and prevents the formation of an alpha helix that is typically formed in the absence of the chemical. Which of the following best describes the effect the chemical has on the structure of the protein?

(A) The primary structure held together by covalent bonds is affected.

(B) The secondary structure held together by hydrogen bonds is affected. ✓

(C) The secondary, tertiary, and quaternary structures are affected.

(D) All levels of protein structure are affected.

Answer B

Correct. Alpha helices are a form of secondary protein structure and are held together by hydrogen bonds. This chemical prevents alpha helices from forming, and therefore, it is most likely that the chemical affects hydrogen bonding within the local region of amino acids.

1 Chemistry of Life MCQ

39.

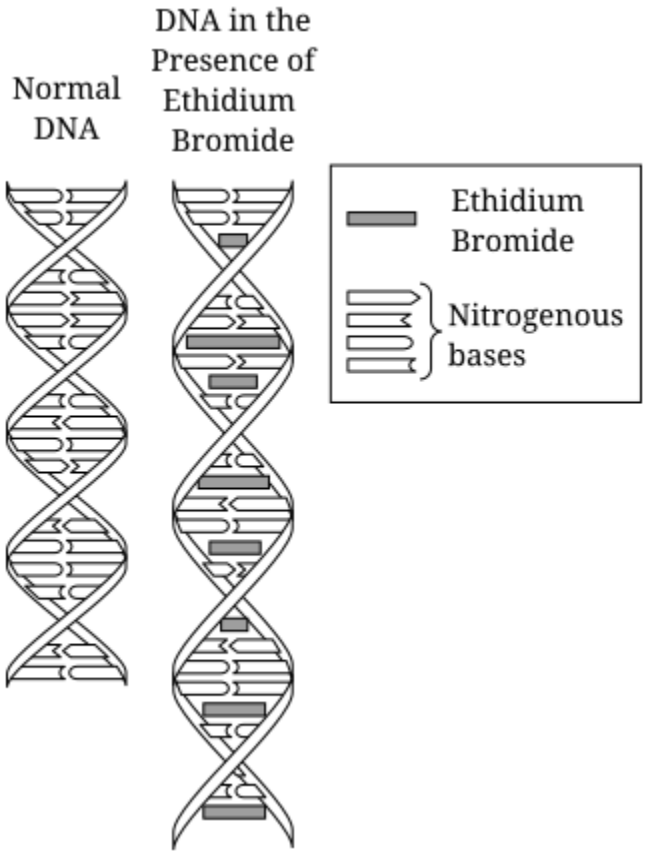


Figure 1. Model of DNA structure in the absence and presence of ethidium bromide

Ethidium bromide is a chemical that disrupts the replication of DNA. Based on the model shown in Figure 1, which of the following best describes the effect of ethidium bromide on DNA structure?

(A) It is inserted between complementary nitrogenous bases, causing the two strands of DNA to denature.

(B) It is inserted between complementary nitrogenous bases, causing hydrogen bonds to form between noncomplementary nitrogenous bases.

(C) It is inserted between sequential nitrogenous base pairs, increasing the length of the DNA molecule. ✓

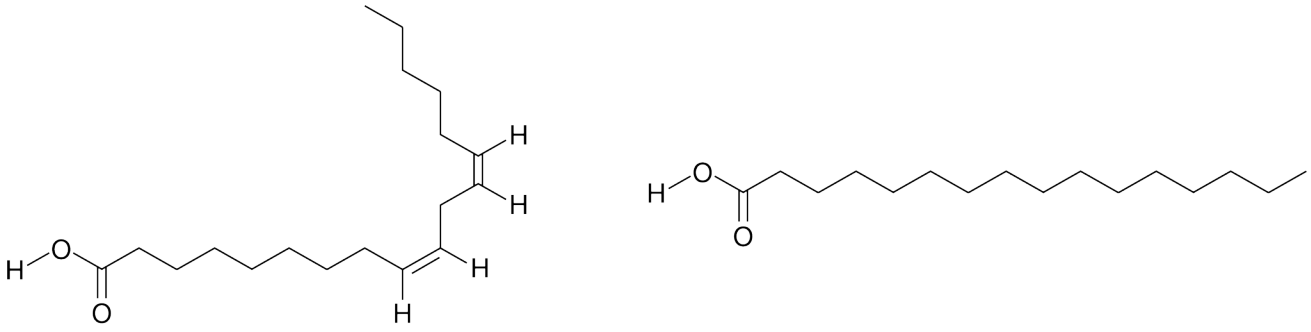
(D) It is inserted in place of certain nitrogenous base pairs, decreasing the length of the DNA molecule.

Answer C

Correct. Ethidium bromide is inserted between sequential base pairs, resulting in a longer DNA molecule.

1 Chemistry of Life MCQ

40. The molecular structures of linoleic acid and palmitic acid, two naturally occurring substances, are shown in the figure.



Linoleic Acid

Palmitic Acid

Based on the molecular structures shown in the figure, which molecule is likely to be solid at room temperature?

(A) Linoleic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together.

(B) Linoleic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

(C) Palmitic acid, because the absence of carbon-carbon double bonds allows the molecules to pack closely together. ✓

(D) Palmitic acid, because the presence of carbon-carbon double bonds prevents the molecules from packing closely together.

Answer C

Correct. Palmitic acid is a saturated fatty acid. The absence of carbon-carbon double bonds in palmitic acid allows the molecules to pack closely together and form a solid at room temperature.

41. The fluid mosaic model of cell membranes describes proteins and steroids associated with a phospholipid bilayer. Which of the following membrane components affect membrane stability?

(A) Embedded proteins give plasma membranes less stability.

(B) Steroids, such as cholesterol, give membranes more stability. ✓

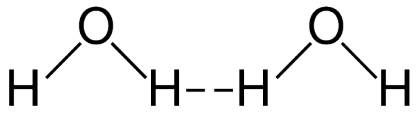
(C) Proteins in the hydrophilic region of the plasma membrane create more stability.

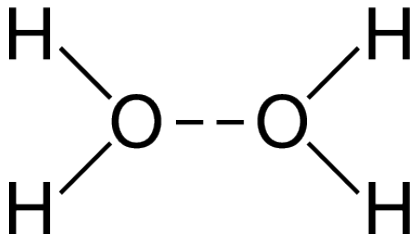
(D) Polar heads of phospholipids give plasma membranes less stability.

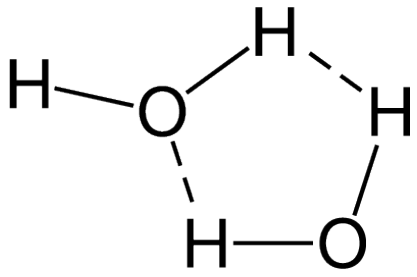
1 Chemistry of Life MCQ

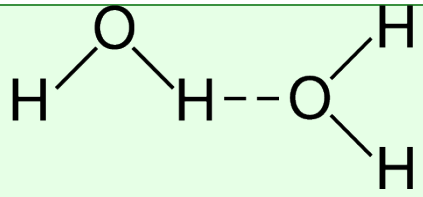
42. Water molecules are polar covalent molecules. There is a partial negative charge near the oxygen atom and partial positive charges near the hydrogen atoms due to the uneven distribution of electrons between the atoms, which results in the formation of hydrogen bonds between water molecules. The polarity of water molecules contributes to many properties of water that are important for biological processes.

Which of the following models best demonstrates the arrangement of hydrogen bonds between adjacent water molecules?

(A) 

(B) 

(C) 

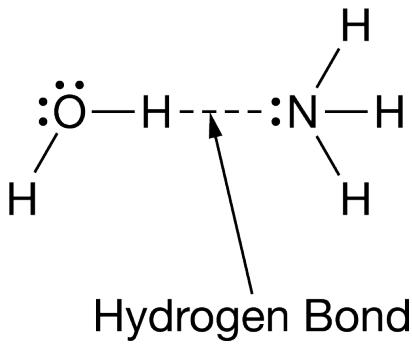
(D) 

Answer D

Correct. The hydrogen bonds between these water molecules correctly show the attractive force between the hydrogen atom of one water molecule and the oxygen atom of the adjacent water molecule.

1 Chemistry of Life MCQ

43. Water and ammonia interact to form hydrogen bonds, as represented in the figure.



Which statement best helps explain the formation of the hydrogen bond represented in the figure?

(A) The oxygen has a partial positive charge, and the nitrogen has a partial negative charge.

(B) The nitrogen has a partial negative charge, and the hydrogen attached to the oxygen has a partial positive charge. ✓

(C) The hydrogen attached to the oxygen has a partial negative charge, and the nitrogen also has a partial negative charge.

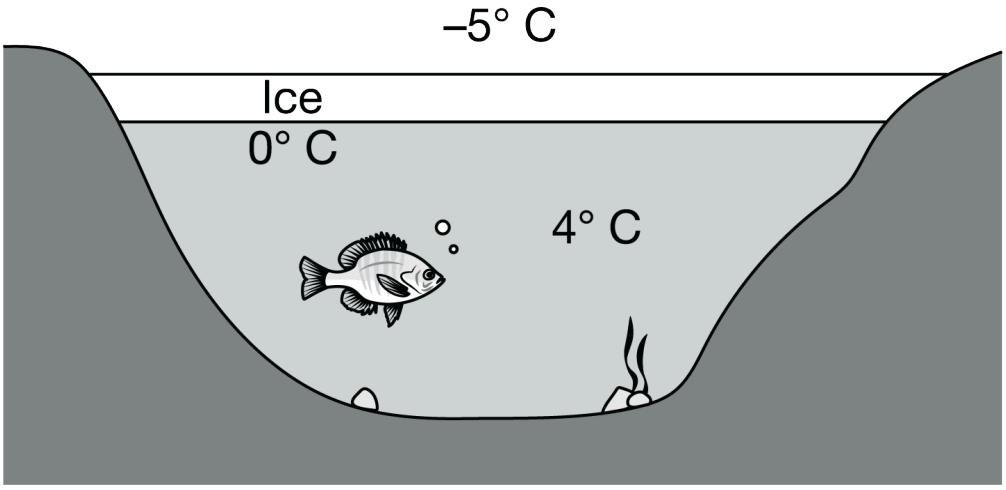
(D) The nitrogen has a partial positive charge, and the hydrogen attached to the oxygen also has a partial positive charge.

Answer B

Correct. The greater electronegativity of oxygen and nitrogen compared with hydrogen makes for unequal sharing of electrons, which results in partial negative charges associated with the oxygen and nitrogen atoms and partial positive charges associated with the hydrogen atoms in both molecules. The attraction between opposite charges results in the formation of the hydrogen bond represented in the figure.

1 Chemistry of Life MCQ

44.



As shown in the diagram, when environmental temperatures drop below freezing, a layer of ice typically forms on the surface of bodies of freshwater such as lakes and rivers.

Which of the following best describes how the structure of ice benefits the organisms that live in the water below?

(A) The water molecules in ice are closer together than those in liquid water, so the ice prevents the passage of air to the water, maintaining a constant gas mixture in the water.

(B) The water molecules in ice are closer together than those in liquid water, so the ice forms a barrier that protects the organisms in the water from the freezing air temperatures.

(C) The water molecules in ice are farther apart than those in liquid water, so the ice floats, maintaining the warmer, denser water at the lake bottom. ✓

(D) The water molecules in ice are farther apart than those in liquid water, so the ice floats, preventing the escape of gases from the liquid water.

Answer C

Correct. The water molecules in ice are farther from each other than are water molecules in liquid water, so ice is less dense than liquid water and floats on its surface, while the denser water at 4 degrees Celsius sinks to the bottom, maintaining a steady temperature all winter.

1 Chemistry of Life MCQ

45. The process of hydrogenation is used in the food industry to convert lipids that are liquids at room temperature into solids at room temperature. The process involves the chemical addition of hydrogen atoms to the fatty acid tails of lipids, breaking some or all of the carbon-carbon double bonds in the fatty acid tails.

Which of the following best explains why hydrogenation converts liquid lipids to solid lipids?

(A) Hydrogenation makes the fatty acids more saturated, enabling them to be more densely packed together. ✓

(B) Hydrogenation makes the fatty acids less saturated, enabling them to be more densely packed together.

(C) Hydrogenation makes the fatty acids more saturated, resulting in fewer lipid molecules fitting within a given space.

(D) Hydrogenation makes the fatty acids less saturated, resulting in fewer lipid molecules fitting within a given space.

Answer A

Correct. Adding hydrogen atoms results in fatty acid tails that are more saturated. The hydrocarbon chains of saturated fatty acids are straighter than those of unsaturated molecules and can therefore pack more closely together. More intermolecular forces result in higher melting points, converting the lipids to solids at room temperature.

46. One form of the genetic disorder lipodystrophy involves the lack of an enzyme that is required for the synthesis of triglycerides, a type of lipid. Triglycerides serve as an energy reserve in fat cells.

Which of the following best describes the most likely effect of lipodystrophy on cell function?

(A) There will be a decrease in phospholipids in cell membranes.

(B) There will be a decrease in cholesterol in cell membranes.

(C) There will be a decrease in fat storage in cells. ✓

(D) There will be a decrease in fatty acid synthesis in cells.

1 Chemistry of Life MCQ

47.

TABLE 1. PERCENT CHEMICAL COMPOSITION OF FOUR MACROMOLECULES

Macromolecule Sample	C (%)	O (%)	H (%)	N (%)	P (%)	S (%)
A	25	25	50	0	0	0
B	15	20	40	20	0	5
C	40	4	45	0	1	0
D	25	20	35	15	5	0

A researcher analyzed four different samples of macromolecules, where all macromolecules in each sample are of the same type. The researcher measured the percent of carbon, oxygen, hydrogen, nitrogen, phosphorus, and sulfur atoms in each sample. The results are shown in Table 1.

Which of the following claims is best supported by the data in Table 1 ?

(A) Sample A contains nucleic acids.

(B) Sample B contains protein. ✓

(C) Sample C contains nucleic acids.

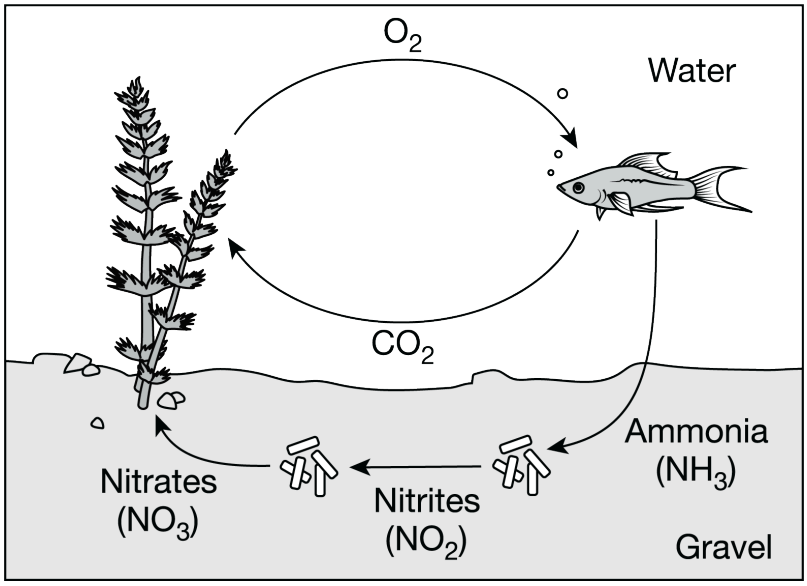
(D) Sample D contains protein.

Answer B

Correct. This sample is consistent with the atoms found in the largest quantities in proteins (C, H, O, N). A small amount of sulfur is also typically found in proteins. This sample does not have phosphorus, which would be indicative of a nucleic acid and not a protein.

1 Chemistry of Life MCQ

48. The figure shows a model of the exchange of matter between the organisms that live together in an aquarium. The model includes matter exchange between plants, fish, and bacteria. The bacteria are represented as rod-shaped organisms living in the gravel at the bottom of the aquarium.



Which of the following statements best describes how molecules released by the fish become nutrients for the plants?

(A) The carbon dioxide molecules released by the fish are converted by the bacteria to oxygen atoms, which are used by the plants to make water molecules.

(B) The oxygen molecules released by the fish are converted by the bacteria to ammonia molecules, which are used by the plants to make lipids and fatty acids.

(C) The nitrites released by the fish are converted by the bacteria to carbon dioxide molecules, which are used by the plants to make carbohydrates.

(D) The ammonia molecules released by the fish are converted by the bacteria to nitrates, which are used by the plants to make proteins and nucleic acids. ✓

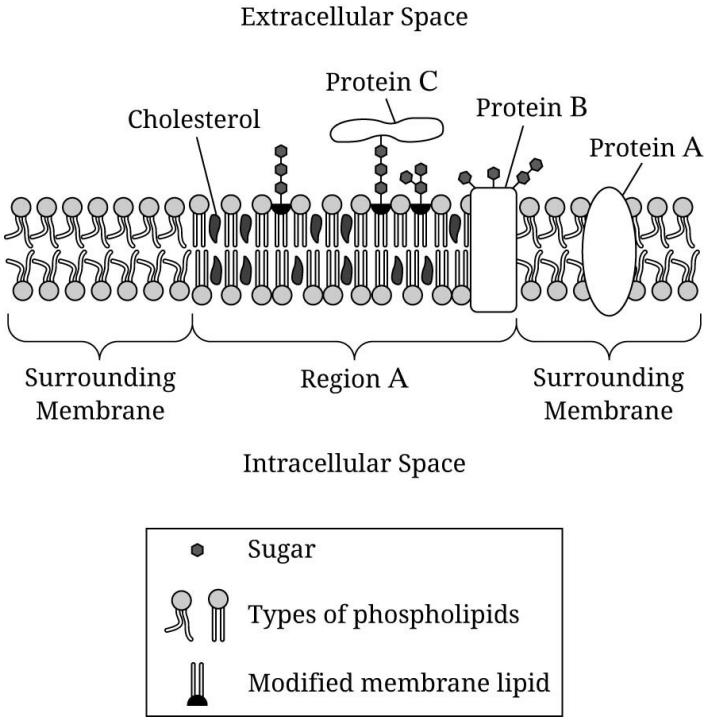
Answer D

Correct. As shown in the figure, the ammonia molecules released by the fish are converted by the bacteria to nitrites and nitrates. The plants absorb the nitrates, which are nitrogen-containing nutrients that the plants use to make the amino acids and nucleotides needed for building proteins and nucleic acids.

1 Chemistry of Life MCQ

49. Researchers have identified regions, including region A in Figure 1, of the cell membrane that have a different density and composition than the surrounding membrane.

Figure 1. Structure of a portion of the cell membrane



Which of the following best describes the chemical composition of the different types of macromolecules that make up the membrane represented in Figure 1?

(A) All the molecules contain carbon and oxygen, but only some contain nitrogen. ✓

(B) All the molecules contain carbon and hydrogen, but only some contain oxygen.

(C) All the molecules contain carbon and oxygen, but only some contain hydrogen.

(D) All the molecules contain carbon and nitrogen, but only some contain oxygen.

1 Chemistry of Life MCQ

50.

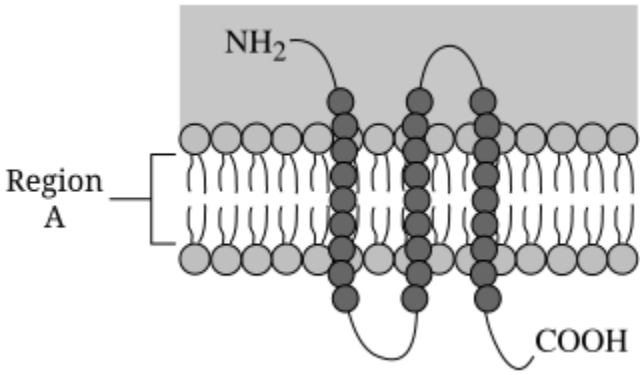


Figure 1. Structure of a membrane protein

Based on Figure 1, the amino acids in region A are most likely to have which of the following characteristics?

(A) Most amino acids will be hydrophobic because they interact most favorably with the phospholipid tails. ✓

(B) Most amino acids will be hydrophilic because they interact most favorably with the phospholipid heads.

(C) Most amino acids will be ionic amino acids because they interact most favorably with the phospholipid tails.

(D) Most amino acids will be polar amino acids because they interact most favorably with the phospholipid heads.

Answer A

Correct. Region A brackets the hydrophobic tails of the membrane phospholipids. In order for a protein to be embedded in the membrane among the phospholipid tails, the amino acids of the portion of the protein in region A must also be hydrophobic.

51. Which of the following statements best describes the structure of complex carbohydrates?

(A) Several polysaccharide chains are linked via hydrogen bonds.

(B) Monosaccharides are connected by covalent bonds. ✓

(C) Several polysaccharide chains are linked via ionic bonds.

(D) Monosaccharides are joined by peptide bonds.

1 Chemistry of Life MCQ

52. A small protein is composed of 110 amino acids linked together in a chain. As shown in Figure 1, the first and last five amino acids in the chain are hydrophobic (have nonpolar and uncharged R-groups), whereas the remaining 100 amino acids are hydrophilic (have charged or polar R-groups). The nature of the R-group determines if the amino acid is hydrophobic or hydrophilic.

A mutation results in the production of a version of the small protein that is only 105 amino acids long, as shown in Figure 2. Five of the hydrophobic amino acids are missing from one end of the chain.

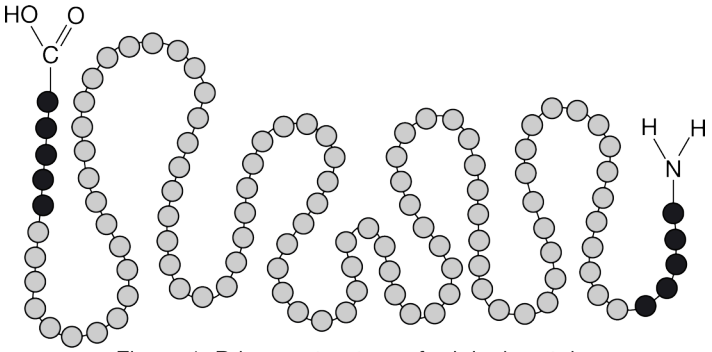
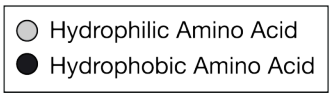


Figure 1. Primary structure of original protein



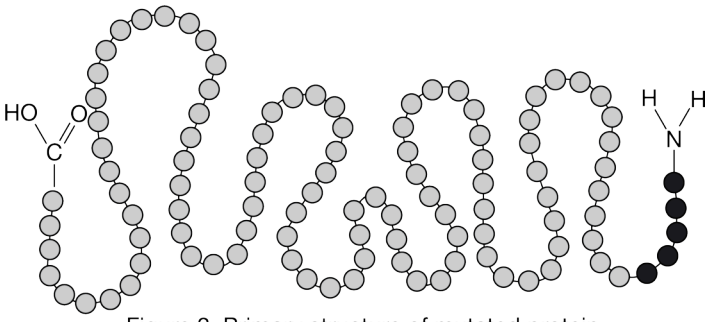
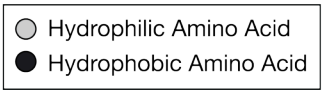
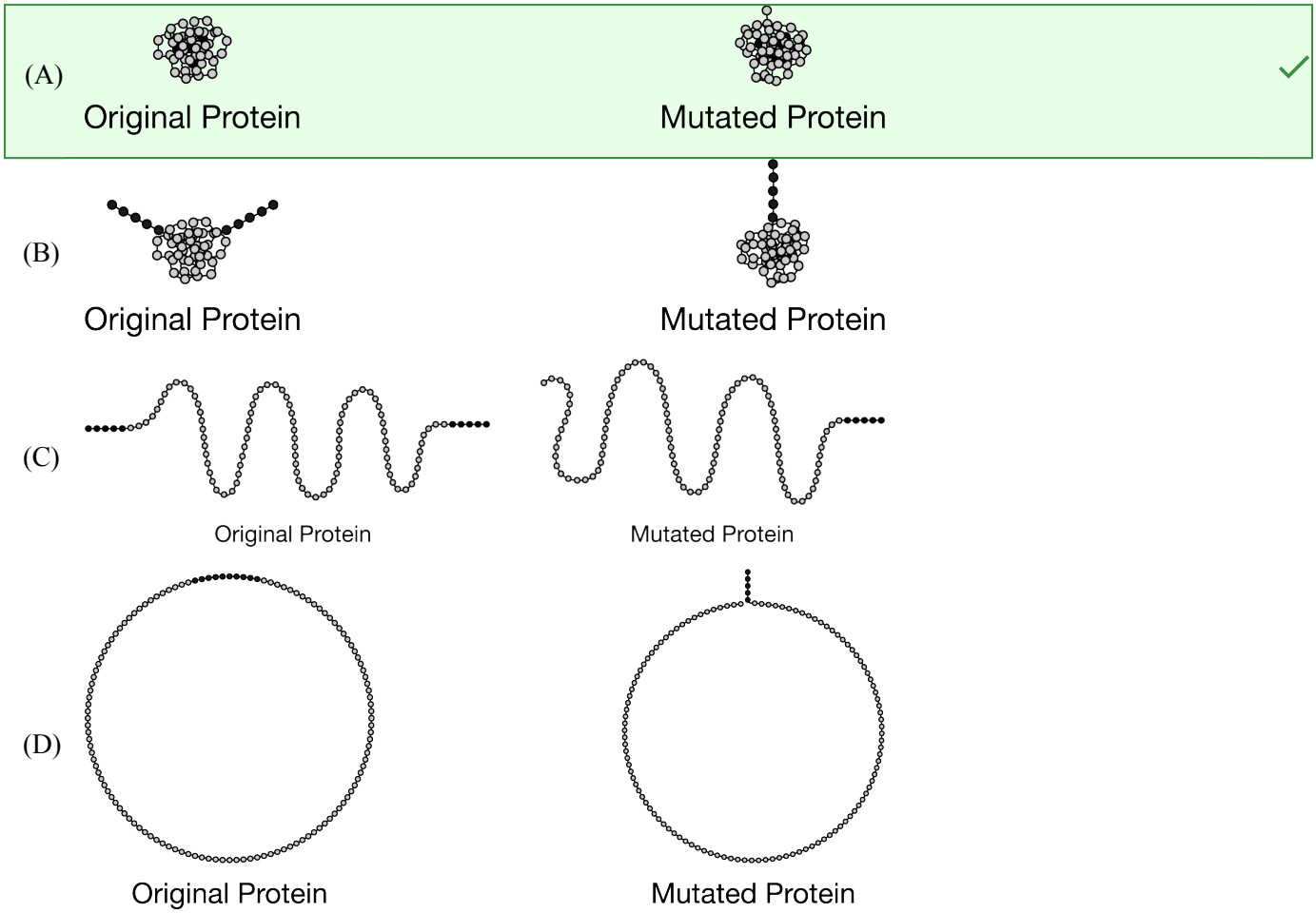


Figure 2. Primary structure of mutated protein



Which of the following best depicts the tertiary structures of the two proteins in water? The diagrams in the options are not drawn to the same scale as those in Figure 1 and Figure 2.

1 Chemistry of Life MCQ

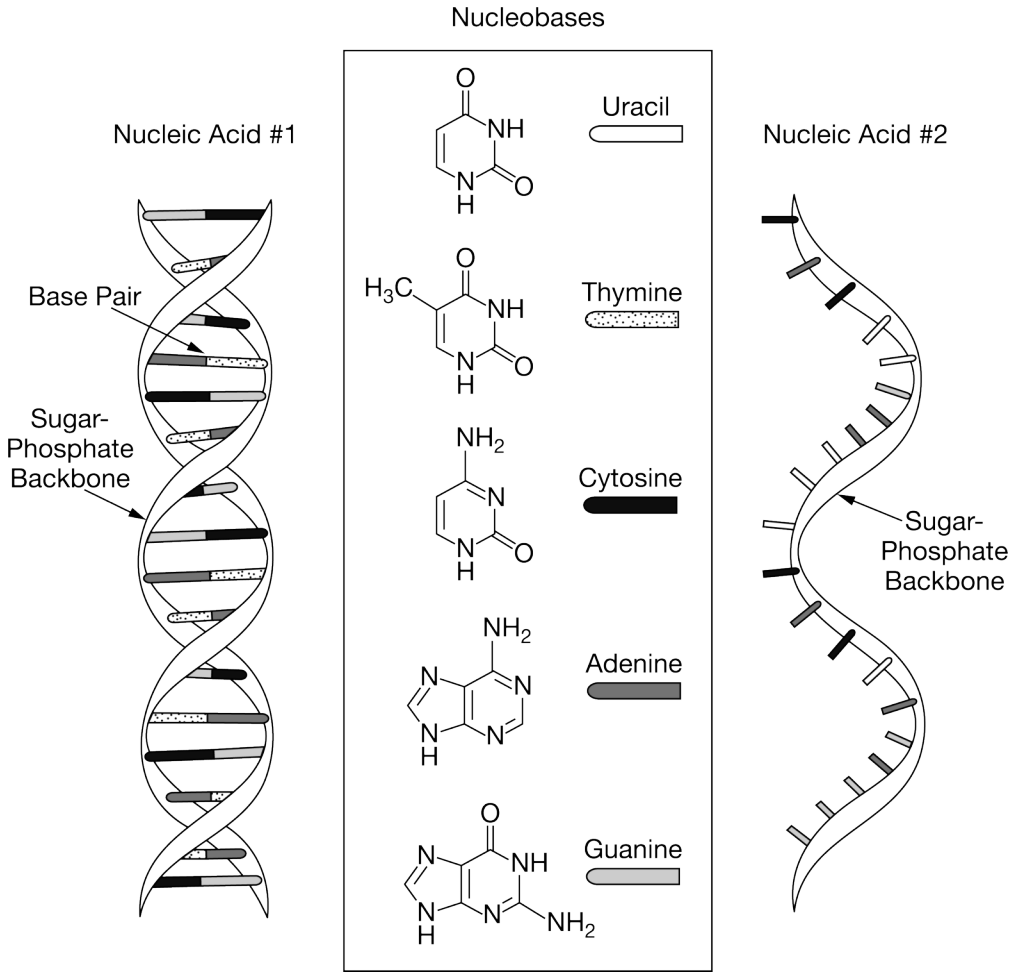


Answer A

Correct. The hydrophilic middle of the protein chain would be attracted to the water molecules, and the hydrophobic ends would be repelled. Hydrophobic portions of proteins in water tend to be found near the center of the tertiary structure, as far from the water as possible.

1 Chemistry of Life MCQ

53.



Which of the following conclusions is most clearly supported by the representations of nucleic acid #1 and nucleic acid #2 ?

(A) Nucleic acid #1 contains only purines, whereas nucleic acid #2 contains only pyrimidines.

(B) Nucleic acid #1 contains the sugar ribose, whereas nucleic acid #2 contains the sugar deoxyribose.

(C) Nucleic acid #1 contains positively charged phosphate groups, whereas nucleic acid #2 does not.

(D) Nucleic acid #1 contains adenine-thymine base pairs, whereas nucleic acid #2 does not. ✓

Answer D

Correct. The representation of nucleic acid #1 shows adenine-thymine base pairs, whereas the representation of nucleic acid #2 shows only uracil in the nucleotide chain and does not show any base pairing.

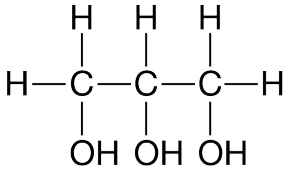
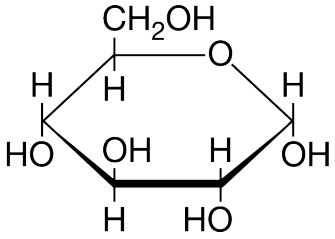
1 Chemistry of Life MCQ

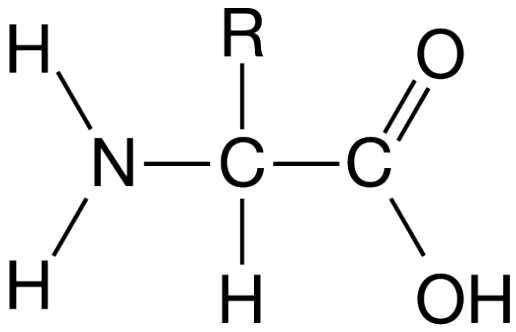
54. Students conducted a controlled experiment to investigate whether sawdust provides enough nutrients to support plant growth. The students separated ten nearly identical sunflower seedlings into two groups. They grew the seedlings in the first group in potting soil and the seedlings in the second group in sawdust composed mostly of cellulose. After twenty days, the students recorded observations about the seedlings in each group. The students' observations are presented in the table.

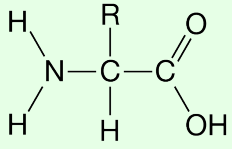
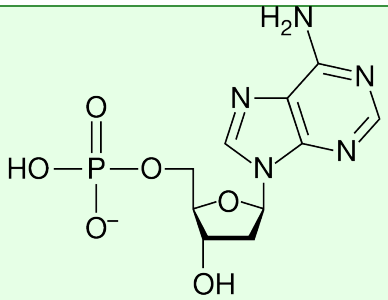
Treatment Group	Mean Seedling Height	Observations
Seedlings in potting soil	18.5 cm	The leaves have a dark green color and are normal in size.
Seedlings in sawdust	4.8 cm	The leaves have a grayish color and are small in size.

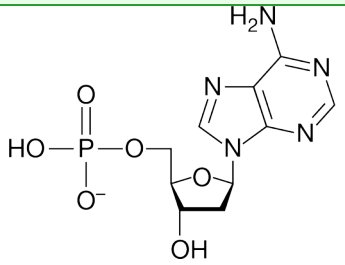
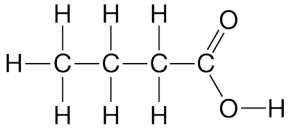
The observed differences between the groups most likely resulted from differences in the ability of the seedlings to produce which of the following monomers?

1 Chemistry of Life MCQ

(A)  and 

(B)  only

(C)  and 

(D)  and 

Answer C

Correct. Sawdust will not provide sufficient amounts of nitrogen and phosphorus, resulting in the seedlings having a reduced ability to produce both of the monomers shown (amino acids and nucleotides).

1 Chemistry of Life MCQ

55. Figure 1 represents a common process that occurs in organisms.

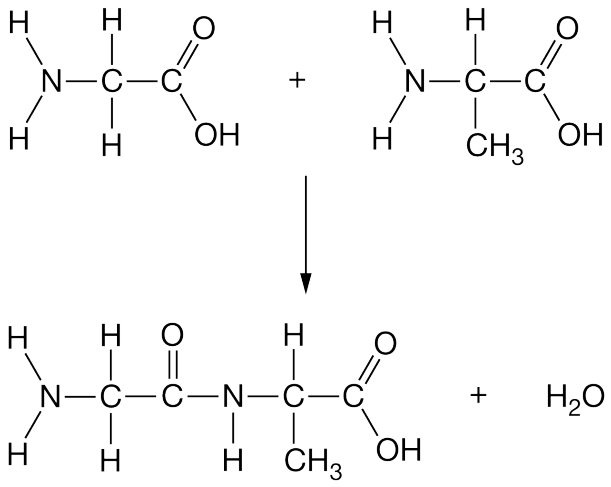


Figure 1. Structural formula for a common biological reaction

Which of the following is an accurate description of the process shown in Figure 1 ?

(A) The linking of amino acids with an ionic bond as an initial step in the protein synthesis process

(B) The formation of a more complex carbohydrate with the covalent bonding of two simple sugars

(C) The hydrolysis of amino acids with the breaking of covalent bonds with the release of water

(D) The formation of a covalent peptide bond in a dehydration synthesis reaction ✓

Answer D

Correct. This reaction combines two smaller amino acids to form a dipeptide (a step in the formation of a protein) with the release of water. The presence of nitrogen is consistent with both amino acids and proteins. The combining of smaller molecules to form a larger molecule with the release of water is a dehydration synthesis reaction.

1 Chemistry of Life MCQ

56.

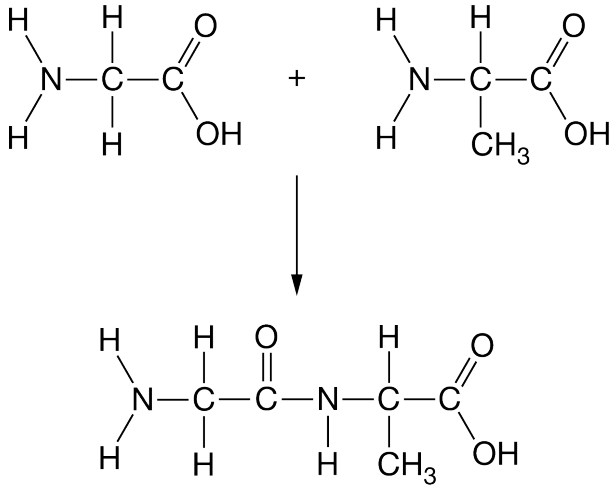


Figure 1. Formation of a peptide bond

Which of the following best describes the formation of the bond shown in Figure 1 ?

(A) An ionic bond is formed between a carbon atom of one amino acid and the nitrogen atom of the other amino acid.

(B) An ionic bond is formed when the negative charge of an OH group is balanced by the positive charge of a hydrogen ion.

(C) A covalent bond is formed between a carbon atom and a nitrogen atom along with the formation of H_2O . ✓

(D) A covalent bond is formed that replaces the hydrogen bond between the OH group and the H atom.

Answer C

Correct. Even though the water molecule that is produced is not shown in Figure 1, it shows the formation of the peptide bond and the missing H and OH.

57. Which of the following best describes the role of phospholipids?

(A) Phospholipids provide structural stability to animal cell membranes.

(B) Phospholipids support physiological functions, including growth.

(C) Phospholipids provide energy storage and support cell function.

(D) Phospholipids form the lipid bilayer found in cell membranes. ✓

1 Chemistry of Life MCQ

58. Phosphorous (P) is an important nutrient for plant growth. Figure 1 shows *Arabidopsis thaliana* plants grown under phosphorus-sufficient (left) and phosphorus-starved (right) conditions for six weeks.

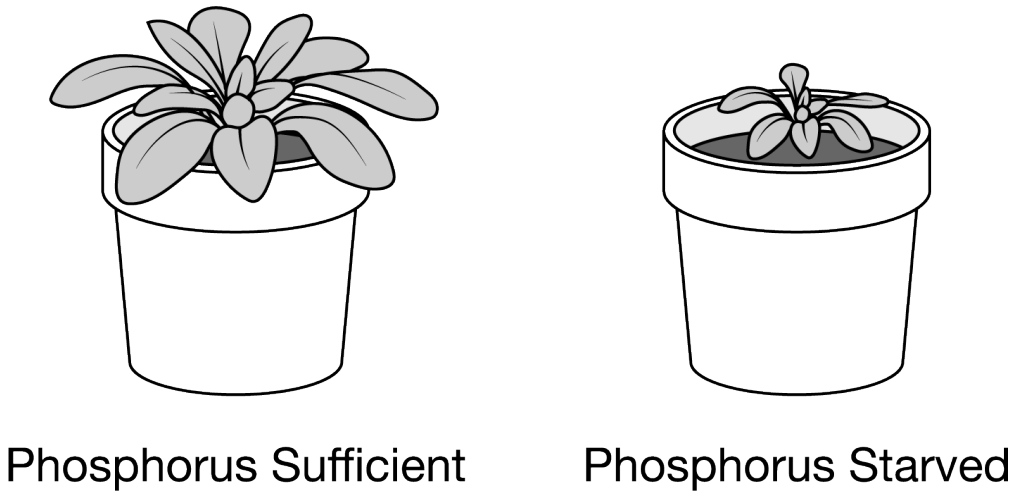


Figure 1. *Arabidopsis thaliana* plants grown for six weeks.

Which of the following is the most likely reason for the difference in leaf growth?

(A) The phosphorus-starved plant was unable to synthesize both the required proteins and lipids, limiting growth.

(B) The phosphorus-starved plant was unable to synthesize both the required proteins and carbohydrates, limiting growth.

(C) The phosphorus-starved plant was unable to synthesize both the required nucleic acids and lipids, limiting growth. ✓

(D) The phosphorus-starved plant was unable to synthesize both the required carbohydrates and nucleic acids, limiting growth.

Answer C

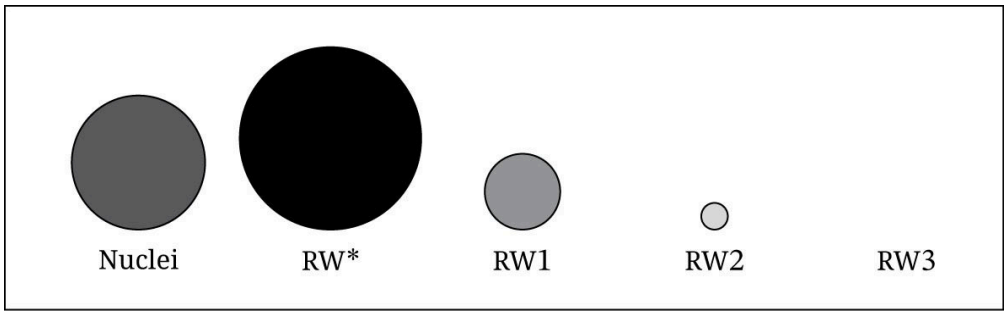
Correct. Phosphorus is used to make nucleic acids and certain lipids. Without phosphorus atoms, nucleic acids and lipids cannot be made for the plant to use for growth.

1 Chemistry of Life MCQ

59. A researcher conducted an experiment in a brightly lit aquarium containing a large population of single-celled green algae. Water containing radioactive phosphorus was added to the aquarium. After 7 days, the researcher took a water sample containing the algae and separated the algae from the radioactive water (RW*).

The researcher added deionized water to rinse the algae and saved the rinse water and then repeated that process twice. The rinse water from each of the three rinses was saved as RW1, RW2, and RW3, respectively. The researcher also removed and kept the nuclei of the algal cells. The nuclei were mixed with deionized water, and a drop of the mixture was placed on a piece of filter paper. A drop of the radioactive water and a drop of each of the rinse water samples were also placed on the same paper. These samples were exposed to X-ray to determine their radioactivity. The results after 24 hours are shown in Figure 1.

Figure 1. Radioactivity of saved algal water samples. Darker color and larger circle size indicate a higher level of radioactivity.



Based on Figure 1, which of the following statements best explains why the nuclei sample had a higher level of radioactivity than any of the rinse water samples?

(A) Phosphorus is used in the building of nucleic acids in the nucleus. ✓

(B) The nuclei sample was contaminated with radioactive water.

(C) Radioactive phosphorus is dissolved from rocks and becomes part of the sugar molecules created during photosynthesis.

(D) Radioactive phosphorus is dissolved from rocks and is used to create cholesterol stored in the nucleus.

60. Which of the following statements best describes the formation of complex carbohydrates?

(A) Monosaccharides are converted to simple sugars.

(B) Monomers of polysaccharides are linked together through peptide bonds.

(C) Nitrogen-containing bases are used to build complex carbohydrates.

(D) Monosaccharides are joined together through covalent bonds. ✓

61. Which of the following best describes the structures of carbohydrates?

1 Chemistry of Life MCQ

(A) They only occur as disaccharides.

(B) They occur as monomers, chains of monomers, and branched structures. ✓

(C) They only occur as long and branched structures.

(D) They occur as chains of monomers that hydrogen bond with complementary chains of monomers.

Answer B

Correct. Carbohydrates occur in many forms. Glucose is an example of a monomer, sucrose is a chain of two carbohydrate monomers, and starch is an example of a carbohydrate with a branched structure.

62. The synthesis of protein or carbohydrate polymers always produces which of the following as a byproduct?

(A) ATP

(B) Oxygen

(C) Carbon dioxide

(D) Urea

(E) Water ✓

63. Which of the following statements best describes a difference between proteins and nucleic acids?

(A) Proteins contain sulfur, while nucleic acids do not. ✓

(B) Proteins contain phosphorus, while nucleic acids contain sulfur.

(C) Both proteins and nucleic acids contain sulfur as a key element.

(D) Both proteins and nucleic acids contain nitrogen and sulfur.

1 Chemistry of Life MCQ

64. Polypeptides are continuously being formed and degraded. One of these processes is shown.

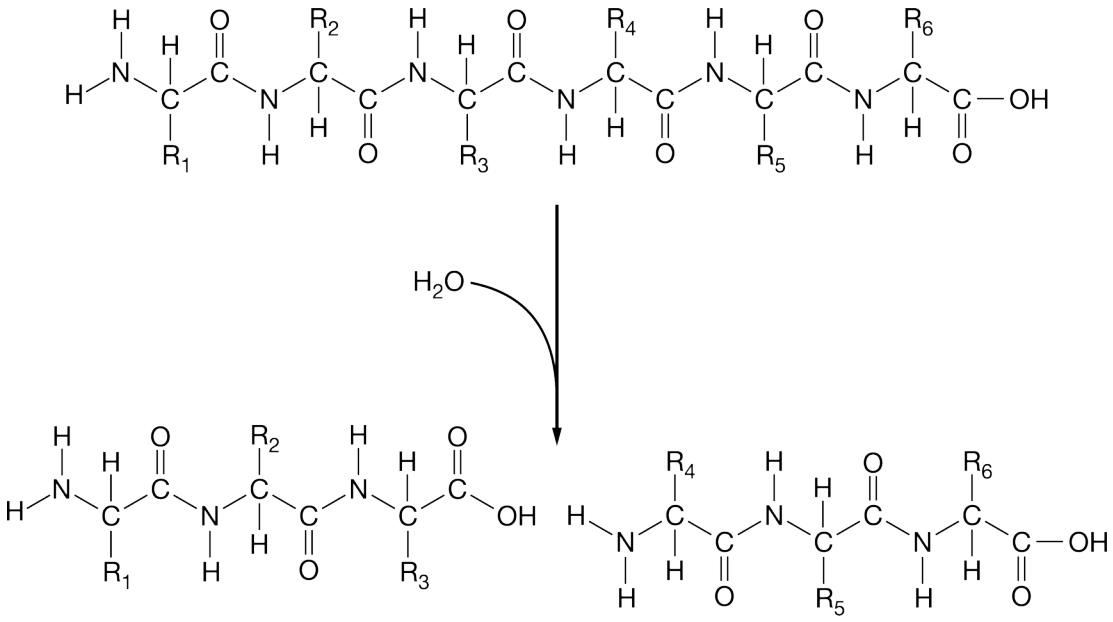


Figure 1. Polypeptide reaction

Which statement is the most accurate description of the reaction shown in Figure 1?

(A) It represents monomers linked by dehydration synthesis.

(B) It represents a polypeptide chain that folds to form the tertiary structure.

(C) It represents a polypeptide chain that is denatured into the primary structure.

(D) It represents a polypeptide chain that is broken down through a hydrolysis reaction. ✓

Answer D

Correct. The reaction shows the polypeptide being split into two smaller units with the addition of a hydroxide ion and a hydrogen ion from water. This facilitates the breaking of the bond holding the polypeptide together.

1 Chemistry of Life MCQ

For following group of questions first study the description of the situation or data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

In the first step of an experiment, rat liver cells were exposed for 5 minutes to amino acids labeled with a radioactive isotope. The cells were then washed to stop any further incorporation of radioactive amino acids. The cells were sampled periodically thereafter, and the radioactivity of a certain protein (protein X) was measured in various cell components, as shown below.

RADIOACTIVE COUNTS (PROTEIN X)

Sampling Time (minutes)	Cell Component						
	Endoplasmic Reticulum	Golgi Apparatus	Cytoplasm	Nucleus	Mitochondria	Lysosomes	Extracellular Space
5	980	20	50	4	2	6	2
20	135	760	20	3	5	120	9
45	50	195	4	6	2	800	3
120	15	30	8	6	8	975	4

65. What is the most likely identity of protein X ?

(A) An enzyme that is involved in transcription

(B) An electron transport protein that participates in respiration reactions

(C) A protein secreted from liver cells into the blood stream that permits proper capillary function

(D) An enzyme that participates in the degradation and recycling of cell components ✓

(E) A building block of the microtubules that are part of the cytoskeleton

66. A researcher is studying a specific type of cellular defect in animal cells. She notices that cells with this defect have reduced energy storage and decreased cell membrane stability.

Which of the following predicts the type of cellular defect that is the most likely cause of the researcher's observation?

(A) A defect in fatty acid synthesis, reducing the formation of fats and phospholipids ✓

(B) A defect in cholesterol synthesis, leading to reduced cholesterol in cell membranes

(C) A defect in steroid hormone production, leading to a depletion of energy reserves

(D) A defect in unsaturated fatty acid production, affecting membrane fluidity

67. Which of the following best describes a characteristic of saturated fatty acids?

1 Chemistry of Life MCQ

(A) They contain at least one double bond between carbon atoms.

(B) They have a kinked carbon chain structure.

(C) They contain only single bonds between carbon atoms. ✓

(D) They are more liquid at room temperature than unsaturated fatty acids

Answer C

Correct. Saturated fatty acids contain only single bonds between carbon atoms. This characteristic allows the carbon chains to pack tightly together, giving saturated fats their solid nature at room temperature.

68.

Segment 1:	5′ - ATATGAGTAGT - 3′ 3′ - TATACTCATCA - 5′	Segment 2:	5′ - GCGCAGACGAC - 3′ 3′ - CGCGTCTGCTG - 5′

The sequences for two short fragments of DNA are shown above. Which of the following is one way in which these two segments would differ?

(A) Segment 1 would not code for mRNA because both strands have T, a base not found in RNA.

(B) Segment 1 would be more soluble in water than segment 2 because it has more phosphate groups.

(C) Segment 1 would become denatured at a lower temperature than would segment 2 because A-T base pairs have two hydrogen bonds whereas G-C base pairs have three. ✓

(D) Segment 1 must be from a prokaryote because it has predominantly A-T base pairs.

Answer C

This option is correct. It demonstrates an understanding of the relationship between structure and function at the molecular level and the ability to make predictions about how change(s) in structure affect functionality. Because G-C base pairs have three hydrogen bonds, they are more stable structurally. Thus, because DNA Segment 2 has more G-C base pairs than DNA Segment 1, Segment 2 is more stable and thus would denature at higher temperatures than Segment 1.

69. Which of the following best explains why surface tension is created on the surface of bodies of water?

1 Chemistry of Life MCQ

(A) Water molecules are attracted to each other because the hydrogen atoms donate electrons to oxygen atoms.

(B) Water molecules are attracted to each other because the sharing of electrons between hydrogen and oxygen is unequal. ✓

(C) Water molecules are attracted to substances dissolved in the water because most dissolved substances are positively charged.

(D) Water molecules are attracted to substances dissolved in the water because most dissolved substances are negatively charged.

Answer B

Correct. The unequal sharing of electrons between hydrogen and oxygen in a water molecule results in a partial positive charge and a partial negative charge on the hydrogen and oxygen atoms, respectively. These partial charges attract other water molecules, resulting in hydrogen bond formation and cohesive forces between the water molecules, especially at the surface of the water.

70. Figure 1 is a diagram of water molecules at the air-water interface at the surface of a pond.

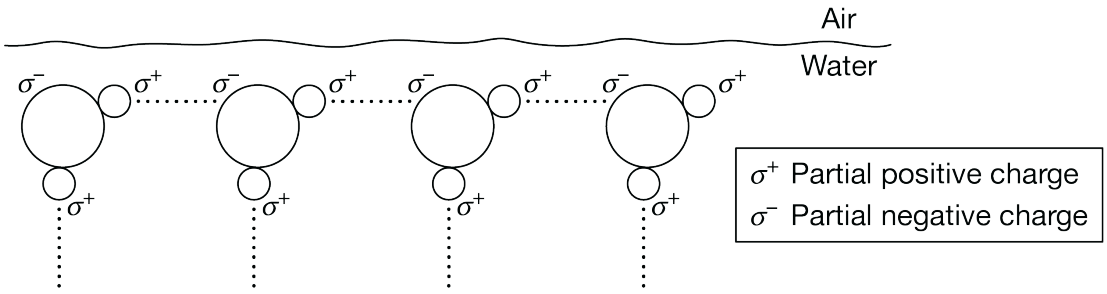


Figure 1. Alignment of water molecules at air-water interface

Based on Figure 1, which of the following best describes how the properties of water at an air-water interface enable an insect to walk on the water's surface?

(A) Covalent bonds between water molecules and the air above provide cohesion, which causes tiny bubbles to form under the feet of the insect.

(B) Ionic bonds between molecules at the surface of the water provide an electric charge, which attracts the feet of the insect, keeping it on the surface.

(C) Polar covalent bonds between molecules at the surface of the water provide adhesion, which supports the weight of the insect.

(D) Hydrogen bonds between molecules at the surface of the water provide surface tension, which allows the water surface to deform but not break under the insect. ✓

1 Chemistry of Life MCQ

Answer D

Correct. Hydrogen bonds form between water molecules due to the partial charges produced as a result of the polarity of the water molecule. The strength of the attraction between the water molecules allows organisms to walk across the surface without breaking the hydrogen bonds.

71. Which of the following best describes how amino acids affect the tertiary structure of a protein?

(A) The number of amino acids determines the tertiary structure of the protein.

(B) The interactions of the different R-groups with other R-groups and with their environment determine the tertiary structure of the protein. ✓

(C) The R-group of the last amino acid that is added to a growing polypeptide chain determines the next amino acid that is added to the chain.

(D) The sequence of the amino acids in the polypeptide chain determines the protein's primary structure but has no effect on its tertiary structure.

Answer B

Correct. The tertiary structure is determined by the attractions, repulsions, and bonding between different amino acids. Changing the sequence of even two amino acids may affect the tertiary shape of the protein.

72. Researchers analyzed the composition of cell membranes of bacteria living in different environments. Thermophiles are bacteria that live at extremely high temperatures, typically at or above $70^{\circ}C$.

Which of the following best predicts the effect of a greater proportion of saturated fatty acids, as compared with unsaturated fatty acids, on the stability of the cell membrane in thermophiles?

(A) A greater proportion of saturated fatty acids has no effect on cell membrane stability because both saturated and unsaturated fatty acids are hydrophobic.

(B) A greater proportion of saturated fatty acids leads to increased membrane stability at lower temperatures because saturated fatty acids increase the stability of the membrane.

(C) A greater proportion of saturated fatty acids leads to decreased membrane stability because saturated fatty acids decrease the fluidity of the membrane.

(D) A greater proportion of saturated fatty acids leads to increased membrane stability at higher temperatures because saturated fatty acids pack tightly together. ✓

Answer D

Correct. Saturated fatty acids pack tightly, which reduces membrane fluidity. This is critical for maintaining membrane stability in thermophiles exposed to extremely high temperatures.

1 Chemistry of Life MCQ

73. In vascular plants, water flows from root to leaf via specialized cells called xylem. Xylem cells are hollow cells stacked together like a straw. A student explains that evaporation of water from the leaf pulls water up from the roots through the xylem, as shown in Figure 1.

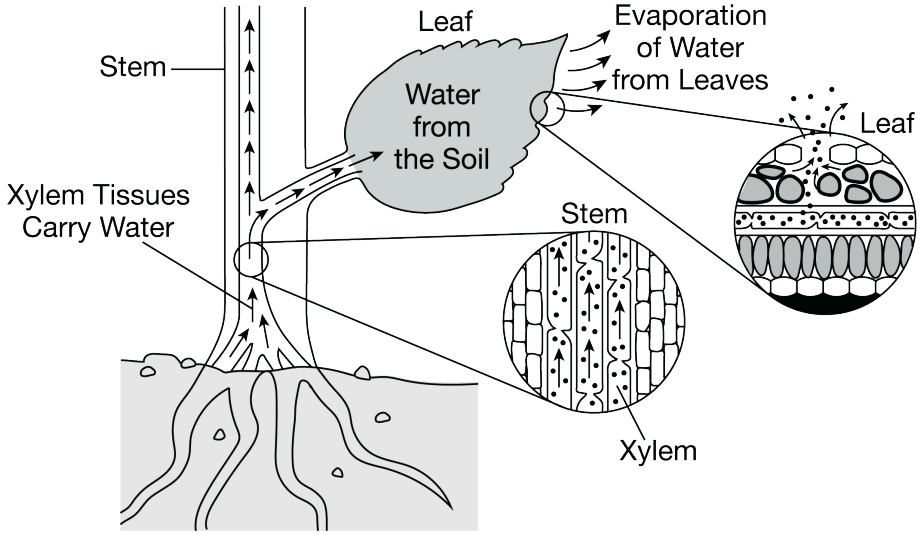


Figure 1. Model of water movement through the xylem, with magnified models of water movement in the stem and leaf.

Which statement describes how water is pulled up through the xylem to the leaves of the plant?

(A) As water exits the leaf, hydrogen bonding between water molecules pulls more water up from below. ✓

(B) As water exits the leaf, signals are sent to the roots to pump more water up to the leaves through the xylem by adhesion.

(C) Evaporation from the leaf decreases the hydrogen bonds that form between the water molecules in the xylem, which helps the water molecules to be pulled up the xylem.

(D) Evaporation of water from the leaf increases the hydrogen bonds that form between water molecules in the air, providing the energy for transport.

Answer A

Correct. As water exits the leaf, hydrogen bonding pulls more water molecules up through the leaf and xylem by cohesion.

74. Which of the following is most directly responsible for water's unique properties?

1 Chemistry of Life MCQ

(A) It contains oxygen atoms.

(B) It contains hydrogen atoms.

(C) It is an ionic compound.

(D) It forms hydrogen bonds. ✓

(E) It is nonpolar.

第二部分 AP Biology Unit 3: 酶相关题目

45 题 ｜ 含答案解析 ｜ 重新编号 1–45

编号对照：原 Q95–126 → 新 1–32；原 Q129–141 → 新 33–45（原 Q127–128 为呼吸作用题，已剔除）

1. Acetylcholinesterase (AChE) is a protein that catalyzes the conversion of acetylcholine to acetate and choline. When the concentration of AChE in an aqueous solution is held constant, the rate of the reaction catalyzed by AChE increases with increasing concentrations of substrate. At low concentrations of acetylcholine, a small increase in the substrate concentration results in a large increase in the reaction rate. At high concentrations of acetylcholine, however, a large increase in the substrate concentration results in only a small increase in the reaction rate.

Which of the following statements correctly explains the observed effect of the acetylcholine concentration on the rate of the enzyme-catalyzed reaction?

(A) The active site of AChE is specific for acetylcholine, and only one substrate molecule can occupy the active site at a time. ✓

(B) AChE begins converting product into substrate as the acetylcholine concentration changes from low to high.

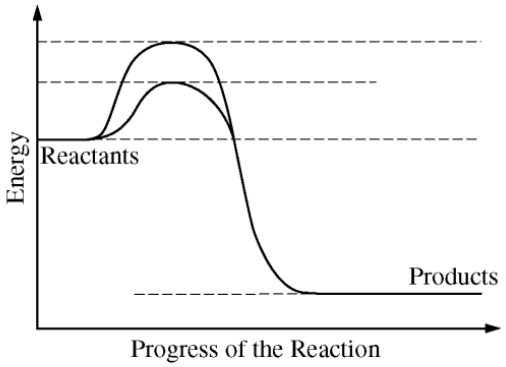
(C) The AChE protein becomes denatured as the acetylcholine concentration changes from low to high.

(D) The substrate specificity of AChE changes as the acetylcholine concentration changes from low to high.

Answer A

Correct. Acetylcholine is the substrate for AChE, and since the shape and charge of the active site is specific for acetylcholine, only one substrate molecule can occupy the active site at a time. Therefore, at high concentrations of acetylcholine, AChE becomes saturated with substrate. When an enzyme is saturated with substrate, adding more substrate to the solution will not lead to an observable increase in the reaction rate.

2. The diagram below shows energy changes in a specific chemical reaction with and without the addition of an enzyme to the reaction.



Which of the following questions can best be answered by the diagram?

(A) Does the addition of an enzyme reduce the activation energy required for a reaction? ✓

(B) Does the addition of an enzyme result in the formation of covalent bonds?

(C) Does the addition of an enzyme produce a greater amount of products?

(D) Does the addition of an enzyme change the pathway for the reaction?

3. A researcher proposes a model of an enzyme-catalyzed reaction in which a reactant is converted to a product. The model is based on the idea that the reactant passes through a transition state within the enzyme-substrate complex before the reactant is converted to the product.

Which of the following statements best helps explain how the enzyme speeds up the reaction?

(A) The enzyme's active site binds to and stabilizes the reactant, which decreases the free-energy change of the reaction.

(B) The enzyme's active site binds to and stabilizes the transition state, which decreases the activation energy of the reaction. ✓

(C) The enzyme's active site binds to and stabilizes the product, which increases the amount of energy released by the reaction.

(D) The enzyme's active site binds to and stabilizes both the reactant and the product at the same time, which increases the reaction's equilibrium constant.

Answer B

Correct. An enzyme typically speeds up a reaction by stabilizing a transition state in the reaction, which decreases the reaction's activation energy. The enzyme's active site binds more tightly to the transition state than to either the substrate or the product, which speeds up the reaction and allows the enzyme to

catalyze the same reaction repeatedly.

4. Amylase is a protein that catalyzes the conversion of starch to simple sugars. Amylase activity in an aqueous solution can be measured by using iodine as a starch indicator. A solution containing iodine and starch will have a dark-blue color, whereas a solution containing iodine but no starch will have a light-brown color. The color change of an iodine solution from dark blue to light brown can be used to measure the rate at which starch is converted to simple sugars.

A student designs an experiment to investigate the effect of environmental pH on amylase function. The design of the experiment is presented in Table 1.

Table 1. An experiment for investigating the effect of pH on amylase function

Test Tube	Environmental pH	Starch Added	Amylase Added	Iodine Added	Pretreated by Boiling
I	5	Yes	Yes	Yes	No
II	6	Yes	Yes	Yes	No
III	7	No	Yes	Yes	No
IV	7	Yes	No	Yes	No
V	7	Yes	Yes	Yes	Yes
VI	7	Yes	Yes	Yes	No
VII	8	Yes	Yes	Yes	No
VIII	9	Yes	Yes	Yes	No

Which of the following statements best justifies the inclusion of test tube V as a control in the experiment?

(A) It will provide a measurement of amylase activity at an acidic pH.

(B) It will provide a measurement of amylase activity at a basic pH.

(C) It will show the color change that occurs in the absence of enzyme activity. ✓

(D) It will show the color change that occurs in the absence of the amylase protein.

Answer C

Correct. Test tube V will contain all the components of the reaction mixture, but the amylase protein will be inactive because it will have been denatured during the pretreatment. Test tube V is an appropriate control in the experiment because it will show the color change that occurs in the absence of amylase activity.

5. A researcher claims that some bacteria contain factors that influence the function of a particular enzyme but other bacteria do not. To test the claim experimentally, the researcher will grow two different bacterial strains in separate liquid cultures and isolate the contents of the cells in each culture. The researcher will add different combinations of cellular contents, substrate, and enzyme to test tubes containing a buffer solution adjusted to the optimal pH of the enzyme and then measure the rate of product formation. The design of the researcher's experiment is presented in Table 1.

Table 1. An experiment for investigating the influence of bacterial factors on enzyme function

Test Tube	Source of Cellular Contents	Cellular Contents Added	Substrate Added	Enzyme Added
1	Strain 1	No	Yes	Yes
2	Strain 1	Yes	No	Yes
3	Strain 1	Yes	Yes	No
4	Strain 1	Yes	Yes	Yes
5	Strain 2	No	Yes	Yes
6	Strain 2	Yes	No	Yes
7	Strain 2	Yes	Yes	No
8	Strain 2	Yes	Yes	Yes

Which of the following statements best justifies the inclusion of test tubes 3 and 7 in the experiment?

(A) They will show whether the isolated cellular contents have enzymatic activity. ✓

(B) They will show whether environmental pH affects the function of the enzyme.

(C) They will show the rate of product formation in the absence of bacterial factors.

(D) They will show the rate of product formation in the absence of the substrate.

Answer A

Correct. The researcher will add cellular contents and the substrate but not the enzyme to test tubes 3 and 7. As such, test tubes 3 and 7 will show whether the cellular contents have enzymatic activity.

6. A researcher designs an experiment to investigate the effect of environmental temperature on the function of an enzyme. For each trial included in the experiment, the researcher will add the enzyme and its substrate to an aqueous buffer solution and then measure the amount of product formed over 20 minutes.

Which of the following must remain the same for all trials of this experiment?

(A) The initial concentration of the substrate ✓

(B) The final concentration of the product

(C) The three-dimensional structure of the enzyme

(D) The temperature of the aqueous buffer solution

Answer A

Correct. Keeping the initial substrate concentration constant allows the researcher to compare the reaction rates under different environmental conditions. The reaction rates will provide the researcher with information about the enzyme's function in different environments.

7. Catalase is an enzyme found in yeast cells that facilitates the chemical breakdown of hydrogen peroxide to water and oxygen gas. An experiment was conducted to determine the effect of pH on catalase function. Five buffer solutions of varying pH (2, 4, 6, 8, and 10) were prepared and added to separate test tubes. Hydrogen peroxide (H_2O_2) was added to each test tube. Yeast was added, and the reactions were timed. After 1 minute the amount of oxygen gas released was determined by measuring the foam layer produced in each test tube. Figure 1 illustrates the experimental setup.

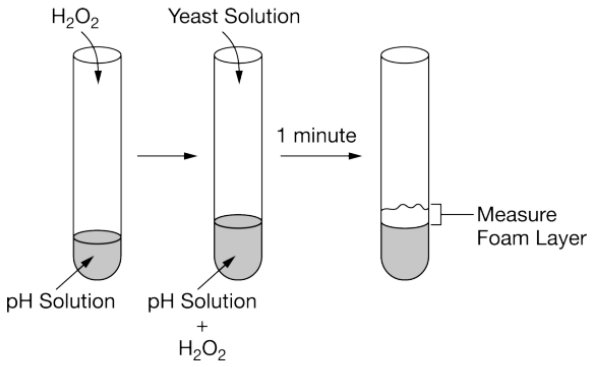


Figure 1. Illustration of experimental procedure

A set of five additional test tubes were prepared and used as controls. Which of the following best describes the contents expected to be contained in one of the five control test tubes?

(A) pH 4 buffer solution and hydrogen peroxide only ✓

(B) All five pH buffer solutions combined and hydrogen peroxide only

(C) Water, hydrogen peroxide, and yeast

(D) Water and yeast only

Answer A

Correct. Since the experiment is studying the effect of pH on catalase in yeast, the control set should determine what would happen without yeast present. Each test tube in the control set should contain hydrogen peroxide and one of the pH solutions.

For following group of questions first study the description of the data and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

To study the actions of the enzyme catalase on hydrogen peroxide, students performed the following experiment. Catalase was extracted from potatoes by blending raw potatoes in a blender with cold distilled water. The filtrate was stored on ice. The following hydrogen peroxide solutions were made: 1 percent, 5 percent, 10 percent, and 15 percent. Filter paper disks were soaked in the catalase filtrate and dropped into beakers containing the various solutions. The activity of the enzyme was measured by the amount of time it took for the disks to float to the surface of the solution on the bubbles produced by the reaction. The following data were obtained.

Hydrogen Peroxide Solution	Average Time, in Seconds, for Disks to Float
1%	30 sec
5%	25 sec
10%	20 sec
15%	10 sec

8. Which of the following experimental designs should the students use as a control for the experiment?

(A) Place a catalase-soaked disk in a beaker of water. ✓

(B) Drop the disks from different heights into the solution.

(C) Poke the floating disks back down to the bottom of the beaker and retime the rise of the disks.

(D) Shake the beakers during the time the disks are rising.

(E) Dip the disks in the 5% solution before putting them in any other solution.

9. Which of the following best describes why ice was used during this experiment?

(A) To increase the activity of the enzyme

(B) To retard the breakdown of the catalase ✓

(C) To purge the solution of excess air trapped during blending

(D) To slow the catalase molecules to increase the probability of contact with the hydrogen peroxide molecules

(E) To increase the size of the active site on the enzyme

---

10. Which of the following best explains why the disks rose to the surface faster in the more concentrated hydrogen peroxide solutions?

(A) There was more enzyme present in the more concentrated solutions.

(B) A greater amount of heat was generated in the more concentrated solutions.

(C) The more concentrated solutions lowered the activation energy of the reaction.

(D) The higher substrate concentrations in the more concentrated solutions speeded the reaction. ✓

(E) The density of the water was lower in the more concentrated solutions.

11. If the potato solution was boiled for 10 minutes and cooled for 10 minutes before being tested, the average time for the disks to float to the surface of the hydrogen peroxide solution would be

(A) less than 1 second

(B) 5 seconds

(C) 10 seconds

(D) 30 seconds

(E) more than 30 seconds ✓

---

12. Which of the following statements best helps explain the reaction specificity of an enzyme?

(A) The free energy of the reactants is greater than the free energy of the products.

(B) The equilibrium constant of the reaction is much greater than 1.

(C) The shape and charge of the substrates are compatible with the active site of the enzyme. ✓

(D) The concentration of the enzyme inside living cells is greater than the concentration of substrate.

Answer C

Correct. The reaction specificity of an enzyme is determined by the compatibility of the substrate with the enzyme's active site. A substrate's shape and charge affect how it fits into an enzyme's active site and whether it forms favorable interactions with the enzyme.

---

13. Gelatin is a protein that is derived from collagen which is found in the bones, skin, and connective tissue of animals. To investigate the ability of various enzymes to digest gelatin, a group of students set up an assay involving camera film. Camera film contains gelatin and appears black when exposed to light but turns clear as the gelatin gets broken down. The students incubated pieces of exposed camera film in test tubes, each containing one of three different enzyme solutions (trypsin, lipase, or amylase) as indicated in Figure 1. The students recorded the time it took for the enzymes to digest the gelatin in each test tube, turning the film from black to clear.

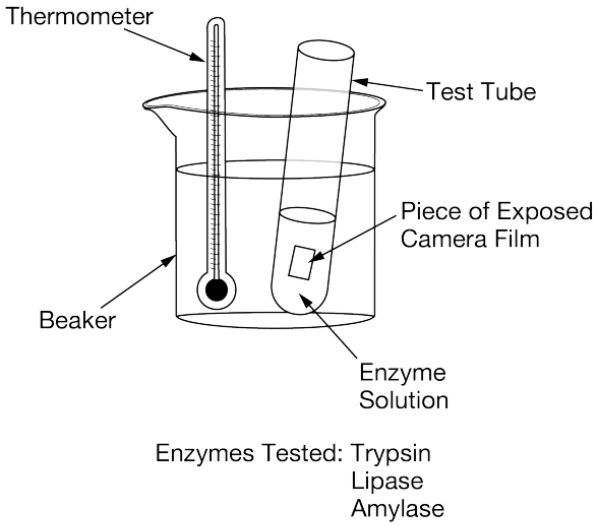


Figure 1. Diagram of experimental setup.

Which of the following would be the most appropriate control for this experiment?

(A) A test tube containing no camera film

(B) A test tube containing only a piece of exposed camera film

(C) A test tube containing a piece of exposed camera film submerged in water ✓

(D) A test tube containing a piece of exposed camera film and all three enzyme solutions

Answer C

Correct. Having one test tube with only water and camera film will show that water is not the substance that causes the film to change color.

14. The enzyme hexokinase catalyzes the conversion of glucose to glucose-6-phosphate, which is an important step in glycolysis. The reaction involves the transfer of a phosphate group from ATP to glucose.

Either a glucose molecule or a water molecule can fit in the active site of hexokinase. The presence of a water molecule in hexokinase's active site would result in the hydrolysis of ATP to ADP instead of the conversion of glucose to glucose-6-phosphate.

Which of the following statements best helps explain the reaction specificity of hexokinase?

(A) Both glucose and water are polar molecules that form favorable interactions with charged and polar amino acid side chains in hexokinase's active site.

(B) Both glucose and water have oxygen atoms that can form covalent bonds with the phosphorus atoms of phosphate groups.

(C) Glucose is an energy-rich organic molecule that can be broken down by glycolysis to produce ATP, whereas water is an inorganic molecule.

(D) Glucose has the right shape and charge to cause hexokinase to undergo a structural change needed for catalysis, whereas water does not. ✓

Answer D

Correct. Hexokinase catalyzes the conversion of glucose to glucose-6-phosphate but not the hydrolysis of ATP. Hexokinase's reaction specificity is explained by the compatibility of glucose and hexokinase's active site, which results in part from glucose's shape and charge. Because of this compatibility, hexokinase undergoes a structural change when it binds to glucose but not to water, and the structural change is necessary for hexokinase to become catalytically active.

15. A student designs an experiment to investigate the influence of temperature on enzyme function. The student's plan is presented in Table 1.

Table 1. An experiment to investigate the influence of temperature on enzyme function

Test Tube	Substrate Added	Enzyme Added	Temperature (°C)
1	Yes	No	10
2	Yes	Yes	10
3	Yes	No	20
4	Yes	Yes	20
5	Yes	No	30
6	Yes	Yes	30
7	Yes	No	40
8	Yes	Yes	40

Which test tubes are controls in the experiment?

(A) Test tubes 1 and 2 only

(B) Test tubes 5 and 6 only

(C) Test tubes 1, 3, 5, and 7 ✓

(D) Test tubes 2, 4, 6, and 8

Answer C

Correct. The test tubes that do not contain the enzyme are negative controls in the experiment. The negative controls help confirm that the results of the experiment are related to enzyme function and not some other factor.

16. Aminolevulinate dehydratase (ALAD) is an enzyme that relies on zinc as a cofactor. A zinc ion binds to the ALAD active site, where it forms favorable interactions with the side chains of three amino acids. Researchers have found that substituting a lead ion for a zinc ion in the ALAD active site causes inhibition of ALAD.

Which of the following statements best helps explain how the lead ion causes inhibition of ALAD?

(A) It changes the shape and charge of the substrate so that it becomes more compatible with ALAD's active site.

(B) It changes the amino acid sequence of the ALAD protein so that the enzyme catalyzes a different reaction.

(C) It changes the three-dimensional structure of the active site so that ALAD is no longer compatible with its substrate. ✓

(D) It changes the enzyme-substrate complex so that the transition state is more stable and the reaction proceeds at a faster rate.

Answer C

Correct. The substitution of a lead ion for a zinc ion most likely changes the three-dimensional structure of the ALAD active site. The three-dimensional structure of the active site plays an important role in the compatibility between an enzyme and its substrate. Changing the structure of the active site will likely interfere with enzyme function by disrupting the enzyme-substrate interaction.

17. Pectinase is a protein that catalyzes the breakdown of pectic polysaccharides in plant cell walls. A researcher designs an experiment to investigate the effect of salinity on the ability of pectinase to lower the activation energy of the reaction involved. The design of the experiment is presented in Table 1. For each test tube, the researcher will measure the amount of product formed over 20 minutes.

Table 1. An experiment to investigate the effect of salinity on pectinase function

Test Tube	Sodium Chloride Concentration (molar)	Temperature (degrees C)	Substrate Added	Pectinase Added
1	0	23	Yes	No
2	0	23	Yes	Yes
3	0.5	23	Yes	No
4	0.5	23	Yes	Yes
5	1.0	23	Yes	No
6	1.0	23	Yes	Yes
7	1.5	23	Yes	No
8	1.5	23	Yes	Yes

Which of the following statements best helps justify the inclusion of test tube 5 in the experiment?

(A) It will act as a control for test tube 4 by showing the effect of the presence or absence of the substrate.

(B) It will act as a control for test tube 4 by showing the effect of a change in environmental temperature.

(C) It will act as a control for test tube 6 by showing the effect of the presence or absence of the enzyme. ✓

(D) It will act as a control for test tube 6 by showing the effect of a change in sodium chloride concentration.

Answer C

Correct. Test tube 5 will act as a control for test tube 6. The solutions in test tubes 5 and 6 are nearly identical except that the researcher will add pectinase to test tube 6 but not to test tube 5. A comparison of the data for test tubes 5 and 6 will show the effect of the presence or absence of the enzyme.

Researchers investigated the influence of environmental pH on the activity of peroxidase, an enzyme that catalyzes the conversion of hydrogen peroxide to water and oxygen gas. In an experiment, the researchers added a hydrogen peroxide solution containing guaiacol to several identical test tubes and adjusted the solution in each test tube to a different pH. The researchers included the guaiacol because it caused the solutions to change color as the reactions proceeded, which the researchers relied on for measuring reaction rates. Finally, the researchers added the same amount of peroxidase to each test tube and measured the rate of each reaction at $23\,^{\circ}\mathrm{C}$. The results of the experiment are represented in Figure 1.

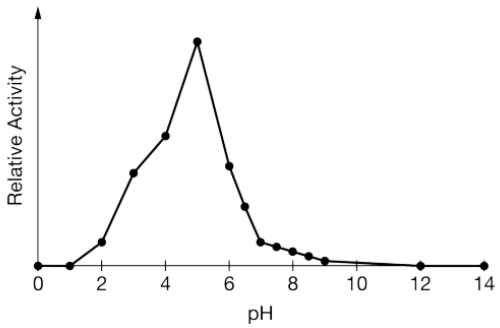


Figure 1. The effect of pH on peroxidase activity

18. One of the researchers proposes using oxygen gas production to measure reaction rates. Which of the following statements best justifies the use of the proposed modification as a way of creating an appropriate control for the investigation?

(A) The experiment can be repeated without hydrogen peroxide, which will help eliminate an uncontrolled variable.

(B) The experiment can be repeated without peroxidase, which will introduce a second independent variable.

(C) The experiment can be repeated without guaiacol, which will reveal the effect of guaiacol on the reaction rates. ✓

(D) The experiment can be repeated without water, which will reveal whether the reaction can occur inside a living cell.

Answer C

Correct. In the original experiment, the researchers used guaiacol as a reaction indicator. By using oxygen gas production to measure reaction rates, the researchers no longer need the guaiacol and can repeat the experiment without it, which will reveal the effect of guaiacol on the reaction rates.

19. Which of the following actions will provide a negative control for the investigation?

(A) Repeating the experiment at $25^\circ C$

(B) Repeating the experiment using twice the amount of hydrogen peroxide

(C) Repeating the experiment using twice the amount of peroxidase

(D) Repeating the experiment using heat-denatured peroxidase ✓

Answer D

Correct. A negative control is a group in which no response is expected. It is the opposite of the positive control, in which a known response is expected. Heat-denatured peroxidase will be inactive. Repeating the experiment using heat-denatured peroxidase will allow the researcher to measure reaction rates in the presence of inactive peroxidase.

20. Based on Figure 1, which of the following statements best predicts the effect that a change from a moderately acidic environment (pH near 6) to a basic environment will have on peroxidase activity?

(A) Peroxidase activity will decrease. ✓

(B) Peroxidase activity will increase.

(C) Peroxidase activity will stay the same.

(D) Peroxidase activity will increase at first and then decrease.

Answer A

Correct. Based on Figure 1, peroxidase's optimal pH is 5. A change from a moderately acidic environment (pH near 6) to a basic environment (pH above 7) will result in a decrease in peroxidase activity.

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21. A researcher designs an experiment to investigate whether soil bacteria trigger the synthesis of defense enzymes in plant roots. The design of the experiment is presented in Table 1. For each group in the experiment, the researcher will determine the average rate of change in the amount of defense enzymes in the roots of the seedlings.

Table 1. An experiment to investigate the effect of soil bacteria on plant defenses

Group	Number of Seedlings	Type of Soil	Treatment Solution
1	10	Sterile potting soil	Contains actively reproducing soil bacteria
2	10	Sterile potting soil	Contains heat-killed soil bacteria
3	10	Sterile potting soil	Contains no soil bacteria

Which of the following statements best helps justify the inclusion of group 2 as one of the controls in the experiment?

(A) It will show whether the changes observed in group 1 depend on the metabolic activity of soil bacteria. ✓

(B) It will show whether the changes observed in group 1 depend on the type of plants used in the experiment.

(C) It will show the average growth rate of seedlings that are maintained in a nonsterile environment.

(D) It will show the changes that occur in the roots of seedlings following an infection by soil bacteria.

Answer A

Correct. The seedlings in group 1 will be treated with a solution that contains actively reproducing bacteria, which will be metabolically active. In contrast, the seedlings in group 2 will be treated with a solution that contains heat-killed bacteria, which will be metabolically inactive. The researcher will be able to compare the results for group 1 with those for group 2. The comparison will help the researcher determine whether the changes observed in group 1 depend on the metabolic activity of soil bacteria.

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22. A researcher claims that increasing the environmental temperature by $10^{\circ}C$ will double the rate of an enzyme-catalyzed reaction. To test the claim, the researcher designs an experiment that uses a particular enzyme isolated from plants. The design of the experiment is presented in Table 1. For each test tube in the experiment, the researcher will measure the rate of product formation.

Test Tube	Temperature ($^{\circ}C$)	Substrate Added	Enzyme Added	Pretreated by Boiling
1	15	Yes	Yes	No
2	25	No	Yes	No
3	25	Yes	No	No
4	25	Yes	Yes	No
5	25	Yes	Yes	Yes
6	35	Yes	Yes	No

Which of the following statements best helps justify the inclusion of test tube 5 as a control in the experiment?

(A) It will provide a measurement of product formation in the absence of the substrate.

(B) It will provide a measurement of product formation in the presence of a denatured enzyme. ✓

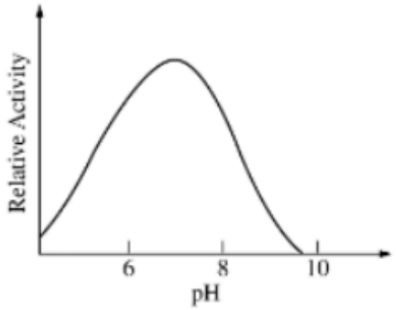
(C) It will show the effect of doubling the amount of substrate on the rate of product formation.

(D) It will show the effect of increased enzyme activity on the rate of product formation.

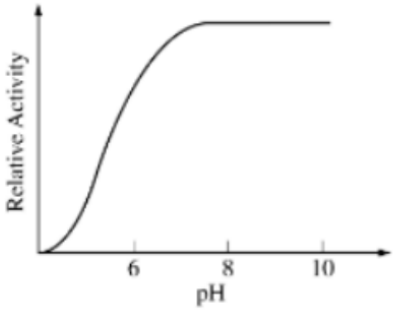
Answer B

Correct. The contents of test tube 5 will be pretreated by boiling. The pretreatment will denature the plant enzyme used in the experiment. As such, test tube 5 will provide a measurement of product formation in the presence of a denatured enzyme.

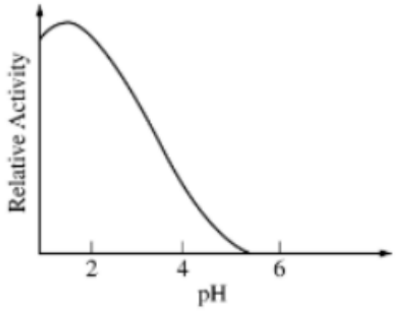
For the following questions: Graphs I-IV depict the effect of pH on the activity of four different hydrolytic enzymes.



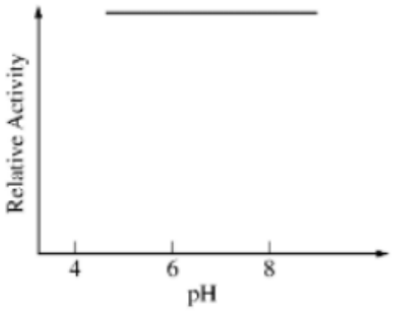
Graph I



Graph II



Graph III



Graph IV

23. Graphs representing enzymes sensitive to changes in pH include which of the following?

(A) I only
(B) IV only
(C) II and III only
(D) III and IV only
(E) I, II, and III only ✓

24. The most likely explanation for the results shown in Graph I is that

(A) pH affects the shape of the active site of the enzyme ✓
(B) pH affects the temperature of the reaction
(C) the enzyme has a quaternary structure
(D) the enzyme has disulfide bonds
(E) pH affects the primary structure of the enzyme

25. Enzymes with their highest activity at an alkaline (basic) pH are represented by which of the following graphs?

(A) I only

(B) II only ✓

(C) III only

(D) I and III only

(E) I and IV only

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26. In an experiment, a researcher prepares a reaction mixture by dissolving a substance in a buffered solution. The substance is the substrate of a certain enzyme. The researcher adds a small amount of the enzyme to the reaction mixture and measures the amount of product that is formed over time. The data are represented in Figure 1.

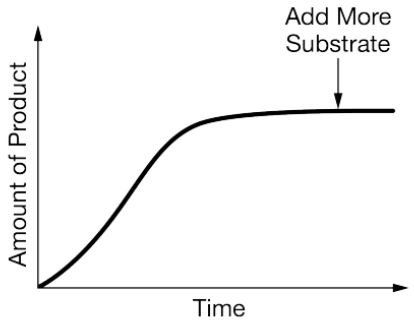


Figure 1. The amount of product formed by an enzyme-catalyzed reaction over time

Which of the following best predicts the immediate result of adding more substrate to the reaction mixture at the point indicated by the arrow in Figure 1?

(A) The amount of product will decrease until the reaction rate goes to zero.

(B) The amount of product will decrease until the reaction reaches its equilibrium point or until the enzyme is been used up by the reaction.

(C) The amount of product will increase until the reaction reaches its equilibrium point or until the substrate is used up by the reaction. ✓

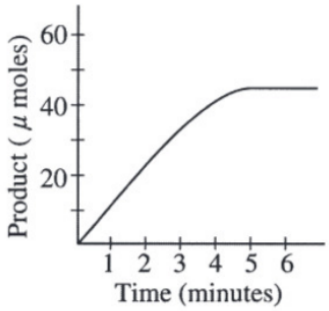
(D) The amount of product will increase without stopping because the enzyme will be unchanged by the reaction.

Answer C

Correct. The graph shown in Figure 1 indicates that the amount of product became constant as a consequence of the reaction reaching its equilibrium point or as a consequence of the substrate being used up. Adding more substrate will likely result in an increase in the amount of product because the reaction catalyzed by the enzyme will convert the added substrate into product.

---

27. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



During which time interval is the reaction rate lowest?

(A) 0-1 minute

(B) 1-2 minutes

(C) 2-3 minutes

(D) 3-4 minutes

(E) 4-5 minutes ✓

The coral species *Acropora millepora* is a marine animal that has a symbiotic relationship with algae species in the genus *Symbiodinium*. An increase in the concentration of dissolved CO_2 in oceans has a negative impact on this relationship and the health of the coral species. Scientists investigated the effect of CO_2 concentrations on the number of algae cells found in the coral (Figure 1) and on both the gross (total) and net rates of photosynthesis of those algae (Figure 2). The net rate of photosynthesis is equal to the gross rate of photosynthesis minus the rate of respiration by the algae.

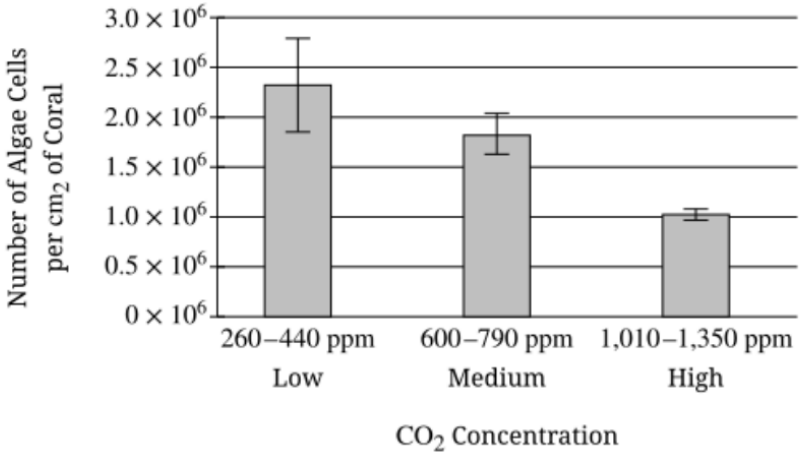


Figure 1. Effect of CO_2 concentration on the number of algae per cm^2 of coral. Error bars represent $\pm 2SE_{\bar{x}}$.

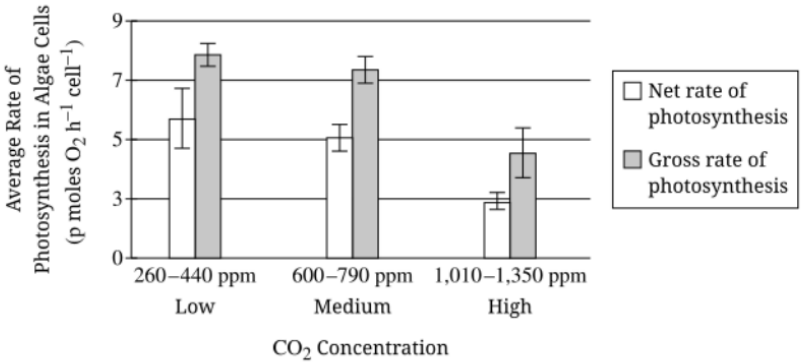


Figure 2. Average gross and net rates of photosynthesis in algae under various concentrations of CO_2. Error bars represent $\pm 2SE_{\bar{x}}$.

28. Researchers observe that the water with the high CO_2 concentration has a lower pH than does the water with the low CO_2 concentration. The researchers claim that lower pH alters the structure of the enzymes of photosynthesis.

Which of the following most likely explains how a lower pH could alter the structure of photosynthetic enzymes?

(A) There will not be enough H^+ ions for the ribosome to correctly form peptide bonds between amino acids during synthesis of the enzymes.

(B) There will not be enough H^+ ions for the necessary hydrogen bonds to form in the secondary, tertiary, and quaternary structures of the enzymes.

(C) There will be additional H^+ ions that can disrupt peptide bonds in the primary structure of the enzymes.

(D) There will be additional H^+ ions that can disrupt hydrogen bonds in the secondary, tertiary and quaternary structures of the enzymes. ✓

Answer D

Correct. Additional H^+ ions compete for hydrogen bonding with amino acid side chains and therefore denature the protein and alter the structure of the photosynthetic enzymes.

29. Based on Figure 2, which of the following best supports the claim that CO_2 concentration and photosynthetic productivity are inversely related?

(A) The net rate of photosynthesis is less than the gross rate of photosynthesis at the high concentration of CO_2.

(B) Both the net and gross rates of photosynthesis at the high concentration of CO_2 are lower than those at the low concentration of CO_2. ✓

(C) The net rate of photosynthesis at the low concentration of CO_2 is lower than the gross rate of photosynthesis at the high concentration of CO_2.

(D) The difference between net and gross rates of photosynthesis is greater at the low concentration of CO_2 than it is at the high concentration of CO_2.

Answer B

Correct. This describes an inverse relationship where one measure increases as the other decreases. Because the rate of photosynthesis decreases as CO_2 concentration increases, this supports the claim that CO_2 concentration is inversely related to photosynthetic rate (productivity).

30. Alcohol dehydrogenase (ADH) is an enzyme that aids in the decomposition of ethyl alcohol ($\mathrm{C_2H_5OH}$) into nontoxic substances. Methyl alcohol acts as a competitive inhibitor of ethyl alcohol by competing for the same active site on ADH. When attached to ADH, methyl alcohol is converted to formaldehyde, which is toxic in the body.

Which of the following statements best predicts the effect of increasing the concentration of substrate (ethyl alcohol), while keeping the concentration of the inhibitor (methyl alcohol) constant?

(A) There will be an increase in formaldehyde because ADH activity increases.

(B) Competitive inhibition will be terminated because ethyl alcohol will bind to methyl alcohol and decrease ADH activity.

(C) The peptide bonds in the active site of the enzyme will be denatured, inhibiting the enzyme.

(D) Competitive inhibition will decrease because the proportion of the active sites occupied by substrate will increase. ✓

Answer D

Correct. The increase in amount of substrate will increase the number of enzyme-ethanol complexes formed, increasing the amount of nontoxic product formed from ethyl alcohol.

31. A researcher proposes a model to explain how enzyme-substrate interactions determine enzyme specificity. The model is based on the idea that substrate molecules form favorable interactions with the amino acid side chains in an enzyme's active site.

Based on the model, which of the following statements best explains an enzyme's specificity for a particular substrate molecule?

(A) A hydrophilic molecule interacts with nonpolar side chains in the enzyme's active site.

(B) A hydrophobic molecule interacts with polar side chains in the enzyme's active site.

(C) A molecule with positive charges interacts with positively charged side chains in the enzyme's active site.

(D) A molecule with negative charges interacts with positively charged side chains in the enzyme's active site. ✓

Answer D

Correct. Because opposite charges attract, a molecule with negative charges will form favorable interactions with positively charged side chains in an enzyme's active site. Based on the model, the favorable interactions helps explain the enzyme's specificity for the molecule as a substrate.

32. A researcher claims that different enzymes exhibit maximal function over different pH ranges. To test the claim, the researcher carries out an experiment that includes three different enzymes: pepsin, salivary amylase, and arginase. The results of the experiment are represented in Figure 1.

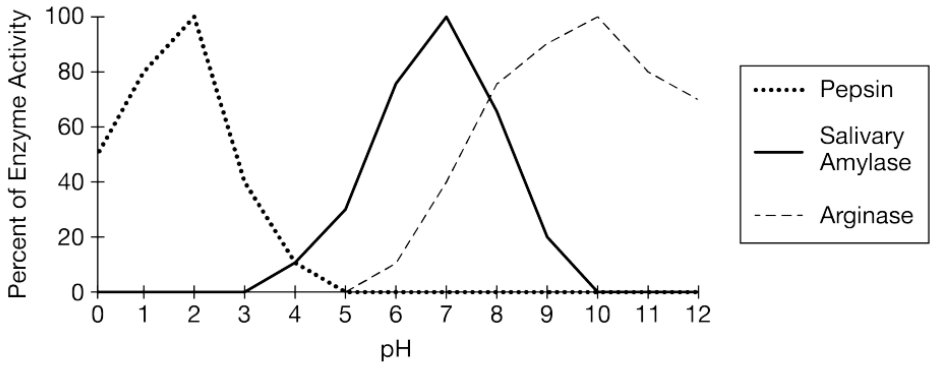


Figure 1. The effect of pH on three different enzymes

Which of the following actions will provide the most appropriate negative control for the experiment?

(A) Repeating the experiment with a fourth enzyme

(B) Repeating the experiment at several different temperatures

(C) Repeating the experiment with denatured enzymes ✓

(D) Repeating the experiment using several different methods for measuring pH

Answer C

Correct. Repeating the experiment with denatured enzymes will provide an appropriate negative control for the experiment. The negative control will show that the changes observed in the experiment result from changes in enzyme function.

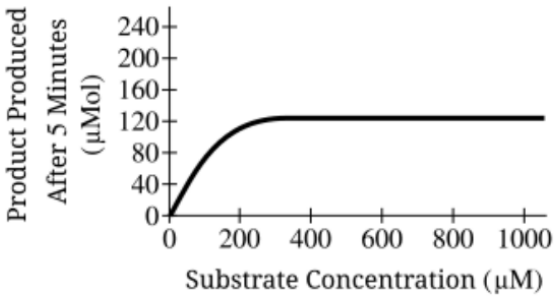
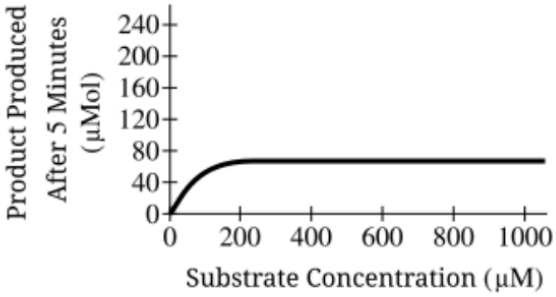


Figure 1. Rate of an enzyme-catalyzed reaction

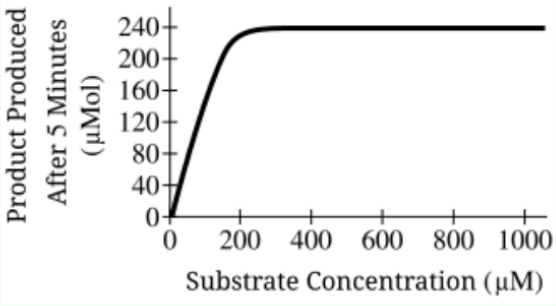
Researchers investigated the dynamics of a reaction catalyzed by an enzyme. The researchers prepared a series of reactions, each with the same concentration of enzyme but with different concentrations of substrate. The researchers measured the amount of product in each reaction mixture after 5 minutes. The results are shown in Figure 1.

33. Which of the following graphs predicts the most likely rate of the enzyme-catalyzed reaction if the concentration of the enzyme in the mixture were doubled?

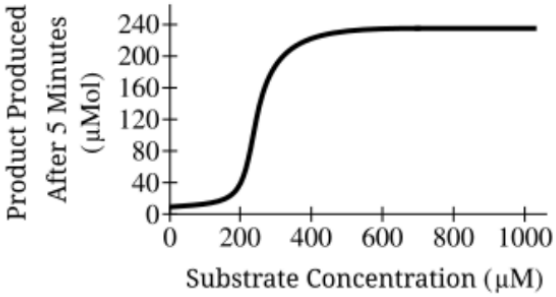
(A)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

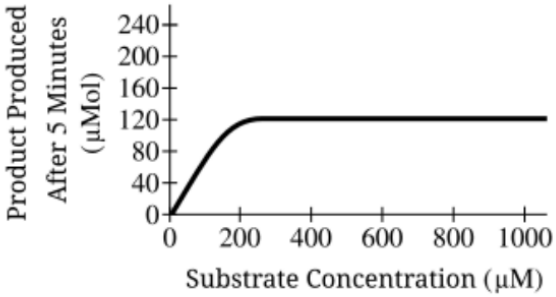
(B)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

(C)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

(D)


Product Produced After 5 Minutes (µMol)
240
200
160
120
80
40
0
0
200
400
600
800
1000
Substrate Concentration (µM)

Answer B

Correct. The rate of the reaction is initially proportional to the concentration of enzyme present. When all active sites are occupied, the rate remains constant at a maximum. Thus, doubling the enzyme concentration should double the rate of the reaction at every substrate concentration, and the maximum rate should also double.

34. Which of the following questions best addresses whether a particular inhibitor is competitive or noncompetitive?

(A) Does the inhibitor bind more tightly to the substrate or to the enzyme?

(B) Can the inhibitor be converted to product by the enzyme?

(C) How much inhibitor is necessary to completely inhibit the activity of the enzyme?

(D) Does the inhibitor bind to an allosteric site or the active site of the enzyme? ✓

Answer D

Correct. Competitive enzyme inhibitors affect the rate of reaction by competing with the substrate for binding to the active site of the enzyme. Noncompetitive inhibitors affect the rate of reaction by binding to an allosteric site of the enzyme, altering the shape and thus the function of the enzyme.

35. Which of the following experimental designs would best allow researchers to investigate the effect of pH on the rate of the enzyme-catalyzed reaction?

(A) Measure reaction rates at different pH levels when the enzyme concentration remains the same and the substrate concentration is 100 µM. ✓

(B) Measure the pH of the reaction mixture when the enzyme concentration remains the same and substrate concentrations vary between 0 µM and 1,000 µM.

(C) Measure reaction rates when the enzyme concentration remains the same but the pH of the reaction mixture increases and the substrate concentration increases from 0 µM to 1,000 µM.

(D) Measure reaction rates when the substrate concentration is 100 µM and both the pH of the reaction mixture and the enzyme concentration are increased.

Answer A

Correct. To investigate the role of pH on the reaction rate, the change in pH must be the only independent variable.

36. Which of the following best describes how an increase in the concentration of substrate in the reaction mixture affected the frequency of enzyme-substrate interactions?

(A) The frequency of enzyme-substrate interactions decreased over all substrate concentrations.

(B) The frequency of enzyme-substrate interactions increased exponentially over all substrate concentrations.

(C) The frequency of enzyme-substrate interactions increased until all of the active sites were interacting with substrate. ✓

(D) The frequency of enzyme-substrate interactions increased until all of the enzyme was depleted.

Answer C

Correct. The frequency of enzyme-substrate interactions is determined by the maximum number of enzyme active sites available to bind substrate and convert it to product. The rate of the reaction increases until the active sites of all enzyme molecules are filled, and then the rate of the reaction remains constant.

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37. Which of the following can be used to determine the rate of enzyme-catalyzed reactions?

(A) Rate of disappearance of the enzyme

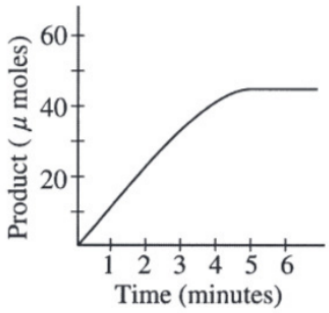
(B) Rate of disappearance of the substrate ✓

(C) Rate of disappearance of the product

(D) Change in volume of the solution

(E) Increase in activation energy

38. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



What is the most likely explanation for the change in the slope of the line between 3 and 5 minutes?

(A) The enzyme had denatured.

(B) The enzyme had achieved its maximum velocity.

(C) A large amount of the substrate had been consumed. ✓

(D) An allosteric inhibitor appeared.

(E) There was a dramatic change in the pH.

39. The enzyme trypsin aids in protein digestion in the small intestine. The relative activity of trypsin at different pH values is shown in Figure 1.

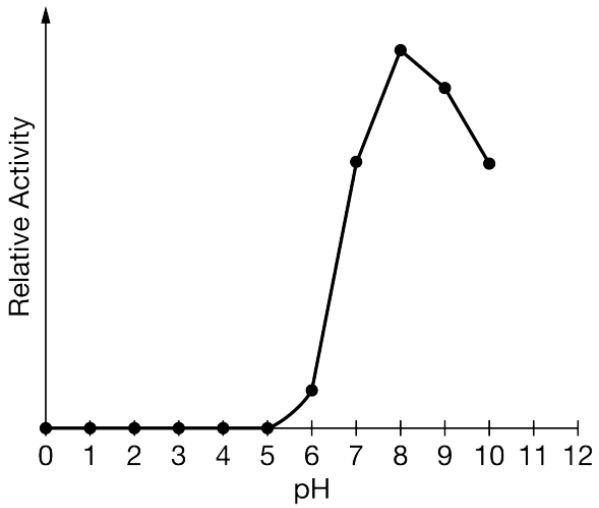


Figure 1. Effect of pH on the activity of trypsin

Which of the following statements best explains the activity levels of trypsin shown in Figure 1 ?

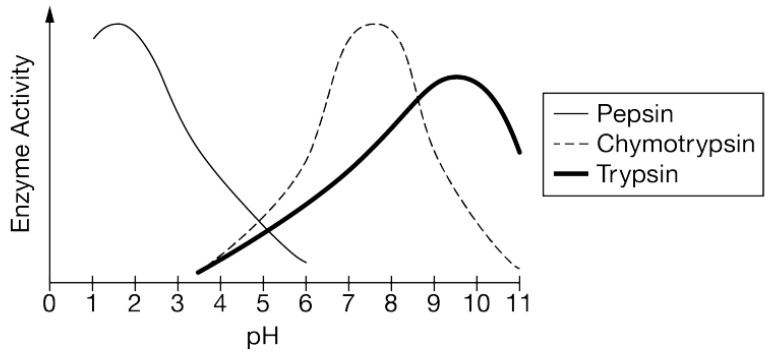
(A) The small intestine releases inhibitor molecules that block the activity of trypsin unless it is at its optimum pH.

(B) The number of effective collisions between trypsin and its substrate increase at higher pH values.

(C) As pH values increase, the substrate concentration decreases, leading to an eventual decline in the rate of the trypsin-catalyzed reaction.

(D) At extremely low pH values, trypsin is denatured and cannot function efficiently. ✓

40. Protein digestion in humans is primarily carried out by three enzymes. Pepsin is found in the stomach $(\mathrm{pH}\ 2)$, where it aids in the breakdown of large proteins into smaller peptides, while trypsin and chymotrypsin are found in the small intestine $(\mathrm{pH}\ 8)$, where they aid in the further breakdown of the proteins into amino acids and dipeptides that can be absorbed into the bloodstream. Graph 1 shows the effect of pH on the activity levels of the three enzymes.



Graph 1. Relative activity of pepsin, trypsin, and chymotrypsin at pH 0 through 11

Which of the following best predicts how the structure and function of pepsin will change as it enters the small intestine?

(A) Pepsin will not change in shape and will continue to break down proteins in the small intestine.

(B) Pepsin will not change in shape but may not work due to the basic environment of the small intestine.

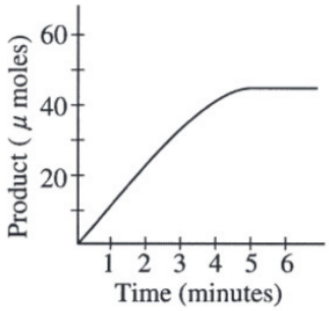
(C) Pepsin will change in shape because of the basic environment of the small intestine; therefore, its enzymatic activity will decrease. ✓

(D) Pepsin will change in shape because of the presence of trypsin and chymotrypsin in the small intestine, both of which act as competitive inhibitors.

Answer C

Correct. The significant change in pH will disturb the hydrogen bonding and cause denaturation of pepsin. As a result, pepsin will stop working in the small intestine.

41. A scientist determined the rate of an enzyme-catalyzed reaction by measuring the amount of product formed over time. The following curve was generated from the data collected.



The rate of the reaction could also be determined by

(A) measuring the change in the amount of enzyme

(B) measuring the change in the amount of substrate ✓

(C) measuring the change in salt concentration

(D) adding more substrate

(E) adding more enzyme

42. Researchers investigated the effect of urea on the three-dimensional structure of a certain enzyme. The researchers dissolved the enzyme in an aqueous buffer solution and added urea to the solution. The enzyme did not appear to have a secondary or tertiary structure. The researchers carefully removed the urea from the solution and determined that the enzyme had the original secondary and tertiary structure again.

Based on the results of the experiment, which of the following statements best predicts the effect of urea on the enzyme's function?

(A) Function will be disrupted by adding the urea and regained by removing the urea. ✓

(B) Function will be disrupted by adding the urea, but it will not be regained by removing the urea.

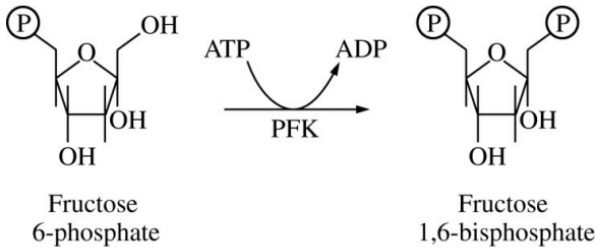
(C) Function will be gained by adding the urea and disrupted by removing the urea.

(D) Function will be unaffected by the addition and removal of the urea.

Answer A

Correct. Based on the information, adding the urea denatures the enzyme, which will disrupt the enzyme's function by disrupting the active site. In contrast, removing the urea causes the enzyme to refold, which will restore the active site and the enzyme's function.

43.



Fructose 6-phosphate

Fructose 1,6-bisphosphate

Figure 1. Reaction catalyzed by phosphofructokinase (PFK) during glycolysis

Phosphofructokinase (PFK) is an enzyme that catalyzes the conversion of fructose 6-phosphate to fructose 1,6-bisphosphate during glycolysis, as represented in Figure 1.

PFK can be allosterically inhibited by ATP at high concentrations. Which of the following is the benefit of regulating glycolysis by the concentration of ATP?

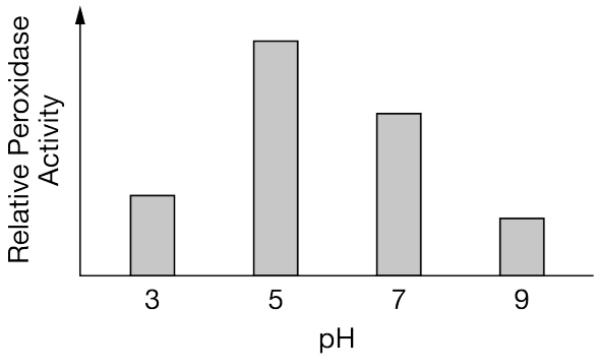
(A) Glycolysis proceeds when the intracellular concentration of ATP is low, which provides ATP to drive cellular reactions. ✓

(B) Glycolysis proceeds when the intracellular concentration of ATP is high and the cell stores ATP for future use.

(C) Glycolysis is inhibited when the intracellular concentration of ATP is low because PFK requires ATP as a substrate for the reaction it catalyzes.

(D) Glycolysis is inhibited when the intracellular concentration of ATP is high because ATP will compete with fructose 1,6-bisphosphate for binding to the active site on the enzyme.

44. The enzyme peroxidase is found in many organisms. It catalyzes the breakdown of hydrogen peroxide into water and oxygen gas. The rate of peroxidase activity at different pH values was assessed by students in the lab. The students' results are shown in graph 1.



Graph 1. Effect of pH on peroxidase activity

If the experiment is repeated at pH 11, the observed activity level of the enzyme will most likely be

(A) the same as the level at pH 7

(B) lower than the level at pH 9 ✓

(C) greater than the level at pH 9

(D) between the levels observed at pH 5 and pH 7.

Answer B

Correct. Peroxidase activity drops off as the pH increases beyond 5, indicating that the optimal level was closer to pH 5. The enzyme is likely to be further denatured by a pH greater than 9. Since this enzyme occurs in many organisms, it is reasonable that the optimal pH is closer to neutral than to a strongly basic or strongly acidic pH.

45. In plants the enzyme EPSP synthase catalyzes a reaction between two different substrates, S3P and PEP, to form a product used in the synthesis of three amino acids. The structure of glyphosate, the active chemical in many herbicides, is very similar to that of PEP.

Which of the following explains how glyphosate most likely functions as an herbicide?

(A) Glyphosate prevents synthesis of the three amino acids by competing with PEP for binding to the active site of EPSP synthase. ✓

(B) Glyphosate prevents synthesis of the three amino acids by binding to an allosteric site of EPSP synthase, increasing the likelihood that PEP will bind to the active site.

(C) Glyphosate causes too much synthesis of the three amino acids by reducing the activation energy of the reaction catalyzed by EPSP synthase.

(D) Glyphosate alters the structure of the three amino acids by binding to both S3P and PEP in the active site of EPSP synthase.

Answer A

Correct. Because the structure of glyphosate is very similar to that of PEP, glyphosate acts as a competitive inhibitor of the EPSP enzyme by competing with PEP for the active site of the enzyme.

第三部分 AP Chemistry Unit 2: Compound Structure and Properties

7 个课时共 115 题（答案版，各课时内沿用原编号）
＋ Progress Check FRQ 1 大题（含评分标准）
Progress Check MCQ 15 题与课时题库全部重复，不再收录

课时 2.1 Types of Chemical Bonds（16 题）

含答案解析 | 沿用原编号（本课时无删改）

2.1

1. The elements C and Se have the same electronegativity value, 2.55. Which of the following claims about the compound that forms from C and Se is most likely to be true?

(A) The carbon-to-selenium bond is unstable.

(B) The carbon-to-selenium bond is nonpolar covalent. ✓

(C) The compound has the empirical formula CSe.

(D) A molecule of the compound will have a partial negative charge on the carbon atom.

Answer B

Correct. The bonding in compounds between elements with the same (or only slightly different) electronegativity values will exhibit covalent bonding (that is, more or less equal sharing of electrons).

2. Two pure elements react to form a compound. One element is an alkali metal, X, and the other element is a halogen, Z. Which of the following is the most valid scientific claim that can be made about the compound?

(A) It has the formula XZ_2.

(B) It does not dissolve in water.

(C) It contains ionic bonds. ✓

(D) It contains covalent bonds.

Answer C

Correct. An alkali metal will readily lose an electron and a halogen will readily gain an electron to form an ionic compound with a 1:1 ratio of X^+: Z^- ions.

3. Which of the following scientific claims about the bond in the molecular compound HF is most likely to be true?

(A) There is a partial negative charge on the H atom.

(B) Electrons are shared equally between the H and F atoms.

(C) The bond is extremely weak.

(D) The bond is highly polar. ✓

2.1

Answer D

Correct. Fluorine is the most electronegative element, and the difference between the electronegativities of H and F is relatively large, resulting in a bond that is highly polar.

4.

Element	Electronegativity
H	2.1
C	2.5
S	2.5
F	4.0
Cl	3.0
Si	1.8

On the basis of the information above, which of the following arranges the binary compounds in order of increasing bond polarity?

(A) $CH_4 < SiCl_4 < SF_4$ ✓

(B) $CH_4 < SF_4 < SiCl_4$

(C) $SF_4 < CH_4 < SiCl_4$

(D) $SiCl_4 < SF_4 < CH_4$

5. Which of the following has the bonds arranged in order of decreasing polarity?

(A) H−F > N−F > F−F ✓

(B) H−I > H−Br > H−F

(C) O−N > O−S > O−Te

(D) Sb−I > Sb−Te > Sb−Cl

6. On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $\mathrm{Mg(NO_3)_2}(s)$?

2.1

(A) There is polar covalent bonding between Mg atoms and N atoms.

(B) There is polar covalent bonding between Mg atoms and O atoms.

(C) There is ionic bonding between N^{5+} ions and O^{2-} ions.

(D) There is ionic bonding between Mg^{2+} ions and NO_3^- ions. ✓

Answer D

Correct. There is ionic bonding in solid $Mg(NO_3)_2$ due to coulombic attractions between Mg^{2+} ions and NO_3^- ions. (There is some polar covalent bonding between N and O within the NO_3^- ion.)

7. Which of the following bonds is most polar?

(A) F-F

(B) O-F

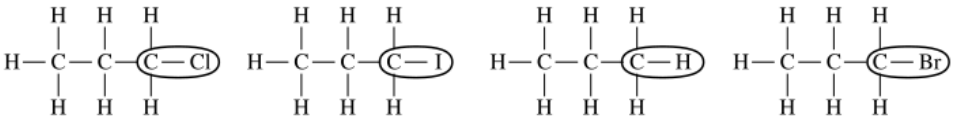
(C) O-I ✓

(D) O-Cl

Answer C

Correct. The polarity of bonds depends on the difference in the electronegativities of the atoms in the molecule; the greater the difference in electronegativity, the more polar the bond. Electronegativity generally decreases going down a column and from right to left across a row. Of the pairs given in the options, oxygen and iodine are farthest apart, have the greatest difference in electronegativity, and therefore form bonds that are the most polar.

8.



Which of the following bonds represented in the diagram above is the most polar?

(A) C–Cl ✓

(B) C–I

(C) C–H

(D) C–Br

2.1

Answer A

Correct. The $C-Cl$ bond is the most polar of the bonds in the responses because Cl is the most electronegative of Cl, I, H and Br. This can be inferred given that C and H have similar electronegativities and the periodic trend is for electronegativity to decrease from the top to the bottom of a column.

9. Which of the following compounds contains both ionic and covalent bonds?

(A) C_2H_5OH

(B) MgF_2

(C) H_2S

(D) NH_4Cl ✓

10. Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?

(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above $1000^\circ C$.

(D) The compound contains strong ionic bonds.

Answer B

Correct. Elements with the same electronegativity are more likely to share electrons and form a nonpolar covalent bond.

11. Which of the following substances has bonds with the greatest ionic character?

(A) CCl_4

(B) $CuCl_2$

(C) $CaCl_2$ ✓

(D) NCl_3

2.1

Answer C

Correct. The degree of ionic character in a chemical bond is a function of the difference in electronegativities of the two atoms that form the bond. Among C, Cu, Ca, and N, the element with the lowest electronegativity is Ca as, it is to the left of the other elements on the periodic table. Thus, the difference between the electronegativity of Ca and the electronegativity of Cl is the largest difference, and the Ca − Cl bond has the greatest ionic character.

12.

$$PCl_3, Cl_2, CaCl_2$$

Which of the following correctly ranks the substances listed above in order of increasing ionic character in their bonding?

(A) Cl_2, PCl_3, $CaCl_2$ ✓

(B) Cl_2, $CaCl_2$, PCl_3

(C) $CaCl_2$, PCl_3, Cl_2

(D) $CaCl_2$, Cl_2, PCl_3

Answer A

Correct. The ionic character of a bond increases with increasing differences between the electronegativities of the bonded elements. The bond in Cl_2 is the least ionic because the two bonded atoms are identical. The bonds in $CaCl_2$ are the most ionic because the bonded atoms (Ca and Cl) have the greatest difference in their electronegativities, which can be inferred from their relative positions in the periodic table. Ca and Cl are on opposite sides of the periodic table, so the difference in their electronegativities is relatively large. The bonds in $CaCl_2$ are more ionic than those in PCl_3 because the difference in electronegativities between Ca and Cl is greater than that between P and Cl.

13. An atom of chlorine and an atom of which of the following elements will form a bond with the most ionic character?

(A) Germanium

(B) Carbon

(C) Copper

(D) Potassium ✓

2.1

Answer D

Correct. The percent ionic character in a bond between two atoms increases with the difference in electronegativity between the bonded atoms. The electronegativity of elements varies with their position in the periodic table, increasing from left to right across a period and from bottom to top in a group. The greatest difference in electronegativity will be between chlorine and potassium because potassium is farther from chlorine on the periodic table than germanium, carbon, or copper are. Thus, the potassium-chlorine bond (K-Cl) will have the most ionic character.

14. Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

Answer D

Correct. A relatively large difference between the electronegativities of the two elements indicates that one element has a tendency to form cations and the other has a tendency to form anions.

15. Of the following compounds, which is the most ionic?

(A) $SiCl_4$

(B) BrCl

(C) PCl_3

(D) $CaCl_2$ ✓

16. Of the following single bonds, which is the LEAST polar?

(A) N—H

(B) H—F

(C) O—F ✓

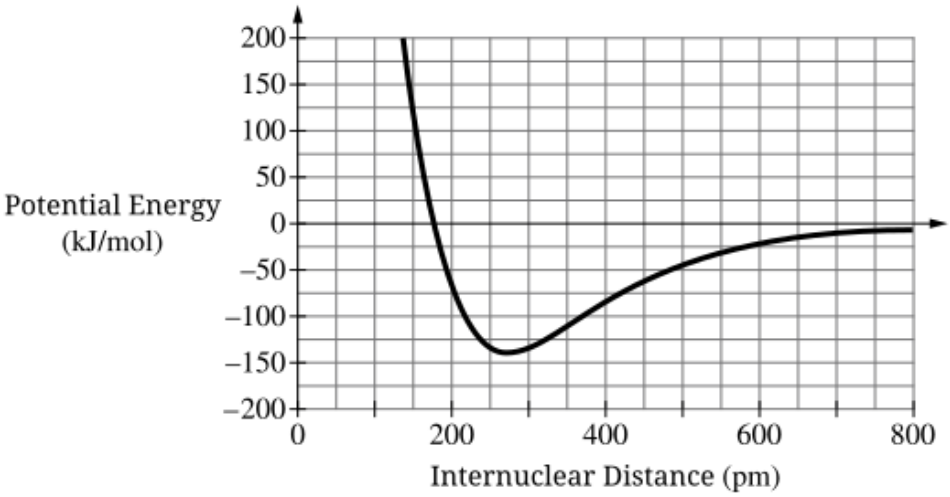
(D) I—F

课时 2.2 Intramolecular Force and Potential Energy（14 题）

含答案解析 | 沿用原编号（本课时无删改）

2.2

The graph shows the potential energy of two atoms as a function of internuclear distance.



1. What is the approximate bond energy?

(A) $140\ \text{kJ/mol}$ ✓

(B) $170\ \text{kJ/mol}$

(C) $200\ \text{kJ/mol}$

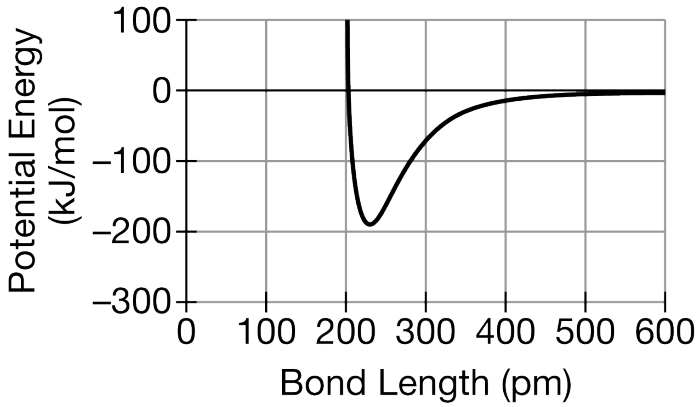
(D) $270\ \text{kJ/mol}$

Answer A

Correct. The minimum in the potential energy curve is $-140\ \text{kJ/mol}$. Bond energy is the energy required to break a bond and has the corresponding positive value, $140\ \text{kJ/mol}$.

2.2

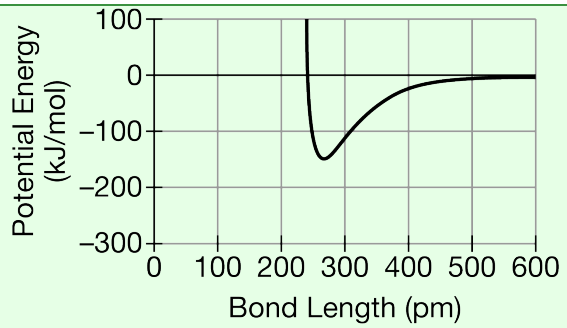
The following graph shows potential energy versus bond length for a Br_2 molecule.



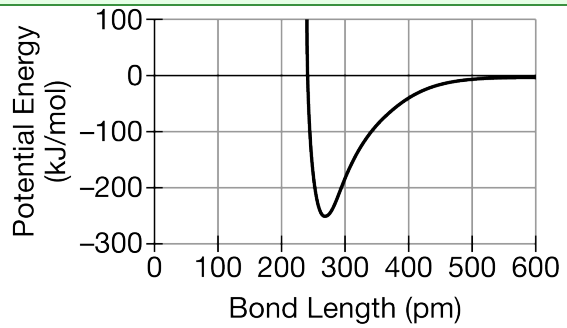
2. Which of the following graphs shows potential energy versus bond length for an I_2 molecule?

2.2

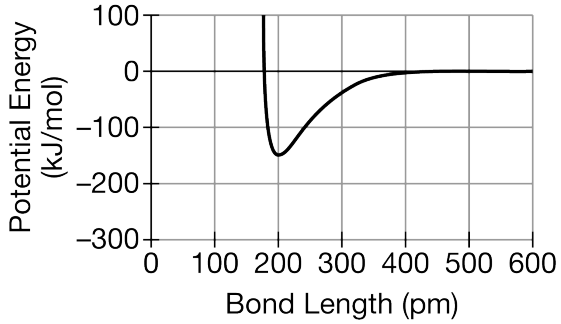
(A)



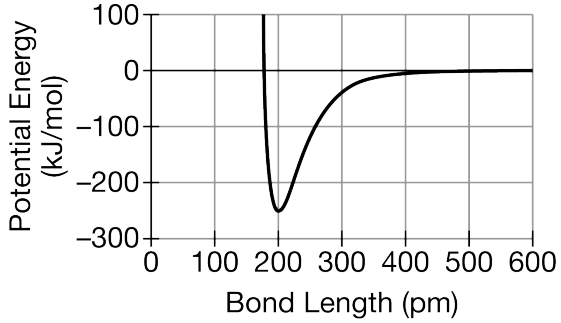
(B)



(C)



(D)

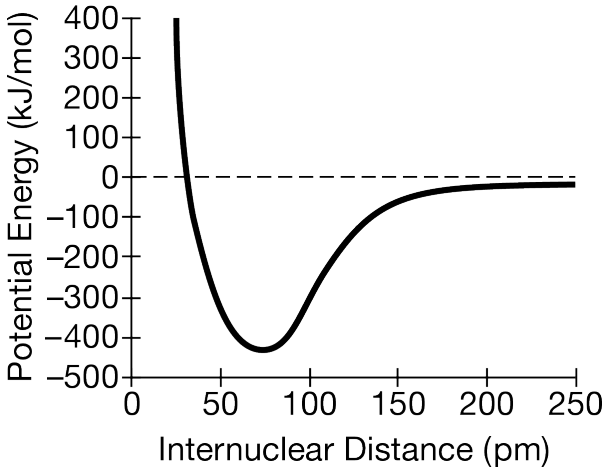


Answer A

Correct. Br and I are in the same group of the periodic table, but I has more occupied shells than Br does, so the I atom has a larger radius than the Br atom and its valence electrons are farther from the nucleus. This means that the $\mathrm{I_2}$ molecule has a longer bond length (the lowest point on the curve is farther to the right) than the $\mathrm{Br_2}$ molecule, and $\mathrm{I_2}$ has a bond energy with a smaller magnitude (the

2.2

lowest point on the curve is not as far down) than Br_2 does.



3. Which of the following can be inferred from the diagram above that shows the dependence of potential energy on the internuclear distance between two atoms?

(A) The atoms form a bond with a bond length of $25\ \text{pm}$.

(B) The atoms form a bond with a bond length of $75\ \text{pm}$. ✓

(C) The net force between the atoms is attractive at $25\ \text{pm}$.

(D) The net force between the atoms is attractive at $75\ \text{pm}$.

Answer B

Correct. The minimum potential energy occurs at an internuclear distance of $75\ \text{pm}$, which corresponds to the length of the stable bond that forms between the two atoms. If the atoms were any closer to each other, the net force would be repulsive. Likewise, if the atoms were farther from each other, the net force would be attractive. At $75\ \text{pm}$, the net force between the two atoms is zero.

---

4. Which of the following compounds has the strongest Coulombic attractions between its ions in the solid state?

(A) MgO ✓

(B) CaO

(C) $MgCl_2$

(D) $CaCl_2$

2.2

Answer A

Correct. The Coulombic force is directly proportional to $\frac{q_1 q_2}{r^2}$, where q_1 and q_2 are the charges of the ions and r is the distance between the ions. Ionic compounds that contain the smallest ions and ions with the greatest charges therefore have the strongest Coulombic attractions between ions. The charge on the cation is $+2$ in all responses. MgO and CaO have a -2 charge on the anion, whereas $\mathrm{MgCl_2}$ and $\mathrm{CaCl_2}$ have a -1 charge on the anion. The $\mathrm{Mg^{2+}}$ ion has one less occupied electron shell and thus is smaller than the $\mathrm{Ca^{2+}}$ ion, so MgO has the strongest Coulombic attractions among the choices given.

5.

Compound	Melting Point ($^\circ\mathrm{C}$)
MgO	2852
NaF	993

The table above shows the melting points of $\mathrm{MgO}(s)$ and $\mathrm{NaF}(s)$. Which of the following best helps explain why the melting point of $\mathrm{MgO}(s)$ is much higher than that of $\mathrm{NaF}(s)$?

(A) The mass of $\mathrm{F^-}$ ions is greater than that of $\mathrm{O^{2-}}$ ions.

(B) The mass of $\mathrm{Mg^{2+}}$ ions is greater than that of $\mathrm{Na^+}$ ions.

(C) The difference between the electronegativity values of Mg and O is smaller than the difference between the values of Na and F.

(D) The charges of $\mathrm{Mg^{2+}}$ and $\mathrm{O^{2-}}$ ions are greater than those of $\mathrm{Na^+}$ and $\mathrm{F^-}$ ions. ✓

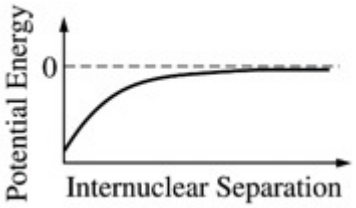
Answer D

Correct. The melting point of an ionic lattice tends to be higher the stronger the force of attraction between the cation and the anion. According to Coulomb's law, the greater the charges, the stronger the force of attraction and the higher the melting point. MgO involves a $+2$ cation and a -2 anion, whereas NaF involves a $+1$ cation and a -1 anion. Hence, MgO has the higher melting point.

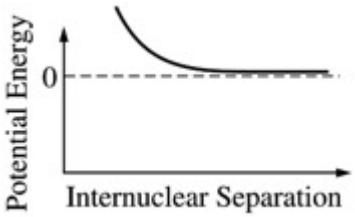
6. Which of the following graphs correctly shows the relationship between potential energy and internuclear separation for two hydrogen atoms?

2.2

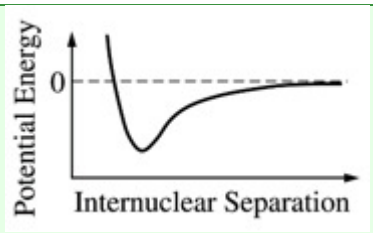
(A)


Potential Energy
0
Internuclear Separation

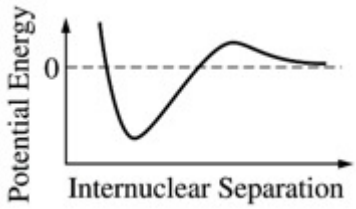
(B)


Potential Energy
0
Internuclear Separation

(C) ✓

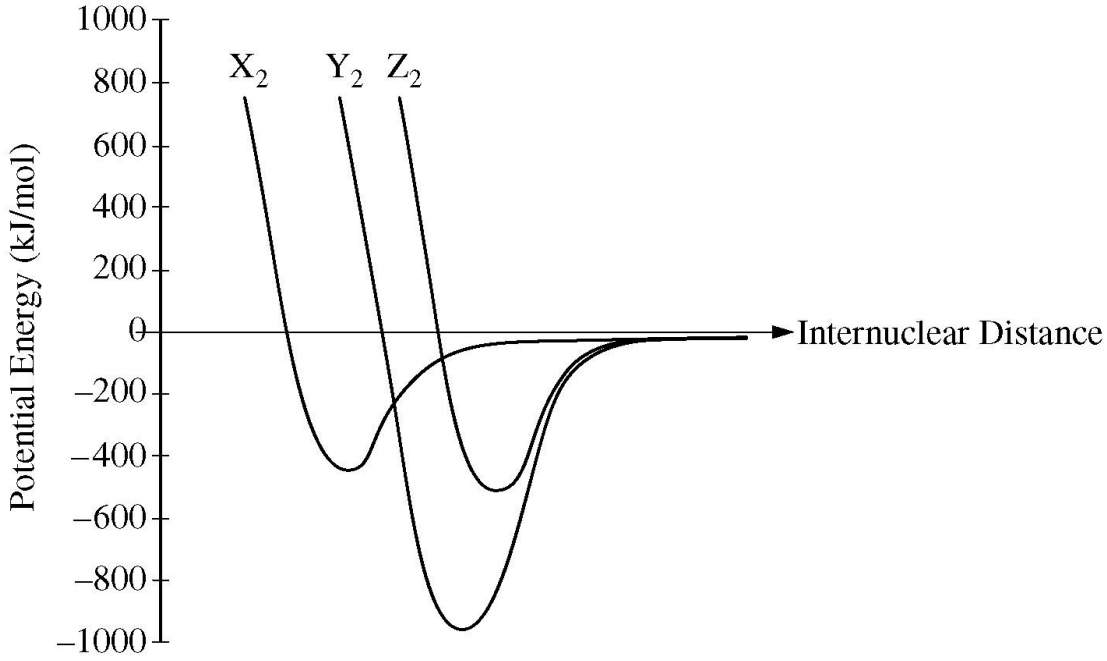

Potential Energy
0
Internuclear Separation

(D)


Potential Energy
0
Internuclear Separation

2.2

7.



The potential energy as a function of internuclear distance for three diatomic molecules, X_2, Y_2, and Z_2, is shown in the graph above. Based on the data in the graph, which of the following correctly identifies the diatomic molecules, X_2, Y_2, and Z_2?

(A) ✓

X_2	**Y_2**	**Z_2**
H_2	N_2	O_2

(B)

X_2	**Y_2**	**Z_2**
H_2	O_2	N_2

(C)

X_2	**Y_2**	**Z_2**
N_2	O_2	H_2

(D)

X_2	**Y_2**	**Z_2**
O_2	H_2	N_2

2.2

8. The lattice energy of a salt is related to the energy required to separate the ions. For which of the following pairs of ions is the energy that is required to separate the ions largest? (Assume that the distance between the ions in each pair is equal to the sum of the ionic radii.)

(A) $Na^+(g)$ and $Cl^-(g)$

(B) $Cs^+(g)$ and $Br^-(g)$

(C) $Mg^{2+}(g)$ and $O^{2-}(g)$ ✓

(D) $Ca^{2+}(g)$ and $O^{2-}(g)$

Lattice enthalpy can be defined as the ΔH° associated with the separation of a solid crystal into gaseous ions. Lattice enthalpies for four compounds are given in the table.

Compound	Lattice Enthalpy (kJ/mol)
LiF	1030
$LiCl$	834
$MgCl_2$	2510
$SrCl_2$	2127

9. Which of the following statements about lattice enthalpy is consistent with the information in the table?

(A) It increases with the increasing radii of the ions in the compound.

(B) It increases with the decreasing radii of the ions in the compound. ✓

(C) It increases with the decreasing charge of the ions.

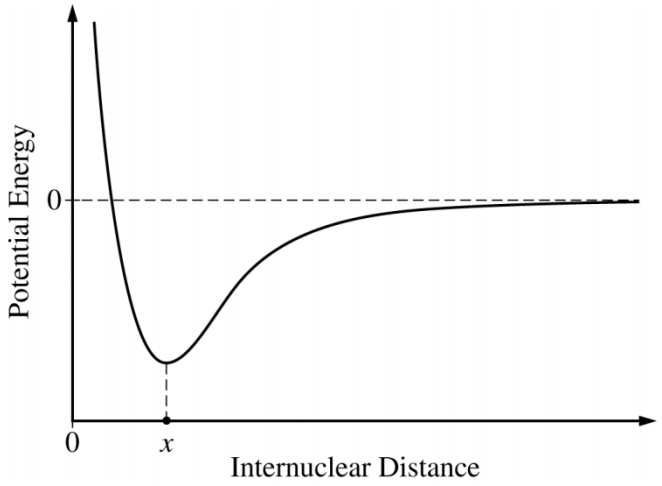
(D) It increases with the decreasing electronegativity of the atoms.

Answer B

Correct. F^- has fewer occupied electron shells and is a smaller ion than Cl^- is, and LiF has a larger lattice enthalpy than $LiCl$ does. Similarly, Mg^{2+} is a smaller ion than Sr^{2+} is, and $MgCl_2$ has a larger lattice enthalpy than $SrCl_2$ does. Therefore, this statement is consistent with the data in the table.

2.2

10.



The potential energy of a system of two atoms as a function of their internuclear distance is shown in the diagram above. Which of the following is true regarding the forces between the atoms when their internuclear distance is x?

(A) The attractive and repulsive forces are balanced, so the atoms will maintain an average internuclear distance x. ✓

(B) There is a net repulsive force pushing the atoms apart, so the atoms will move further apart.

(C) There is a net attractive force pulling the atoms together, so the atoms will move closer together.

(D) It cannot be determined whether the forces between atoms are balanced, attractive, or repulsive, because the diagram shows only the potential energy

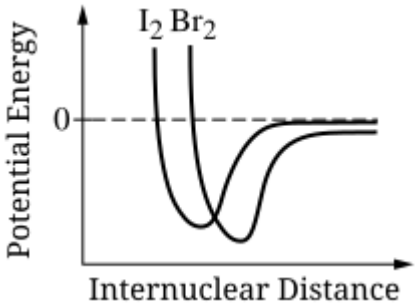
11.

Molecule	Bond Dissociation Energy (kJ / mol)
Br_2	193
I_2	151

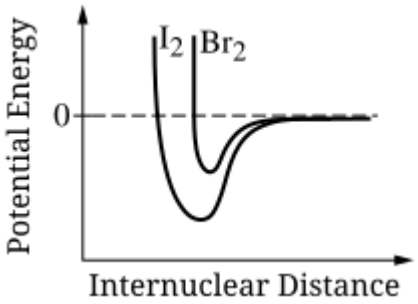
The table above shows the bond dissociation energies for Br_2 and I_2 at $298\ \text{K}$. Which of the following correctly shows the potential energy as a function of internuclear distance for two bromine atoms and two iodine atoms?

2.2

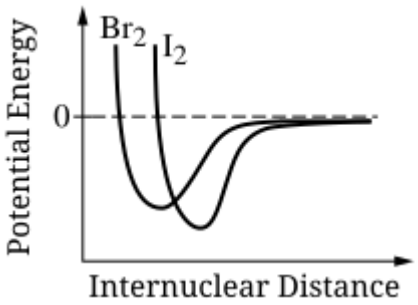
(A)



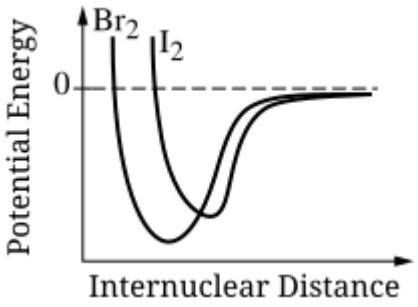
(B)



(C)



(D)

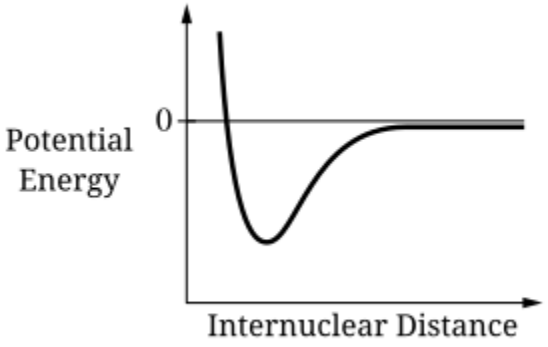


Answer D

Correct. The bond dissociation energy for Br_2 is given to be greater than that for I_2. In addition, Br is smaller than I, so the equilibrium internuclear distance for Br_2 should be less than that for I_2.

2.2

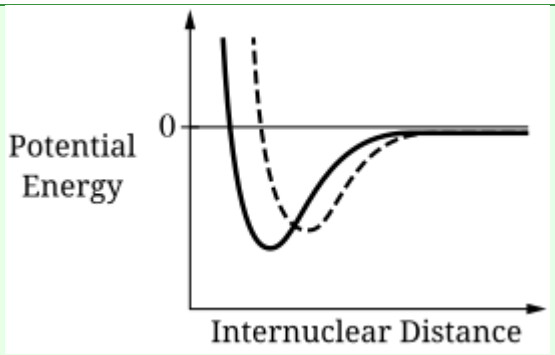
12.



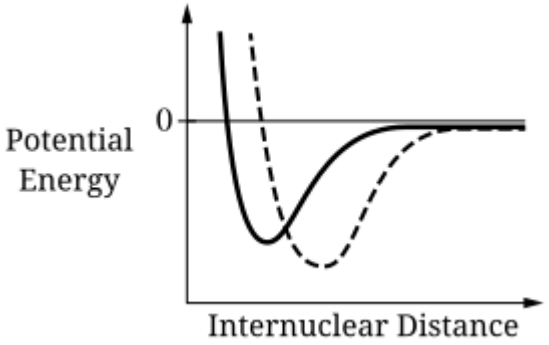
The graph of potential energy versus internuclear distance for the HCl molecule is shown. Which of the following graphs includes the plot for the HBr molecule, shown as a dashed line?

2.2

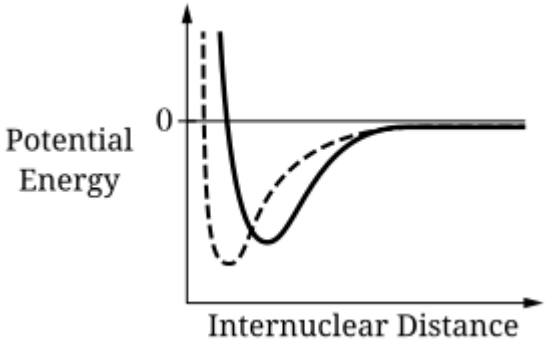
(A)



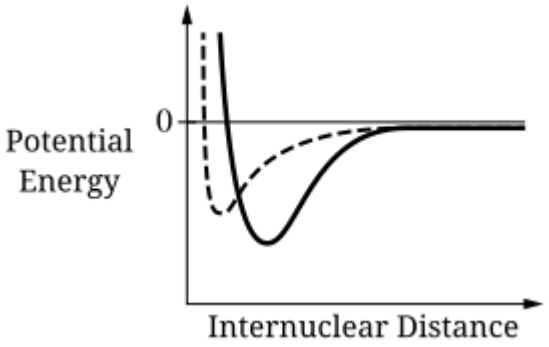
(B)



(C)



(D)



Answer A

Correct. HBr will have an equilibrium internuclear distance that is greater than that for HCl. In addition, HBr will have a smaller bond strength (depth of well) than HCl. The graph satisfies both these

2.2

statements.

Gaseous hydrogen and gaseous chlorine can react to form hydrogen chloride gas, as represented by the following equation.

$$H_2(g) + Cl_2(g) \rightarrow 2\ HCl(g)$$

13. Which of the following correctly ranks the internuclear distances for the H_2, Cl_2, and HCl molecules from shortest to longest?

(A) $H_2 < Cl_2 < HCl$

(B) $H_2 < HCl < Cl_2$ ✓

(C) $Cl_2 < HCl < H_2$

(D) $Cl_2 < H_2 < HCl$

Answer B

Correct. The atomic radius of a Cl atom is much greater than that of an H atom because Cl has more shells of electrons than H does. The electron configuration of an H atom is $1s^1$, and the electron configuration of a Cl atom is $1s^2 2s^2 2p^6 3s^2 3p^5$. Because of the respective atomic sizes of H and Cl, H_2 should have the shortest bond distance and Cl_2 should have the longest bond distance of the three molecules.

---

14. The melting point of MgO is higher than that of NaF. Explanations for this observation include which of the following?

I. Mg^{2+} is more positively charged than Na^{+}.

II. O^{2-} is more negatively charged than F^{-}.

III. The O^{2-} ion is smaller than the F^{-} ion.

(A) I and II only ✓

(B) I and III only

(C) II and III only

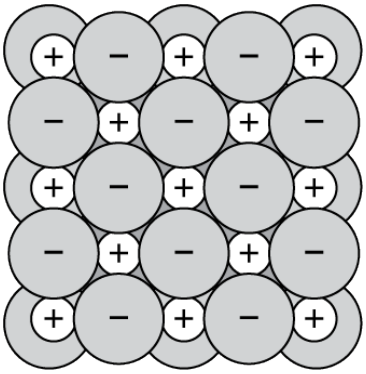
(D) I, II, and III

课时 2.3 Structure of Ionic Solids（9 题）

含答案解析 | 沿用原编号（本课时无删改）

2.3

1.



Which of the following correctly indicates whether the solid represented by the particulate model shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative ions.

(B) It conducts electricity because it is made of particles of different sizes.

(C) It does not conduct electricity because its ions cannot move freely within the solid. ✓

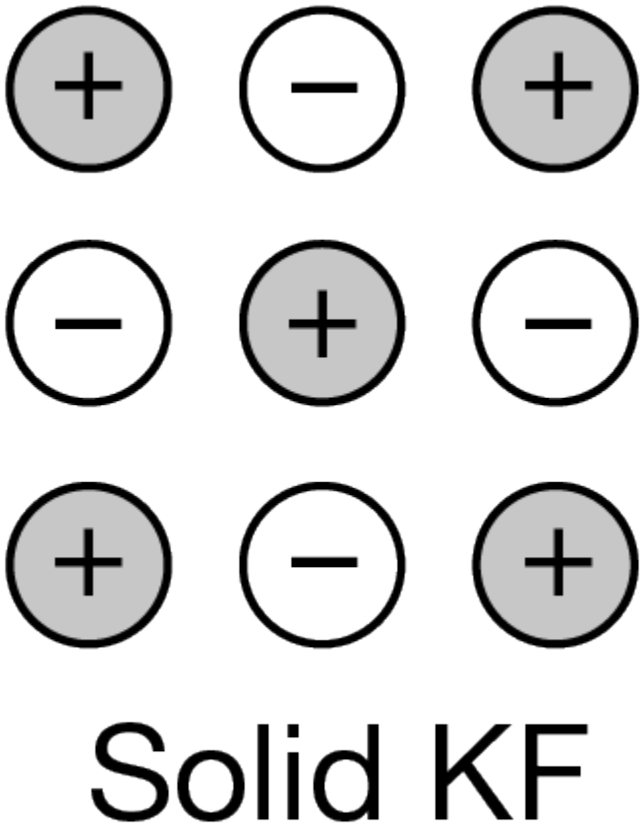
(D) It does not conduct electricity because there are small spaces between the particles.

Answer C

Correct. In the solid state, the ions are held by strong coulombic forces in fixed positions within the lattice; thus they cannot move or migrate through the lattice.

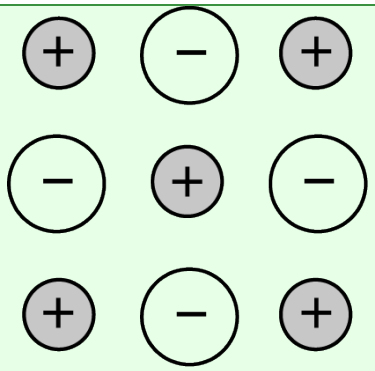
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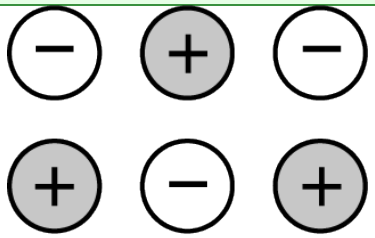
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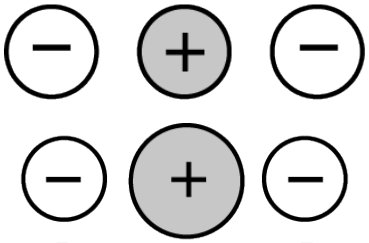


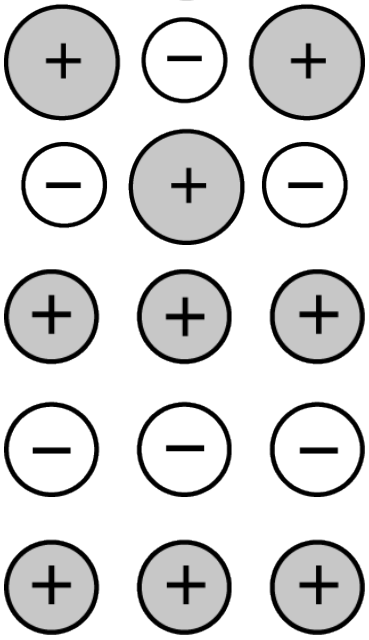
The particle-level diagram above represents the structure of solid KF. Although the molar mass of KCl is greater than that of KF, the density of KCl is actually less than that of KF. Which of the following representations of the structure of KCl best helps to explain this phenomenon?

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(A) 

(B) 

(C) 

(D) 

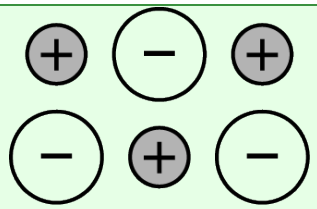
Answer A

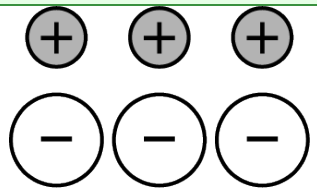
Correct. The diagram shows that the chloride ions are larger than the fluoride ions, which causes the

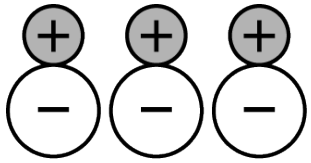
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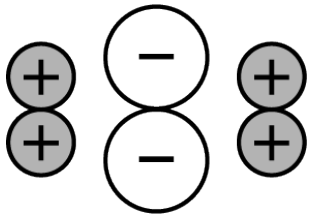
interionic distances in the solid to increase to the point where the effect of the larger spacing among ions in KCl is greater than the effect of the larger mass of the chlorine on the density.

3. The particles in solid KI, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid KI?

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. This diagram shows both types of charged ions being surrounded by ions of the opposite charge, thus maximizing attraction and minimizing repulsion.

2.3

4.

Compound	Lattice Energy $(\mathrm{kJ/mol})$
NaF	930
$\mathrm{MgF_2}$	2978

The energy required to dissociate an ionic solid into gaseous ions (lattice energy) for the compounds NaF and $\mathrm{MgF_2}$ is shown in the table above. On the basis of Coulomb's law, which of the following best helps to explain the large difference between the lattice energies of NaF and $\mathrm{MgF_2}$?

(A) The solubility of $\mathrm{MgF_2}$ is less than that of NaF.

(B) The electronegativity of Mg is greater than that of Na.

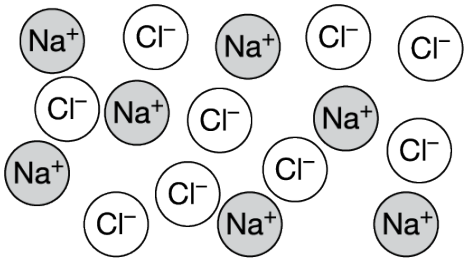
(C) The mass of the Mg cation is greater than that of the Na cation.

(D) The charge of the Mg cation is larger than that of the Na cation. ✓

Answer D

Correct. The coulombic attractions between the Mg cation and F anions are stronger because of the $+2$ charge on Mg as opposed to the $+1$ charge on Na. Stronger interionic attractions lead to a higher lattice energy.

5.



Molten (liquid) NaCl is represented by the particulate diagram shown above. Which of the following indicates whether $\mathrm{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because Na is a metal.

(B) It conducts electricity because ions are free to move. ✓

(C) It does not conduct electricity because Cl is a nonmetal.

(D) It does not conduct electricity because there are no free electrons.

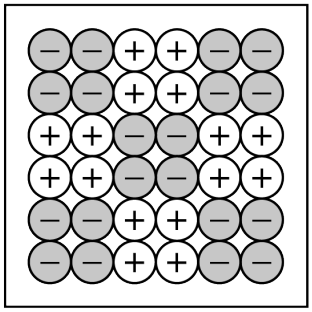
2.3

Answer B

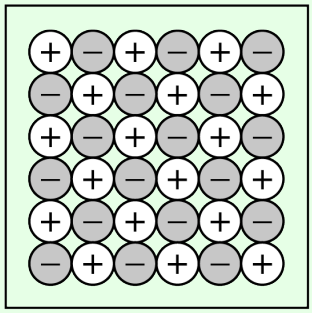
Correct. For a substance to conduct electricity, it must contain charged particles that are mobile. In molten salts such as $NaCl$, those charged particles are ions. (Metals conduct electricity via mobile electrons within a lattice of metal atoms.)

6. Of the following diagrams, which best represents the structure of solid KF ?

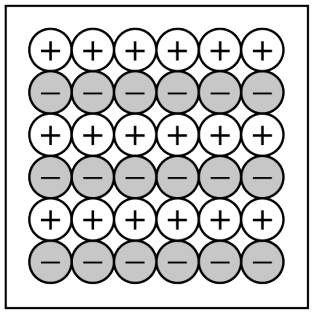
(A)



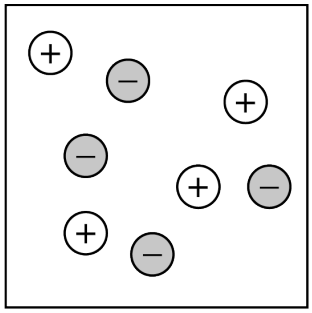
(B) ✓



(C)



(D)



2.3

Answer B

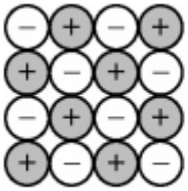
Correct. This diagram shows ions as being tightly packed surrounded by ions of the opposite charge, which is true in a crystal of an ionic compound such as KF. This arrangement represents the lowest energy state attainable by the constituent ions.

7. Which of the following diagrams best illustrates how a displacement in an ionic crystal results in cleavage and brittleness?

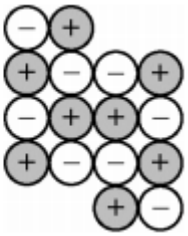
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(A) ✓

Before Displacement

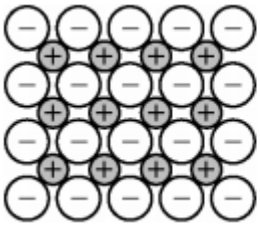


After Displacement

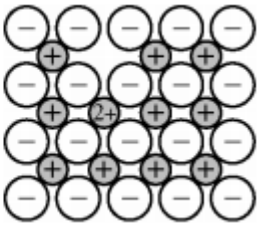


(B)

Before Displacement

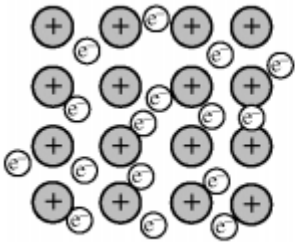


After Displacement

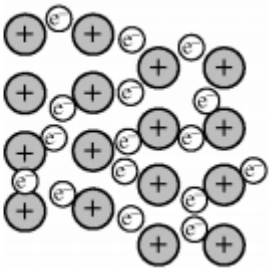


(C)

Before Displacement

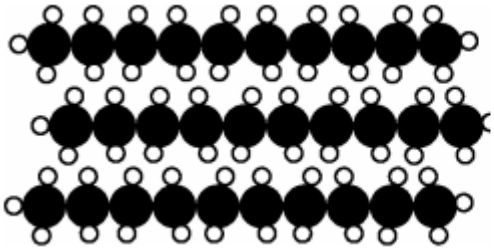


After Displacement

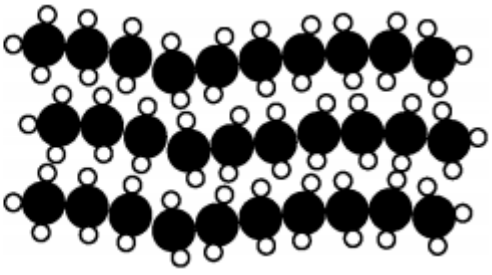


(D)

Before Displacement

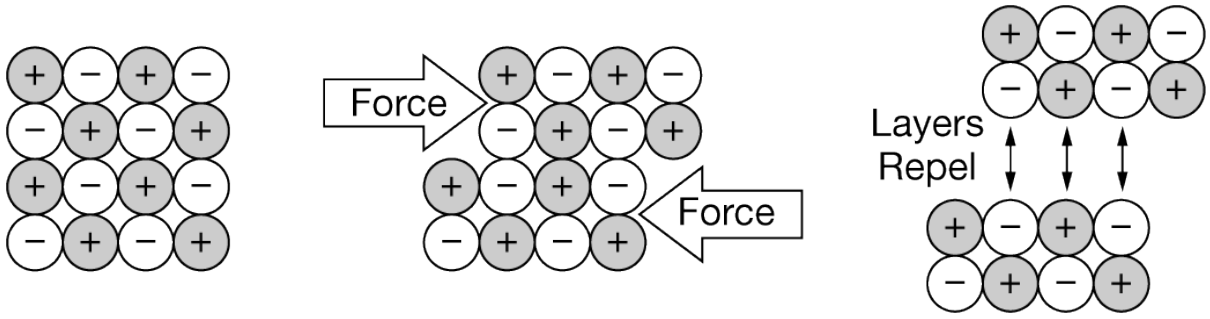


After Displacement



2.3

8.



The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

(A) Density

(B) Brittleness ✓

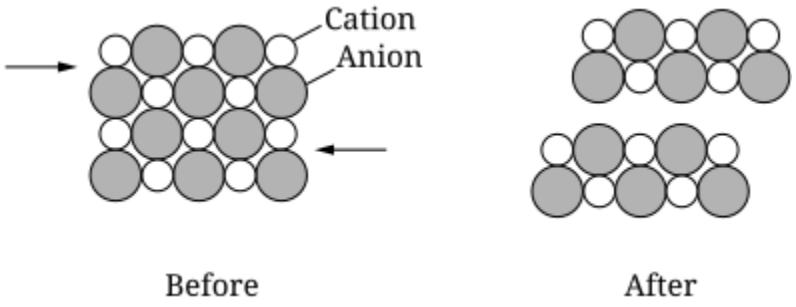
(C) Malleability

(D) Conductivity

Answer B

Correct. The brittleness of ionic solids is due to the internal repulsion of ion layers within the crystal lattice that occurs when the layers are displaced by an outside force parallel to the layers and they move relative to adjacent layers.

9.



When a stress is applied to an ionic crystal, the planes of the ions slide and separate due to the alignment of like charges, as shown in the diagram above. Which of the following characteristics of the crystal is represented by the observation?

(A) Luster

(B) Brittleness ✓

(C) High melting point

(D) Solubility in water

2.3

Answer B

Correct. Brittleness is the tendency of a material to break when pressure is applied.

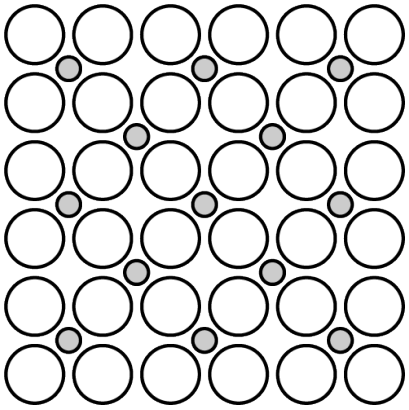
课时 2.4 Structure of Metals and Alloys（11 题）

含答案解析 | 沿用原编号（本课时无删改）

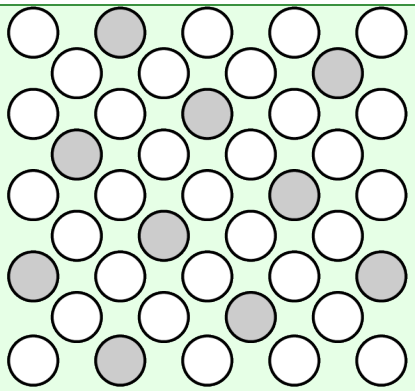
2.4

1. Copper atoms and zinc atoms have the same atomic radius, 135 picometers. Based on this information, which of the following diagrams best represents an alloy containing only copper and zinc atoms?

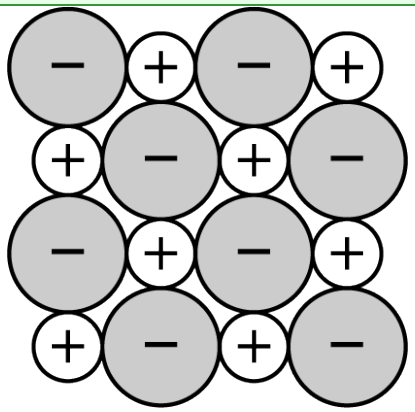
(A)



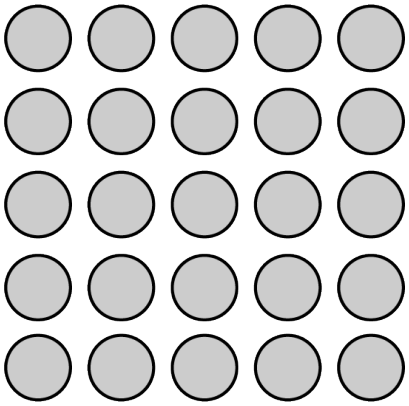
(B) ✓



(C)



(D)

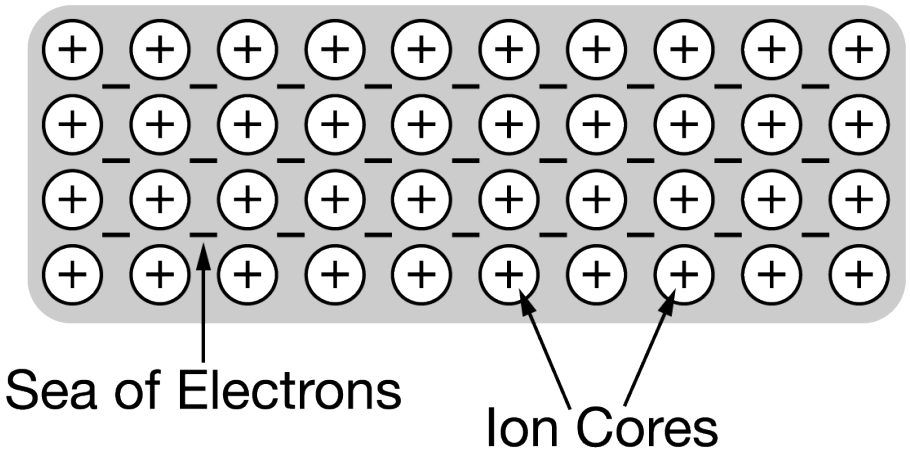


2.4

Answer B

Correct. Being the same size, copper atoms and zinc atoms would form a substitutional alloy in which the atoms substitute for one another in a lattice of positive cores immersed in a sea of electrons.

2.



A particle-level diagram of a metallic element is shown above. Typically, metals are both malleable and ductile. The best explanation for these properties is that the electrons involved in bonding among metal atoms are

(A) unequally shared and form nondirectional bonds

(B) unequally shared and form highly directional bonds

(C) equally shared and form nondirectional bonds ✓

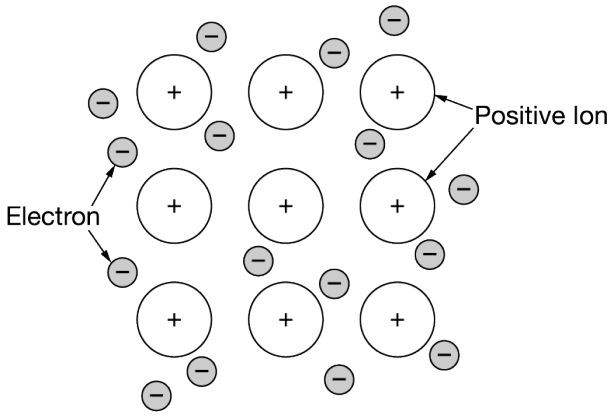
(D) equally shared and form highly directional bonds

Answer C

Correct. In metals, electrons are free to move as described by the electron sea model. Electrons are shared equally among atoms, and the structure is based on simple close-packed structures that contain no fixed directional bonds.

2.4

3.



Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative particles.

(B) It conducts electricity because electrons are free to move through the substance. ✓

(C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.

(D) It does not conduct electricity because the positive particles are not free to move through the substance.

Answer B

Correct. The diagram shows the structure of a metallic solid, which consists of a lattice of positive ions immersed in a sea of electrons in which electrons can move freely.

4.

Element	Metallic Radius (pm)	Melting Point (°C)	Common Oxidation State
Au	144	1064	1+, 3+
Cu	128	1085	1+, 2+
Ag	144	961	1+

To make Au stronger and harder, it is often alloyed with other metals, such as Cu and Ag. Consider two alloys, one of Au and Cu and one of Au and Ag, each with the same mole fraction of Au. If the Au/Cu alloy is harder than the Au/Ag alloy, then which of the following is the best explanation based on the information in the table above?

(A) Cu has two common oxidation states, but Ag has only one.

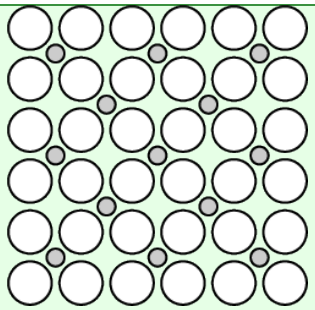
(B) Cu has a higher melting point than Au has, but Ag has a lower melting point than Au has.

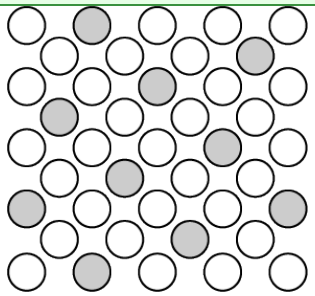
(C) Cu atoms are smaller than Ag atoms, thus they interfere more with the displacement of atoms in the alloy. ✓

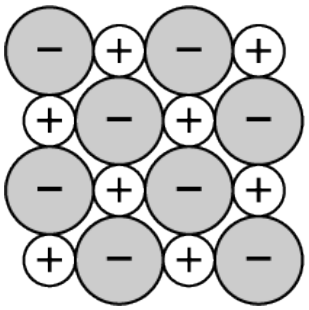
(D) Cu atoms are less polarizable than are Au or Ag atoms, thus Cu has weaker interparticle forces.

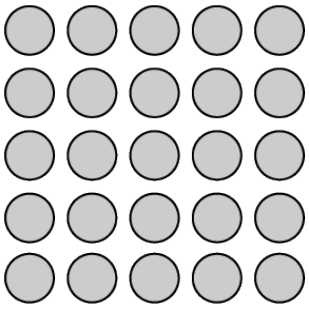
2.4

5. Which of the following particulate-level diagrams best represents an interstitial alloy?

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. In an interstitial alloy, atoms of one of the constituents are significantly smaller than the atoms of the other constituent and occupy the smaller spaces (interstices) among the larger atoms.

2.4

Atomic radii and electronegativity values for some elements are given in the table.

Element	Radius	Electronegativity
Ag	165 pm	1.93
Au	174 pm	2.54
Cu	145 pm	1.90
N	56 pm	3.04
Ni	149 pm	1.91
W	193 pm	2.36

6. Which of the following names a pair of elements that would likely form a substitutional alloy and explains why?

(A) Ni and Cu, because Ni atoms would fit in the spaces between Cu atoms in the solid lattice.

(B) Ni and Cu, because Ni atoms would take the place of Cu atoms in the solid lattice. ✓

(C) W and N, because N atoms would fit in the spaces between W atoms in the solid lattice.

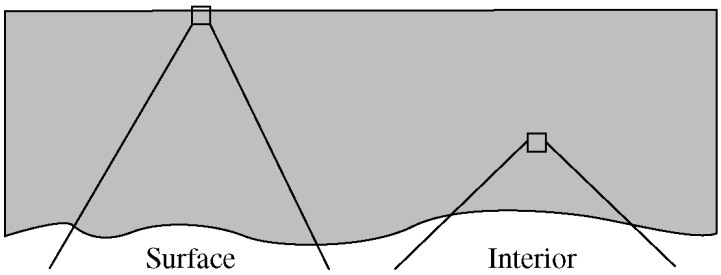
(D) W and N, because N atoms would take the place of W atoms in the solid lattice.

Answer B

Correct. Substitutional alloys form between elements with atoms of comparable radius, where atoms of one element substitute for atoms of another element in the solid lattice. Ni and Cu have similar radii.

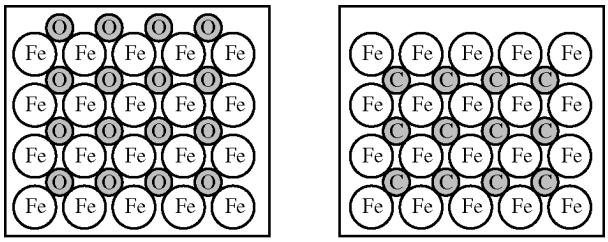
2.4

7. Steel is an alloy consisting of Fe with a small amount of C. Elemental Cr can be added to steel to make the steel less likely to rust; Cr atoms react with oxygen in the air to form a nonreactive layer of chromium oxide on the surface of the steel, preventing the oxidation of underlying Fe atoms. A sample of steel-chromium alloy contains 15 percent Cr by mass. Which of the following diagrams best shows a particle-level view of a surface section and an interior section of the alloy represented below at the left? (The atomic radii of the atoms involved are given in the table below at the right.)

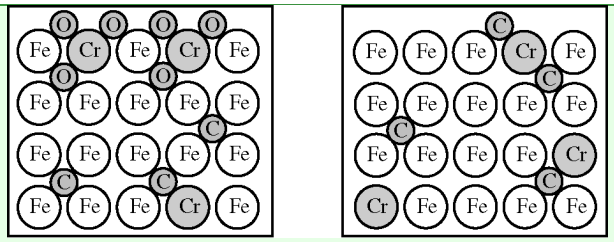


Element	Molar Mass (g/mol)	Atomic Radius (pm)
Fe	55.85	125
Cr	52.00	127
C	12.01	77
O	16.00	73

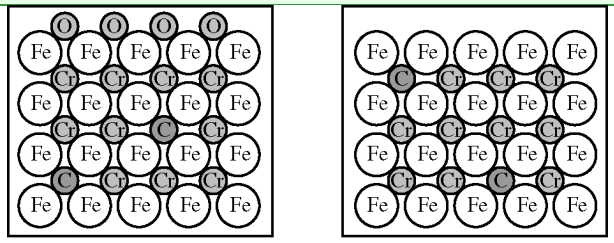
(A)



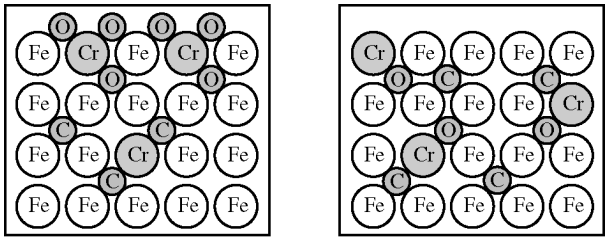
(B) ✓



(C)

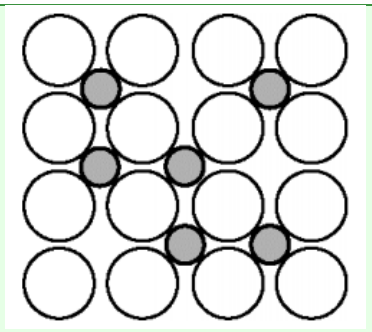


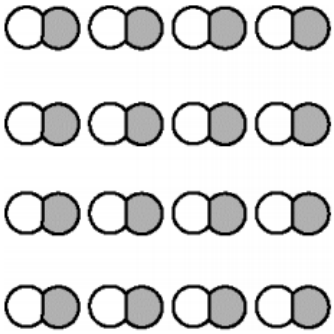
(D)

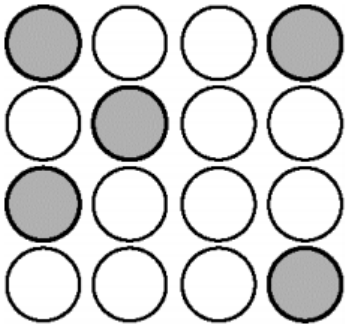


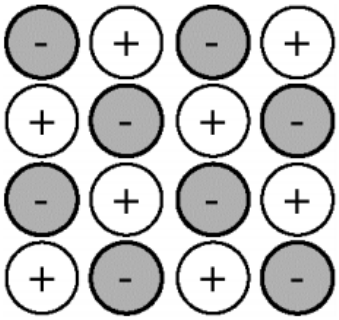
8. Which of the following diagrams best depicts an alloy of Ni and B?

2.4

(A) 

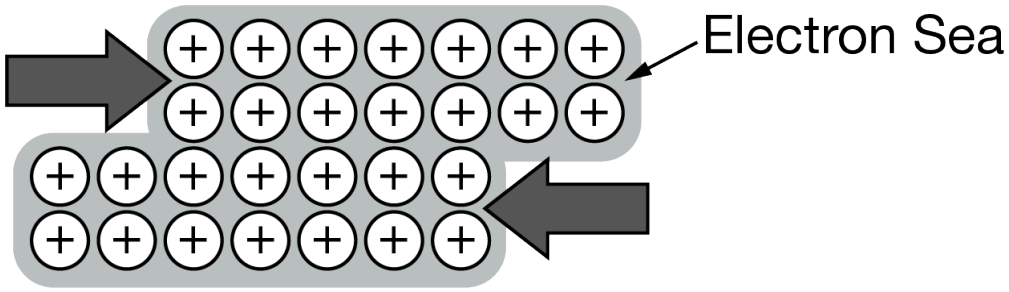
(B) 

(C) 

(D) 

2.4

9.



The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

(A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.

(B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓

(C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.

(D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.

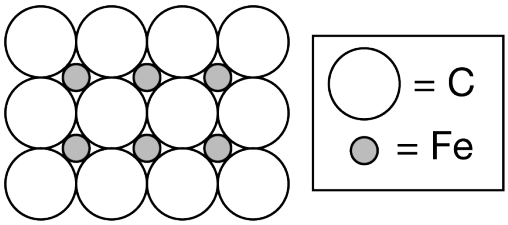
Answer B

Correct. The diagram shows the structure of a metallic solid, which has adjacent layers of positive ions sliding past one another (as indicated by the arrows) while staying within the shading of the electron sea.

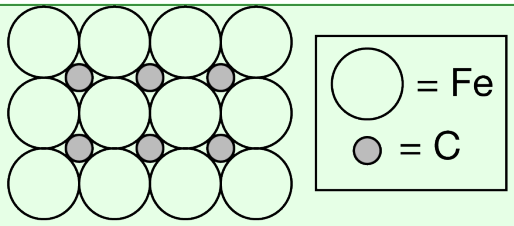
10. Steel is an alloy containing Fe atoms and C atoms. Which of the following diagrams best represents the particle-level structure of steel?

2.4

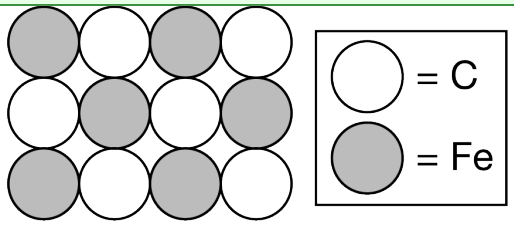
(A)



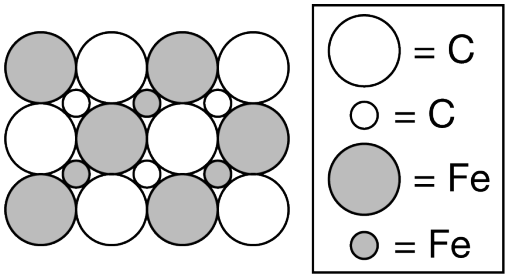
(B)



(C)



(D)



Answer B

Correct. C is in the second period and has two electron shells, whereas Fe is in the fourth period and has four electron shells; thus C atoms are smaller than Fe atoms. The diagram shows how the smaller C atoms occupy the spaces (interstices) between the larger Fe atoms in an interstitial alloy.

11.

Type of Steel	% Carbon	Characteristics	Uses
Low-carbon steel	< 0.2 %	Malleable and ductile	Chains and nails
High-carbon steel	0.6 – 1.5 %	Hard and brittle	Cutting tools

The table above provides some information about two types of steel, both of which are alloys of iron and carbon. Which of the following best helps to explain why high-carbon steel is more rigid than low-carbon steel?

2.4

(A) Elemental carbon is harder than elemental iron.

(B) The additional carbon atoms within the alloy make the high-carbon steel less dense.

(C) The additional carbon atoms within the alloy increase the thermal conductivity of the high-carbon steel.

(D) The additional carbon atoms within the alloy make it more difficult for the iron atoms to slide past one another. ✓

课时 2.5 Lewis Diagrams（8 题）

含答案解析 | 沿用原编号（本课时无删改）

2.5

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4

1. Has a central atom with less than an octet of electrons

(A) H_2O
(B) NH_3
(C) BH_3 ✓
(D) CH_4

2. Has two lone pairs of electrons

(A) H_2O ✓
(B) NH_3
(C) BH_3
(D) CH_4

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

Refer to the following compounds.

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) CH_3COCH_3
(D) CH_3COOH

3. Is isomeric with CH_3CH_2CHO

(A) $CH_3CH_2CH_2CH_3$
(B) $CH_3CH_2CH_2OH$
(C) CH_3COCH_3 ✓
(D) CH_3COOH

2.5

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

4. The molecule with only one double bond

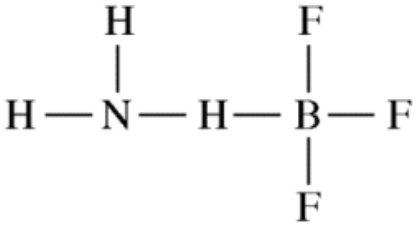
(A) CO_2
(B) H_2O
(C) C_2H_4 ✓
(D) PH_3

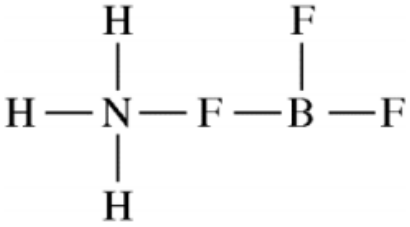
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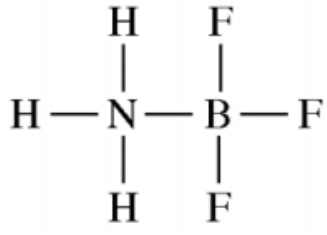
5. NH_3 reacts with BF_3 to form a single species. Which of the following structural diagrams is the most likely representation of the product of the reaction?

2.5

(A) 

(B) 

(C) 

(D)  ✓

6. The electron-dot structure (Lewis structure) for which of the following molecules would have two unshared pairs of electrons on the central atom?

(A) H_2S ✓

(B) NH_3

(C) HCN

(D) CO_2

7. Which of the following molecules contains only single bonds?

(A) CH_3COOH

(B) C_2H_6 ✓

(C) C_6H_6

(D) HCN

8. Which of the following is an isomer of CH_3OCH_3 ?

2.5

(A) CH_3CH_3

(B) CH_3COOH

(C) CH_3CH_2OH ✓

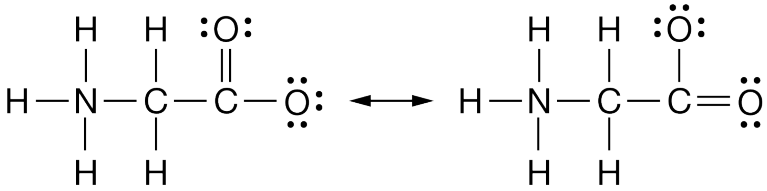
(D) $CH_3CH_2OCH_2CH_3$

课时 2.6 Resonance and Formal Charge（12 题）

含答案解析 | 沿用原编号（本课时无删改）

2.6

1.



Based on the resonance structures shown above, what are the bond orders of the two carbon-oxygen bonds?

(A) 1 and 1.5

(B) 1 and 2

(C) 1.5 and 1.5 ✓

(D) 2 and 2

Answer C

Correct. When two resonance structures are equivalent, they contribute equally to the average structure, and the average structure is the preferred representation of the molecule. Bond orders are obtained by taking the average of the bond orders in the resonance structures. Therefore, in this case both $C-O$ bonds have bond orders equal to the average of 1 and 2, or 1.5.

2. For which of the following molecules are resonance structures necessary to describe the bonding satisfactorily?

(A) SO_2 ✓

(B) CO_2

(C) OF_2

(D) PF_3

Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) SO_2

(C) N_2

(D) F_2

2.6

3. Is best represented by two or more resonance forms

(A) $BeCl_2$

(B) SO_2 ✓

(C) N_2

(D) F_2

A solution is prepared by combining $1.0\ \text{L}$ of $0.50\ M\ \text{HClO}_2$ with $1.0\ \text{L}$ of $0.50\ M\ \text{KClO}_2$. The pH of the solution is 2.0.

4. Consider the Lewis electron-dot diagram for the ClO_2^- ion that minimizes formal charge. Which of the following is true about the hybridization of the Cl atom in the ion?

(A) The Cl atom is not hybridized, and the ion contains one single $\text{Cl}-\text{O}$ bond and one double $\text{Cl}-\text{O}$ bond.

(B) The Cl atom is sp hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ double bonds.

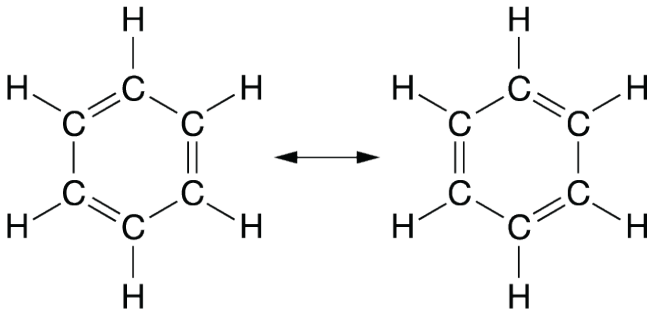
(C) The Cl atom is sp^2 hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds.

(D) The Cl atom is sp^3 hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds. ✓

Answer D

Correct. A valid Lewis electron-dot diagram of the ClO_2^- ion has two nonbonding pairs of electrons around the central Cl atom. Since there are four electron domains around the Cl atom, it is sp^3 hybridized.

5.



The diagram above shows two resonance structures for a molecule of C_6H_6. The phenomenon shown in the diagram best supports which of the following claims about the bonding in C_6H_6 ?

2.6

(A) In the C_6H_6 molecule, all the bonds between the carbon atoms have the same length. ✓

(B) Because of variable bonding between its carbon atoms, C_6H_6 is a good conductor of electricity.

(C) The bonds between carbon atoms in C_6H_6 are unstable, and the compound decomposes quickly.

(D) The C_6H_6 molecule contains three single bonds between carbon atoms and three double bonds between carbon atoms.

Answer A

Correct. Because of resonance, all the bonds between carbon atoms in C_6H_6 are equivalent and have the same strength and equal lengths.

6.

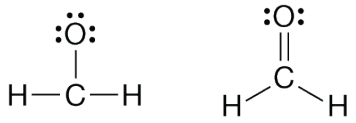


Diagram 1 Diagram 2

Which of the diagrams above best represents the CH_2O molecule, and why?

(A) Diagram 1, because all bond angles are 180°.

(B) Diagram 1, because all atoms have a formal charge of 0.

(C) Diagram 2, because the molecule has a trigonal pyramidal shape.

(D) Diagram 2, because all atoms have a formal charge of 0. ✓

Answer D

Correct. Diagram 2 represents a molecule in which both C and O have a complete octet and all atoms have a formal charge of zero.

2.6

7.

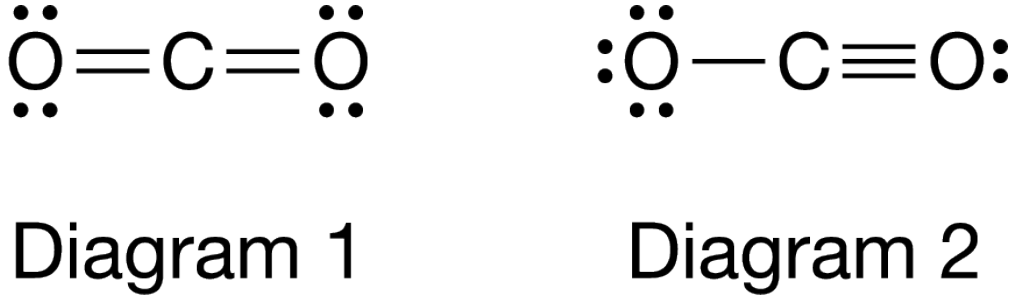


Diagram 1 Diagram 2

Which of the Lewis diagrams shown above is the more likely structure of CO_2, and why?

(A) Diagram 1, because all the atoms have a formal charge of 0. ✓

(B) Diagram 1, because double bonds are stronger than triple bonds.

(C) Diagram 2, because all the atoms have a formal charge of 0.

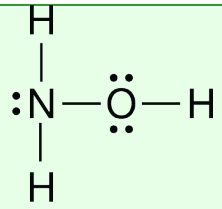
(D) Diagram 2, because triple bonds are stronger than double bonds.

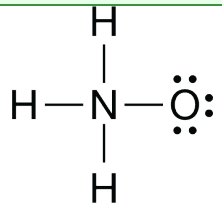
Answer A

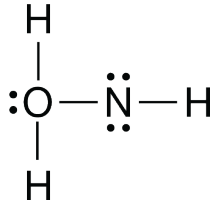
Correct. When the formal charge of each atom in diagram 1 is determined using the rule $\left(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - \left(\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}\right)\right)$, the results are zero for each of the three atoms. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero. And in diagram 2, the results for the three atoms (going left to right) are -1, 0, and $+1$, respectively.

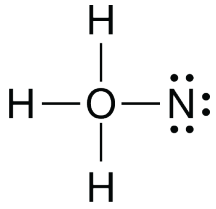
8. Based on formal charges, which of the following is the best Lewis electron-dot diagram for H_3NO?

2.6

(A)  ✓

(B) 

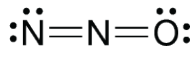
(C) 

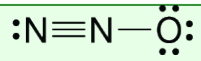
(D) 

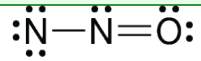
Answer A

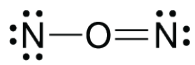
Correct. When the formal charge of each atom in this diagram is determined using the rule (formal charge of an atom = the number of valence electrons of the atom − (the number of electrons in lone pairs + half the number of electrons in bonds with other atoms)), the sum of the formal charges on all the atoms in H_3NO is equal to zero. A sum of formal charges on all atoms equal to zero is preferable.

9. Which of the following Lewis diagrams best represents the bonding in the N_2O molecule, considering formal charges?

(A) 

(B)  ✓

(C) 

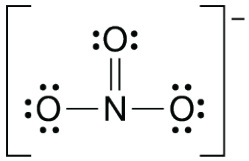
(D) 

2.6

Answer B

Correct. When the formal charge of each atom in this diagram is determined using the rule: $(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - (\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}))$. The sum of the formal charges on all the atoms in N_2O is equal to zero. The most favored structure considers the minimization of the formal charges and the electronegativities of the atoms. Oxygen is more electronegative than nitrogen; therefore, the Lewis diagram that shows a negative formal charge on the oxygen atom is the best for depicting the molecule.

10.



Which of the following statements, if true, would support the claim that the NO_3^- ion, represented above, has three resonance structures?

(A) The NO_3^- ion is not a polar species. ✓

(B) The oxygen-to-nitrogen-to-oxygen bond angles are 90°.

(C) One of the bonds in NO_3^- is longer than the other two.

(D) One of the bonds in NO_3^- is shorter than the other two.

Answer A

Correct. If the NO_3^- ion did NOT have three equivalent structures, then the ion would be polar because of the unequal distribution of charge within the ion. Equal contributions of the three resonance forms to the structure of the ion result in a planar ion with three equivalent polar nitrogen-oxygen bonds. Due to the trigonal planar symmetry of the NO_3^- ion, the negative dipoles cancel out.

11. Which of the following molecules has an angular (bent) geometry that is commonly represented as a resonance hybrid of two or more electron-dot structures?

(A) CO_2

(B) O_3 ✓

(C) BeF_2

(D) OF_2

2.6

12. Resonance is most commonly used to describe the bonding in molecules of which of the following?

(A) CO_2

(B) O_3 ✓

(C) H_2O

(D) SF_6

课时 2.7 VSEPR and Hybridization（45 题）

含答案解析 | 沿用原编号（本课时无删改）

2.7

1. According to the VSEPR model, the progressive decrease in the bond angles in the series of molecules CH_4, NH_3, and H_2O is best accounted for by the

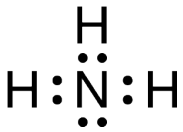
(A) increasing strength of the bonds

(B) increasing electronegativity of the central atom

(C) increasing number of unshared pairs of electrons ✓

(D) decreasing repulsion between hydrogen atoms

2.



Based on the Lewis diagram for NH_3, shown above, the H-N-H bond angle is closest to which of the following?

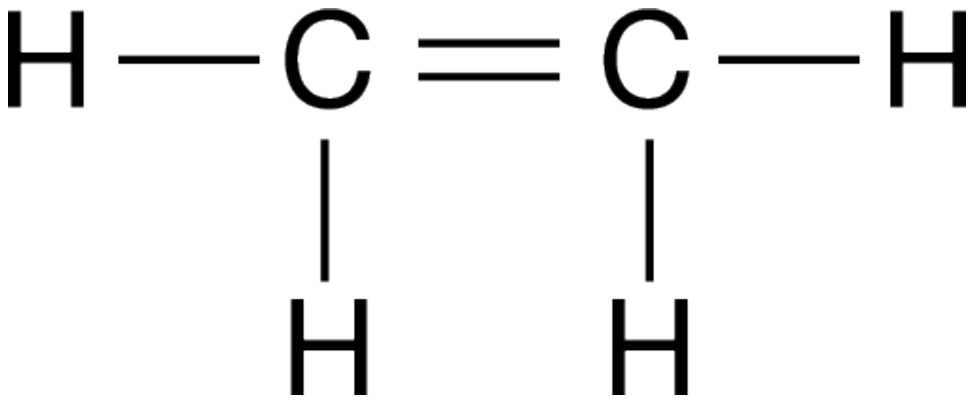
(A) 60°

(B) 90°

(C) 109.5° ✓

(D) 120°

3.



A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the H-C-H bond angles are closest to

(A) 90°

(B) 109.5°

(C) 120° ✓

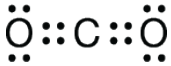
(D) 180°

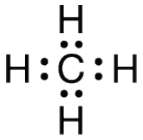
2.7

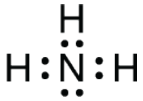
Answer C

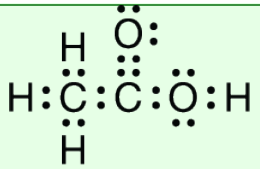
Correct. With three electron domains around each C atom, their geometry is trigonal planar (which minimizes repulsion among the three electron pairs). Therefore the H-C-H bond angle is approximately 120°.

4. Which of the following complete Lewis diagrams represents a molecule containing a bond angle that is closest to 120°?

(A) 

(B) 

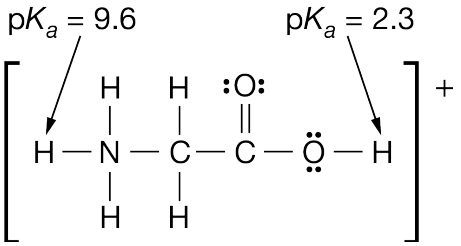
(C) 

(D) 

Answer D

Correct. The carbon atom attached to the oxygen atoms exhibits sp^2 hybridization, thus the bond angles around it are close to 120°. Another explanation is, the carbon atom attached to the two oxygen atoms is surrounded by three electron domains, which, according to VSEPR theory, will repel one another to produce a trigonal planar geometry around that atom.

2.7



The structural formula of the glycinium cation is shown above. Arrows indicate the $\mathrm{p}K_a$ values for the labile protons in the molecule.

5. What is the approximate $\mathrm{H-O-C}$ bond angle in the glycinium cation?

(A) 180°

(B) 120°

(C) 105° ✓

(D) 90°

Answer C

Correct. The oxygen atom in the $\mathrm{H-O-C}$ bond has two lone pairs of electrons, so the geometry of the electron domains around the oxygen atom is roughly tetrahedral. The bond angle is thus close to 109°.

6. Which of the following is true about the geometry of the glycinium cation?

(A) The leftmost C atom and all the atoms directly bonded to it lie in the same plane.

(B) Both C atoms and both O atoms lie in the same plane. ✓

(C) The $\mathrm{N-C-C}$ bond angle is 180°.

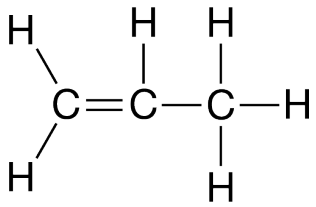
(D) The geometry around the N atom is planar.

Answer B

Correct. The carbon atom bonded to two oxygen atoms and another carbon atom does not have any nonbonding pairs of electrons, thus the central carbon atom has three electron domains. According to VSEPR theory, all four atoms are in the same plane (sp^2 hybridization).

2.7

7.



Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?

(A) The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓

(B) The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is 180°.

(C) The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.

(D) The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

Answer A

Correct. Double bonds between two atoms are always stronger than single bonds between those same two atoms because in a double bond there are more electrons between the two positive nuclei of the atoms, which results in a stronger bond.

Reaction 1: $CaC_2(s) + 2\ H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(s)$

Reaction 2: $NaOCl(aq) + 2\ HCl(aq) \rightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$

Reaction 3: $C_2H_2(g) + Cl_2(g) \rightarrow C_2H_2Cl_2(g)$

8. When Reaction 3 occurs, does the hybridization of the carbon atoms change?

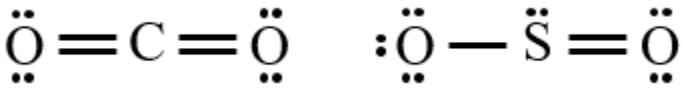
(A) Yes; it changes from *sp* to sp^2. ✓

(B) Yes; it changes from *sp* to sp^3.

(C) Yes; it changes from sp^2 to sp^3.

(D) No; it does not change

2.7

9. 

Lewis electron-dot diagrams for CO_2 and SO_2 are given above. The molecular geometry and polarity of the two substances are

(A) the same because the molecular formulas are similar

(B) the same because C and S have similar electronegativity values

(C) different because the lone pair of electrons on the S atom make it the negative end of a dipole

(D) different because S has a greater number of electron domains (regions of electron density) surrounding it than C has ✓

Answer D

This option is correct. Both the C-O and S-O bonds are polar, due to the different electronegativities of the bonded atoms. However, the CO_2 molecule is linear and so the bond dipoles cancel and the molecule has no net dipole. In SO_2, the molecule is bent and so the bond dipoles do not cancel and a net dipole is present. The polarity of the two substances differs because of the differences in geometry, which is a direct consequence of the number of electron domains around the central atom (2 in CO_2 and 3 in SO_2), as stated in this option.

Refer to the following diatomic species

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2

10. Contains 1 sigma (σ) and 2 pi (π) bonds

(A) B_2

(B) N_2 ✓

(C) O_2

(D) F_2

2.7

11. Has a bond order of 2

(A) B_2

(B) N_2

(C) O_2 ✓

(D) F_2

12. Has the largest bond-dissociation energy

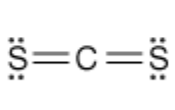
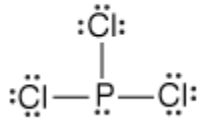
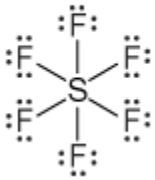
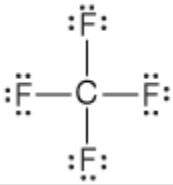
(A) B_2

(B) N_2 ✓

(C) O_2

(D) F_2

13.

CS_2	PCl_3	SF_6	CF_4
			

Based on the diagrams shown above, which of the following would have the greatest dipole moment, and why?

(A) CS_2, because the double bonds create an area of higher electron density around the C atom.

(B) PCl_3, because the Cl atoms draw electron density away from the P atom and the bond dipoles do not cancel one another. ✓

(C) SF_6, because it has the greatest number of polar bonds.

(D) CF_4, because the large electronegativity difference between C and F results in very polar bonds.

Answer B

Correct. PCl_3 is predicted by Valence Shell Electron Pair Repulsion (VSEPR) theory to have a trigonal pyramidal molecular shape. The central P atom has four electron domains around it, giving a tetrahedral electronic geometry with one of the positions occupied by a lone pair of electrons. The bond dipoles would not cancel, and PCl_3 would have a nonzero dipole moment.

2.7

Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) SO_2

(C) N_2

(D) F_2

14. Is the molecule in which the intramolecular forces are strongest

(A) $BeCl_2$
(B) SO_2
(C) N_2 ✓
(D) F_2

15. Is a polar molecule

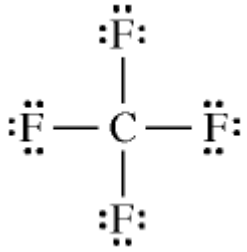
(A) $BeCl_2$
(B) SO_2 ✓
(C) N_2
(D) F_2

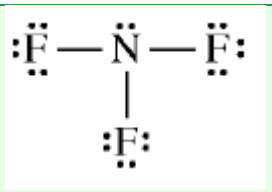
16. The geometry of the SO_3 molecule is best described as

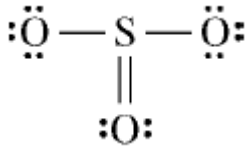
(A) trigonal planar ✓
(B) trigonal pyramidal
(C) bent
(D) tetrahedral

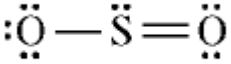
17. Which of the following Lewis electron-dot diagrams represents the molecule that contains the smallest bond angle?

2.7

(A) 

(B)  ✓

(C) 

(D) 

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The question refers to the following species.

(A) H_2O
(B) NH_3
(C) BH_3
(D) CH_4

18. Has a trigonal-pyramidal molecular geometry

(A) H_2O
(B) NH_3 ✓
(C) BH_3
(D) CH_4

19. Is predicted to have the largest bond angle

2.7

(A) H_2O
(B) NH_3
(C) BH_3 ✓
(D) CH_4

---

20. Which of the following has molecules with a pyramidal shape?

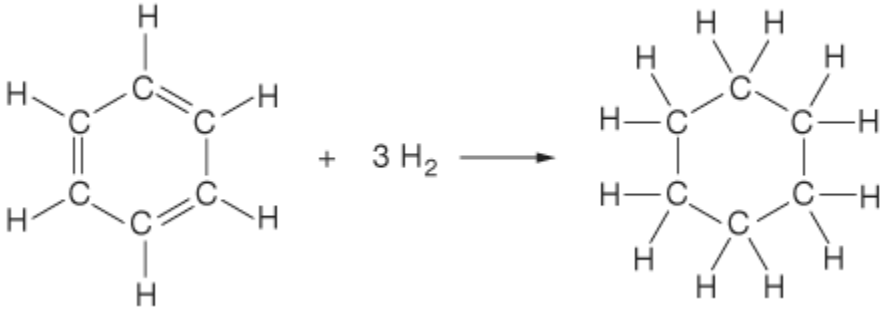
(A) $NH_3(g)$ ✓
(B) $BH_3(g)$
(C) $H_2S(g)$
(D) HBr(g)

21. H - C ≡ C - H

What is the hybridization of the carbon atoms in a molecule of ethyne, represented above?

(A) *sp* ✓
(B) sp^2
(C) sp^3
(D) dsp^2

22.



In the reaction represented above, which of the following sets correctly identifies the hybridization of the C atoms before and after the reaction occurs?

Set	Before	After
A	sp^2	sp^2
B	sp^2	sp^3
C	sp^3	sp^2
D	sp^3	sp^3

2.7

(A) Set A

(B) Set B ✓

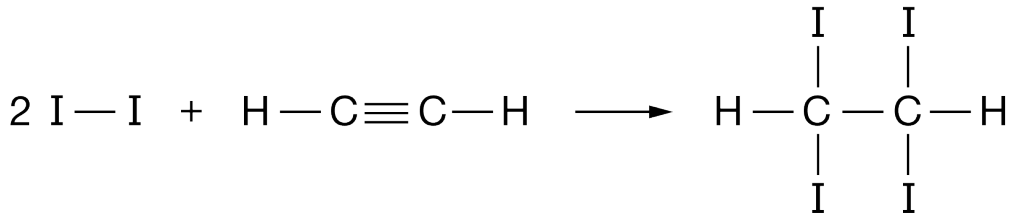
(C) Set C

(D) Set D

Answer B

Correct. In the C_6H_6 molecule, each carbon atom is surrounded by three electron domains in a trigonal planar geometry, and the hybridization of the orbitals of each carbon atom is sp^2. In the C_6H_{12} molecule, each carbon atom is surrounded by four electron domains in a tetrahedral geometry, and the hybridization of the orbitals of each carbon atom is sp^3.

23. The reaction between I_2 and C_2H_4 is represented by the following equation.



How does the hybridization of the carbon atoms change from reactants to products?

(A) sp to sp^2

(B) sp to sp^3 ✓

(C) sp^2 to sp

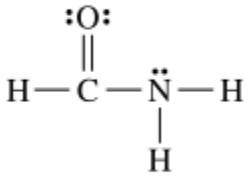
(D) sp^3 to sp

Answer B

Correct. Each C atom in C_2H_4 has two electron domains, so they have sp hybridization. Each C atom in $C_2H_2I_4$ has four electron domains, so they have sp^3 hybridization. The hybridization of the carbon atoms therefore changes from sp to sp^3.

2.7

24.



The complete Lewis electron-dot diagram for $HCONH_2$ is shown above. Identify the hybridization of the valence orbitals for the carbon atom and the approximate hydrogen-carbon-nitrogen bond angle according to VSEPR theory.

(A) sp hybridized and 180°

(B) sp^2 hybridized and 120° ✓

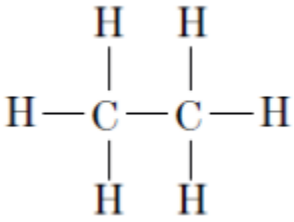
(C) sp^2 hybridized and 180°

(D) sp^3 hybridized and 120°

Answer B

Correct. The carbon atom has three electron domains: the $C-H$ bond, the $C-N$ bond, and the $C=O$ bond. According to VSEPR theory, three domains around an atom achieve maximum separation and greatest stability if the domains are arranged in a trigonal planar fashion, with 120° bond angles. When there are three electron domains, the valence orbitals are sp^2 hybridized.

25.

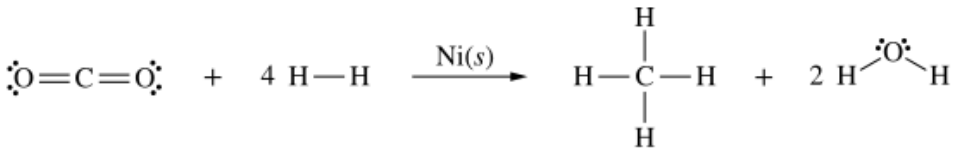


The hybridization of the carbon atoms in the molecule represented above can be described as

2.7

(A) sp
(B) sp^2
(C) sp^3 ✓
(D) dsp^2

In the gas-phase reaction represented by the following equation, methane (CH_4) and water (H_2O) are produced from carbon dioxide (CO_2) and hydrogen (H_2) at high temperature and pressure.



26. Which of the following identifies the hybridization of the valence orbitals of the central atoms of the methane and water molecules?

	CH_4	H_2O
A	sp^2	sp
B	sp^2	sp^2
C	sp^3	sp^2
D	sp^3	sp^3

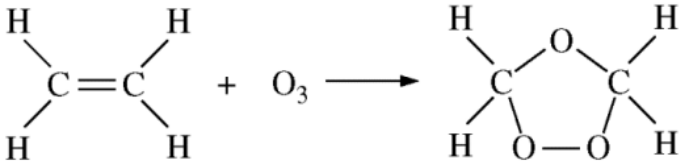
(A) A
(B) B
(C) C
(D) D ✓

Answer D

Correct. The hybridization of the valence orbitals of the central atoms of CH_4 and H_2O are both sp^3 because both have four electron domains around the central atom.

2.7

27.



In the reaction represented above, what is the hybridization of the C atoms before and after the reaction occurs?

(A)

Before	After
sp	sp^2

(B)

Before	After
sp	sp^3

(C)

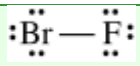
Before	After
sp^2	sp

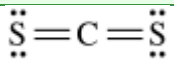
(D) ✓

Before	After
sp^2	sp^3

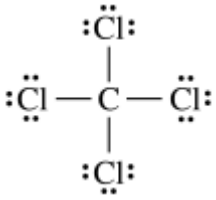
28. Which of the following molecules has the largest dipole moment?

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(A)  ✓

(B) 

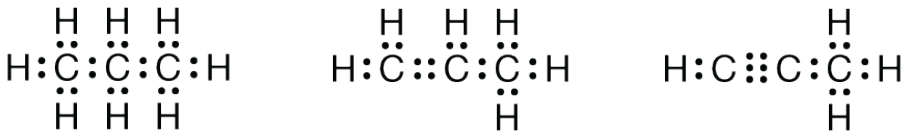
(C) 

(D) 

Answer A

Correct. The molecule with the largest dipole moment has bond dipoles that do not cancel as a result of the molecular shape. The BrF molecule has a bond dipole and the molecule is not symmetrical, so the bond dipoles do not cancel.

29.



Molecule 1 Molecule 2 Molecule 3

Lewis diagrams of molecules of three different hydrocarbons are shown above. Which of the following claims about the molecules is best supported by the diagrams?

(A) All the atoms in molecule 1 lie in one plane.

(B) All the molecules have the same empirical formula.

(C) The C-C-C bond angle in molecule 2 is close to 180°.

(D) The strongest carbon-to-carbon bond occurs in molecule 3. ✓

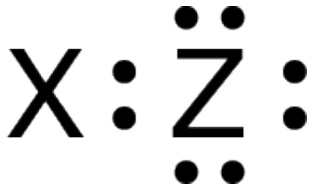
Answer D

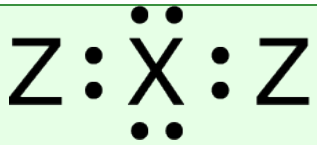
Correct. Molecule 3 contains a triple bond between two carbon atoms. Between two of the same kind of

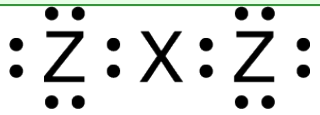
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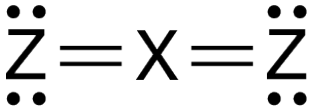
atom, triple bonds are stronger than double bonds, which in turn are stronger than single bonds.

30. In the following diagrams, elements are represented by X and Z, which form molecular compounds with one another. Which diagram represents a molecule that has a bent molecular geometry?

(A) 

(B)  ✓

(C) 

(D) 

Answer B

Correct. There are four electron domains around the central atom X, and their geometry is tetrahedral, which minimizes the repulsion between the four electron pairs. The Z-X-Z bond angle is approximately 109.5°, and the molecular geometry is bent.

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) CO_2
(B) H_2O
(C) CH_4
(D) C_2H_4
(E) PH_3

31. The molecule with the largest dipole moment

2.7

(A) CO_2

(B) H_2O ✓

(C) C_2H_4

(D) PH_3

32. The molecule that has trigonal pyramidal geometry

(A) CO_2

(B) H_2O

(C) C_2H_4

(D) PH_3 ✓

---

33. Which of the following molecules contains exactly three sigma (σ) bonds and two pi (π) bonds?

(A) C_2H_2 ✓

(B) CO_2

(C) HCN

(D) SO_3

34. Of the following molecules, which has the largest dipole moment?

(A) CO

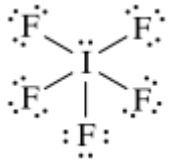
(B) CO_2

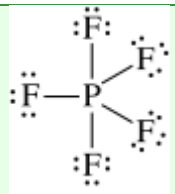
(C) HF ✓

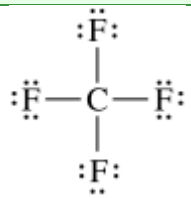
(D) F_2

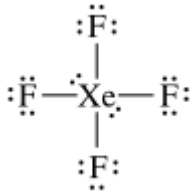
35. Which of the following represents a molecule with a trigonal bipyramidal geometry?

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(A) 

(B)  ✓

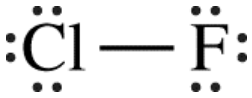
(C) 

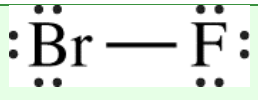
(D) 

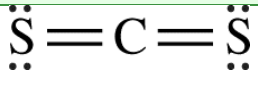
Answer B

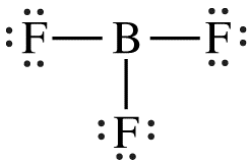
Correct. The Lewis diagram for a molecule having trigonal bipyramidal geometry must have five bonding pairs and no nonbonding pairs of electrons around the central atom. This response has the required arrangement of bonding and nonbonding pairs around the central P atom.

36. Which of the following Lewis electron-dot diagrams represents the molecule that is the most polar?

(A) 

(B)  ✓

(C) 

(D) 

2.7

37. Which of the following molecules is nonpolar but has polar covalent bonds?

(A) H_2O_2

(B) H_2O

(C) CCl_4 ✓

(D) CH_2Cl_2

38. Which of the following arranges the molecules N_2, O_2, and F_2 in order of their bond enthalpies, from least to greatest?

(A) $F_2 < O_2 < N_2$ ✓

(B) $O_2 < N_2 < F_2$

(C) $N_2 < O_2 < F_2$

(D) $N_2 < F_2 < O_2$

39. Pi (π) bonding occurs in each of the following species EXCEPT

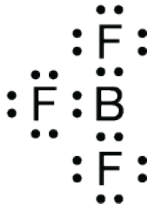
(A) CO_2

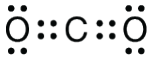
(B) CN^-

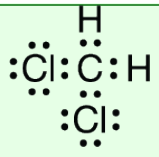
(C) C_6H_6

(D) CH_4 ✓

40. Which of the following Lewis diagrams represents a molecule that is polar?

(A) 

(B) 

(C)  ✓

(D) 

Answer C

Correct. In the CH_2Cl_2 molecule, the H and Cl atoms are in a tetrahedral arrangement around the

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central C atom. The two electronegative chlorine atoms create a partial negative charge on their side of the carbon atom, leaving a partial positive charge on the opposite (hydrogen) side of the carbon atom. This separation of charge constitutes a dipole, and the molecule is polar.

An equimolar mixture of SO_2 and O_2 is placed in a rigid container at $298\ K$. The initial pressure in the container is $0.20\ atm$. A catalyst is added to the mixture, and the reaction represented below occurs.

$$2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g) \qquad \Delta H^\circ = -200\ kJ/mol_{rxn}$$

41. Which of the following gives the shape of the SO_2 molecule and best helps explain why it has that shape?

(A) It is linear, because there are two electron domains around the sulfur atom.

(B) It is linear, because there are three electron domains around the sulfur atom.

(C) It is bent, because there are two electron domains around the sulfur atom.

(D) It is bent, because there are three electron domains around the sulfur atom. ✓

Answer D

Correct. A Lewis electron-dot diagram for SO_2 can be drawn with two equivalent resonance structures. S is the central atom, and there is one single bond and one double bond, with resonance between the two bonds. The S atom has one nonbonding pair of electrons. Since there are two bonds and one nonbonding pair of electrons on the S atom, the molecule is bent.

---

42. The BF_3 molecule is nonpolar, whereas the NF_3 molecule is polar. Which of the following statements accounts for the difference in polarity of the two molecules?

(A) In NF_3, each F is joined to N with multiple bonds, whereas in BF_3 , each F is joined to B with single bonds.

(B) N — F bonds are polar, whereas B — F bonds are nonpolar.

(C) NF_3 is an ionic compound, whereas BF_3 is a molecular compound.

(D) Unlike BF_3, NF_3 has a nonplanar geometry due to an unshared pair of electrons on the N atom. ✓

43. Which of the molecules represented below contains carbon with sp^2 hybridization?

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(A) CH_2Cl_2

(B) C_2H_6

(C) $C_2H_2Cl_2$ ✓

(D) $C_2H_4Cl_2$

44. Which of the following has a zero dipole moment?

(A) HCN

(B) SO_2

(C) NO_2

(D) PF_5 ✓

45. Which of the following species is NOT planar?

(A) CO_3^{2-}

(B) NO_3^-

(C) ClF_3

(D) PCl_3 ✓

Unit 2 Progress Check FRQ

1 大题（含评分标准），与课时题库无重复

Unit 2 Progress Check: FRQ

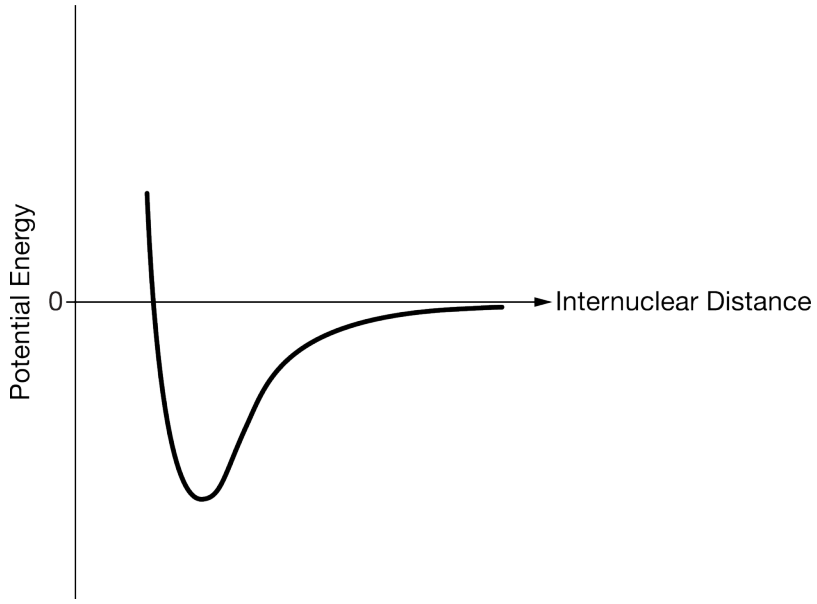
1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about N_2 and N_2H_4.

(a) In the box below, draw the complete Lewis electron-dot diagram of N_2 .

(b) Based on the Lewis electron-dot diagram that you drew, is the N_2 molecule polar? Explain.

The following graph shows the potential energy of two nitrogen atoms versus the distance between their nuclei.



(c) On the graph, indicate the distance that corresponds to the bond length of the N_2 molecule by placing an X on the horizontal axis.

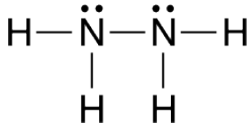
(d) On the graph, which shows the potential energy curve of two N atoms, carefully sketch a curve that corresponds to the potential energy of two O atoms versus the distance between their nuclei.

Unit 2 Progress Check: FRQ

N_2 can react with H_2 to form the compound N_2H_4.

(e) A sample of N_2H_4 has a mass of 25 g. Identify the numerical quantity that is needed to convert the number of grams of N_2H_4 to the number of moles of N_2H_4. (You do not need to do the actual calculation.)

(f) The Lewis electron-dot diagram of N_2H_4 is shown below.



(i) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4, compare the length of the nitrogen-to-nitrogen bond in N_2 with the length of the nitrogen-to-nitrogen bond in N_2H_4.

(ii) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4, compare the strength of the nitrogen-to-nitrogen bond in N_2 with the strength of the nitrogen-to-nitrogen bond in N_2H_4.

(iii) Identify the hybridization of the N atoms in N_2H_4.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The response meets both of the criteria below:

- ☐ The diagram shows a total of ten electrons.
- ☐ The diagram shows three pairs of electrons between the N two atoms.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the N_2molecule is not polar, because the three bonding pairs of electrons are shared equally between the two N atoms (they have the same electronegativity), and each N atom has a nonbonding electron pair (the molecule is completely symmetrical).

Part (c)

Unit 2 Progress Check: FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response shows an X drawn on the axis and located directly above the lowest part (minimum) of the potential energy curve. The lowest potential energy on the curve corresponds to the most stable internuclear distance (i.e., bond length).

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The sketched curve meets both of the criteria below:

- ☐ The curve has its minimum to the right of where the given curve has its minimum (because the length of the O-O double bond is greater than that of the N-N triple bond).
- ☐ The minimum value of the curve is higher than the minimum of the given curve (because the energy of the O-O double bond is lower than that of the N-N triple bond).

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the molar mass of $\mathrm{N_2H_4}$ (or $32\ \mathrm{g/mol}$) is needed to solve the problem.

Part (f)(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the nitrogen-to-nitrogen bond in $\mathrm{N_2}$ is shorter than the nitrogen-to-nitrogen bond in $\mathrm{N_2H_4}$. (No explanation is required, however, the length of a triple bond is shorter than that of a single bond.)

Unit 2 Progress Check: FRQ

Part (f)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the nitrogen-to-nitrogen bond in N_2 is stronger than the nitrogen-to-nitrogen bond in N_2H_4 . (No explanation is required, however, the strength of a triple bond is greater than that of a single bond.)

Part (f)(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response indicates that the hybridization of the N atoms is sp^3 . (No explanation is required, however, the N atoms in N_2H_4 are surrounded by four electron domains, one of which is a lone pair. This situation corresponds to sp^3 , hybridization.)